A Randomized, Double-blind, Placebo-controlled, Multicenter Phase III Clinical Study on Toripalimab Combined with Platinum-based Doublet Chemotherapy in Subjects with Resectable, Stage II-III Non-small Cell Lung Cancer

Protocol Number: JS001-029-III-NSCLC
Study Phase: Phase III
Investigational Product: Toripalimab Injection (JS001)

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Principal Investigator: Professor Shun Lu
Leading study site: Chest Hospital, Shanghai Jiaotong University

Protocol Version No.: Version 6.0
Version Date: August 23, 2022

Sponsor: Shanghai Junshi Biosciences Co., Ltd.

Confidentiality Statement

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Summary of Amendments

File	Version date	Summary of Changes and Reason for Amendment
Version 1.0	March 20, 2019	N/A
Version 2.0	June 28, 2019	The protocol was modified to address comments from the Center for Drug Evaluation (CDE) in China. Major changes included: 1. Addition of event-free survival (EFS) as a co-primary endpoint and inclusion of an independent radiological committee (IRC) for evaluation of tumor-based radiologic endpoints. 2.An adjuvant treatment period was added. 3.Statistical analysis plan was modified to correspond to the addition of the co-primary endpoint of EFS and IRC evaluation of tumor-based radiologic endpoints.
Version 3.0	July 3, 2020	In accordance with clinical practice, emerging data from external studies and evolving clinical practice guidelines, the following major modifications were made: 1.Inclusion criteria modified to allow enrollment of subjects with resectable, stage IIIB NSCLC and clarification of the criteria for unresectable disease; 2.Revision to allow confirmation of mediastinal lymph node involvement by PET-CT or biopsy by endobronchial ultrasound (EBUS) or mediastinoscopy; 3. Included specifications for the acceptable procedures for acquisition of tumor biopsy specimens (e.g., core needle biopsy); 4.Eligibility criteria for subjects with active hepatitis B were modified; 5.Sample size was increased and statistical analysis corresponding changes in the study population and sample size were modified and the plan for interim analysis of major pathologic response (MPR) was removed;

		6. Plan for Type I error control for multiple comparisons modified from a hierarchical testing procedure to a weighted Bonferroni test with recycling of alpha; 7. The numbers of events required for the interim and final analyses of EFS were revised based on changes to the Type I error control for multiple comparisons.
Version 4.0	October 26, 2021	According to clinical needs and known beneficial study population, after the CDE approval, Stage II population was added and the corresponding sample size and analysis plan was modified; meanwhile, clarify some sections in the original protocol, and mainly update the following: Study population: Inclusion criteria modified to include subjects with stage II NSCLC; ● Sample size: Increased from 406 to 500 subjects; ● Stratification factors for randomization: modified from IIIA vs. IIIB to II vs. IIIA vs. IIIB ● Primary study endpoints: Rate of major pathologic response (MPR) assessed by BIPR in Stage III NSCLC, investigator-evaluated event-free survival (EFS) were revised to rate of MPR assessed by BIPR in Stage III population, rate of MPR assessed by investigator in Stage III population, rate of MPR assessed by BIPR in Stage II-III population, and rate of EFS assessed by investigator in Stage II-III population; ● Inclusion criterion 3: Modified to include subjects with resectable, Stage II NSCLC; ● Statistical hypothesis: According to the primary study objective, the original 2 study hypotheses were adjusted to 4 study hypotheses;

		● Statistical analysis: Modified to provide new and revised analysis plans for the ITT and the Stage III NSCLC populations, respectively. ● Interim and final analyses: A final analysis of MPR, an interim analysis of EFS, and a final analysis of EFS were revised to a final analysis of MPR in Stage III subgroup, an interim analysis of EFS in Stage II-III population (final analysis of MPR in the Phase II-III population and interim analysis of EFS in the Phase III population will be performed at the same time), and a final analysis of EFS in Stage II-III population; ● Multiplicity comparison: reformulated the multiplicity comparison strategy for the 4 statistical hypotheses as the statistical hypotheses were changed to 4 ● Exclusion criteria 3: Revised for clarification to unequivocally state that non-squamous NSCLC subjects with EGFR-sensitive tumor mutations are not eligible for this study. Exclusion criterion 16: Revised to clarify that recent myocardial infarction, cerebral infarction or current hepatic cirrhosis are exclusion criteria; Study flow chart and screening visit updated to state that if the tumor specimen does not provide an evaluable result for PD-L1, the subject does not need to undergo a second biopsy and the subject will be stratified in the group with PD-L1 <1% or non-evaluable PD-L1 ● The window period for study visits clarified in the flowchart and text; ● Modified to clarify that pulmonary function and virology testing performed during screening the period do not need to be repeated and that an ECOG PS score and

		physical examination conducted within 72 hours prior to the first dose may be used for the C1D1 evaluation; ● Description of the process and requirements for unblinding of treatment assignment were expanded to provide more detail; ● Clarified that subjects who have completed the maintenance treatment period should not initiate new anti-tumor therapy prior to documentation of disease progression. ● Modified to collect toripalimab PK and ADA samples and include plan for analyses of this data. ● Based on new requirements from NMPA, the section on SAE reporting includes new subsections describing reporting requirement for events of deaths on study, second malignancy and overdose.
Version 5.0	May 23, 2022	● Interim Analysis and Statistical Analysis Plan: Defined the final analysis of MPR assessed by BIPR in Stage III population; revised the requirements for the interim analysis and final analysis of investigator-evaluated EFS in Stage III population; and the requirements for the final analysis of investigator-evaluated EFS in Stage II-III population. ● Sample size: revised the number of events required for final analysis of investigator-evaluated EFS for Stage III and Stage II-III population ● Multiple comparison: adjusted multiple comparison strategy; ● Adjusted text description, e.g., PET-CT was clarified as whole-body PET-CT ● Added allowable dose volume error in 5.2.2 - Dose Tracking to meet clinical practice;

		● Added description of combination phase at 5.3.2.1 - Toripalimab Dose Adjustment to clarify the scope of application of dose adjustment principles;
Version 6.0	August 23, 2022	● Sample size calculation: Updated (initial allocation) significance level and (nominal) power values. ● Interim analysis and analysis plan: Deleted the plan for the independent final analysis of MPR assessed by BIPR in Stage III population; ● Multiple comparison: adjusted multiple comparison strategy; ● Criteria for discontinuation of treatment and assessment of efficacy: Deleted the parenthesis descriptions in "Progression of disease after surgery (R2 resection)" and "Local recurrence or distant metastasis (R0/R1 resection)" for reasonable data entry based on actual clinical situation. ● Efficacy assessments: Added instructions to align image assessment rules between sites and between investigator IRCs.

Sponsor Signature Page

Protocol title:	A Randomized, Double-blind, Placebo-controlled, Multicenter Phase III Clinical Study on Toripalimab Combined with Platinum-based Doublet Chemotherapy in Subjects with Resectable, Stage II-III Non-small Cell Lung Cancer
Protocol Number:	JS001-029-III-NSCLC
Version number and version date:	Version 6.0, August 23, 2022

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Shanghai Junshi Biosciences Co., Ltd.

Protocol leader	Signature	Date
Weihua Wang	_____	_____

Investigator's Statement

I have read the clinical study protocol (A Randomized, Double-blind, Placebo-controlled, Multicenter Phase III Clinical Study of Toripalimab Combined with Platinum-based Doublet Chemotherapy in Subjects with Resectable, Stage II-III Non-Small Cell Lung Cancer version 6.0, dated August 23, 2022). I hereby agree to conduct the clinical study in accordance with all the clauses outlined in the study protocol, current regulations, and ethical principles stated in the Declaration of Helsinki.

I agree to comply with the guideline of Good Clinical Practice and other applicable NMPA regulations/guidelines.

I agree that the confidential information contained in this document shall not be used for other purposes than the evaluation or conduct of the clinical study, without prior written consent by Shanghai Junshi Biosciences Co., Ltd.

Principal investigator's name:___ (Print)

Signature:_____ Date:_____DD/MM/YYYY

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Protocol Synopsis

Study Title	A Randomized, Double-blind, Placebo-controlled, Multicenter Phase III Clinical Study on Toripalimab Combined with Platinum-based Doublet Chemotherapy in Subjects with Resectable, Stage II-III Non-small Cell Lung Cancer
Protocol No.	JS001-029-III-NSCLC
Version No.	V6.0
Sponsor	Shanghai Junshi Biosciences Co., Ltd.
Study Site	Chest Hospital, Shanghai Jiaotong University
Principal investigator	Professor Shun LU
Participating Sites	Approximately 56
Study Objectives	**Primary objectives:** • To compare the event-free survival (EFS), as evaluated by investigators, of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage III non-small cell lung cancer (NSCLC). • To compare the major pathologic response (MPR) rates, as evaluated by a central blinded independent pathologic review (BIPR) committee, of neoadjuvant toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage III NSCLC. • To compare the EFS, as evaluated by investigators, of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage II-III NSCLC. • To compare the MPR rates, as evaluated by a central BIPR committee, of neoadjuvant toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage II-III NSCLC.

	Secondary objectives: **Efficacy:** To compare the BIPR- and investigator-evaluated pathologic complete response (pCR) rates, IRC-evaluated EFS, disease-free survival (DFS), 2-year and 3-year overall survival (OS) rates, and OS of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage III and stage II-III NSCLC. **Safety:** To evaluate the safety, tolerability, feasibility of surgical resection, and incidence of pre- and postoperative complications in subjects with resectable stage II-III NSCLC treated with toripalimab combined with platinum-based doublet chemotherapy.
	Pharmacokinetics: To evaluate the pharmacokinetic properties of toripalimab. **Immunogenicity:** To evaluate the immunogenicity of toripalimab and its impact on pharmacokinetics, safety and efficacy. **Exploratory objective:** To explore potential predictive biomarkers for efficacy in tumor tissue specimens and peripheral blood samples (including PD-L1 expression in tissue specimens, tumor mutation burden [TMB], whole exome sequencing [WES], and ctDNA levels).
Study Endpoints	**Primary study endpoints:** • EFS as evaluated by investigators in Stage III NSCLC • MPR rate as evaluated by BIPR in Stage III NSCLC • EFS as evaluated by investigators in Stage II-III NSCLC • MPR rate as evaluated by BIPR in Stage II-III NSCLC **Secondary study endpoints:** • pCR rate as evaluated by BIPR and investigators in stage III NSCLC • EFS as evaluated by IRC in Stage III NSCLC • pCR rate as evaluated by BIPR and investigators in stage II-III NSCLC • EFS as evaluated by IRC in Stage II-III NSCLC

	• DFS as evaluated by IRC and investigators in Stage III NSCLC • DFS as evaluated by IRC and investigators in Stage II-III NSCLC • OS in Stage III NSCLC, and Stage II-III NSCLC • 2- and 3-year OS rates in Stage III NSCLC population, and Stage II-III NSCLC population • Safety: Adverse events (AEs), abnormal laboratory results, and treatment-related serious adverse events (SAEs) graded according to NCI CTCAE v5.0; feasibility of surgical resection, percentage of surgery delays or cancellations, change in type of surgery, and duration of surgery Pharmacokinetics: toripalimab blood concentrations Immunogenicity: incidence of antidrug antibodies (ADA), ADA titer and incidence of neutralizing antibodies (Nab) (in subjects who are ADA-positive). **Exploratory study endpoints:** Exploratory analysis of potential biomarkers related to efficacy (including PD-L1 expression in tissue specimens, TMB, WES, and change of ctDNA in peripheral blood samples).
Study Population	Histologically confirmed resectable stage II, stage IIIA and IIIB (N2) NSCLC (based on the American Joint Committee on Cancer [AJCC] Cancer Staging Manual, version 8), without epidermal growth factor receptor (EGFR) mutations or anaplastic lymphoma kinase (ALK) translocation for non-squamous NSCLC, male or female, aged 18 to 70 years, and Eastern Cooperative Oncology Group (ECOG) performance status score 0 to 1. Approximately 500 subjects are to be enrolled.
Study Design	This is a randomized, double-blind, placebo-controlled, multicenter phase III clinical study designed to evaluate the efficacy and safety of toripalimab (JS001) combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet in subjects with resectable, stage II-III NSCLC. Subjects who meet all the inclusion criteria but do not meet any exclusion criteria are randomized into 2 groups at a ratio of 1:1 according to the stratification factors below: • Disease stage: II vs. IIIA vs. IIIB

- PD-L1 status: PD-L1 expression ≥1% vs. PD-L1<1% or non-evaluable
- Planned surgical type: pneumonectomy vs. lobectomy
- Histopathologic subtype: squamous cell carcinoma vs. non-squamous cell carcinoma

Neoadjuvant treatment should start within 3 days after randomization.

Experimental arm neoadjuvant phase: toripalimab 240 mg IV in combination with platinum-based doublet chemotherapy on Day 1 of each 21-day cycle for 3 cycles of neoadjuvant treatment.

Control arm neoadjuvant phase: placebo combined with platinum-based doublet chemotherapy on Day 1 of each 21-day cycle for 3 cycles of neoadjuvant treatment.

Subjects will undergo preoperative radiological evaluation 3-5 weeks after receipt of the last dose of neoadjuvant treatment on C3D1.

Subjects who remain operable will undergo radical resection as defined by the International Association for the Study of Lung Cancer 4-6 weeks after the last dose of neoadjuvant treatment. The pathologic specimen will be staged in accordance with the AJCC Cancer Staging Manual (8th edition). Resected tumor specimens will be assessed by pathologists at the study site for surgical margins and extent of resection (R0, R1, or R2); tumor tissue will also be submitted to the designated central laboratory for blinded assessment of pathologic response (MPR and pCR).

Experimental arm: following surgical resection, subjects will receive 1 cycle of toripalimab 240 mg plus adjuvant platinum-based chemotherapy 30 days after surgery.

Control arm adjuvant phase: following surgical resection, subjects will receive 1 cycle of placebo plus adjuvant platinum-based chemotherapy 30 days after surgery. For subjects receiving adjuvant chemotherapy without radiotherapy, maintenance treatment will begin 3 weeks after administration of adjuvant chemotherapy. If adjuvant radiotherapy is indicated, maintenance treatment will begin 30 days after the end of radiotherapy.

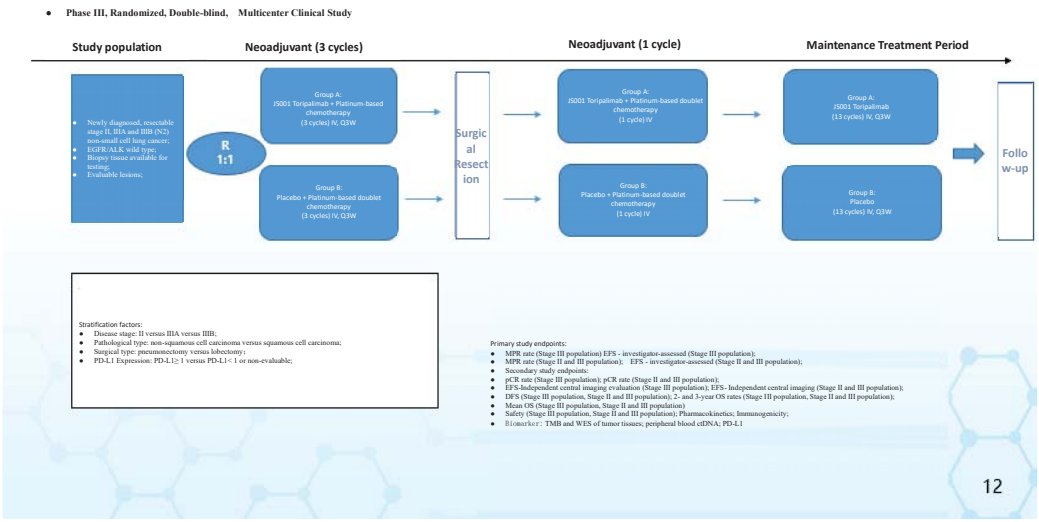
Experimental arm maintenance phase: toripalimab 240 mg IV once every 3 weeks for a total of 13 cycles.

Placebo arm maintenance phase: placebo once every 3 weeks for a total of 13

cycles.

Adverse events (AEs) will be monitored throughout the study and the severity will be graded according to the guidelines listed in the NCI CTCAE version 5.0. Subjects receiving study treatment who discontinue early will be followed up for safety. After discontinuation of study treatment, all subjects will be followed up for survival until death, withdrawal of consent, loss to follow-up, or termination of the study by the Sponsor, whichever occurs first.

Study Schema:



Study Treatment	Toripalimab: • With every 3 weeks as a dosing cycle, toripalimab 240 mg will be administered on Day 1 of each cycle via IV infusion over approximately 20-60 minutes. Platinum-based doublet chemotherapy: squamous cell carcinoma: paclitaxel or docetaxel + cisplatin or carboplatin; non-squamous cell carcinoma: pemetrexed + cisplatin or carboplatin. Taxanes: • With every 3 weeks as a dosing cycle, docetaxel 60-75 mg/m^2 or paclitaxel 175 mg/m^2 will be administered on Day 1 of each cycle via IV infusion. Pemetrexed: • With every 3 weeks as a dosing cycle, pemetrexed 500 mg/m^2 will be administered on Day 1 of each cycle via IV infusion for over 10 minutes. Carboplatin:

	• With every 3 weeks as a dosing cycle, AUC = 5 will be administered on Day 1 of each cycle via IV infusion for over 30-60 minutes or in accordance with local standard practices. Cisplatin: • With every 3 weeks as a dosing cycle, cisplatin 75 mg/m^2 will be administered on Day 1 of each cycle via IV infusion for over 60 minutes (-5 minutes / +10 minutes) or in accordance with local standard practices.
Inclusion Criteria	Subjects must meet all of the following inclusion criteria to be eligible for enrollment in this study: 1. Have sufficient understanding of this study and voluntarily sign the Informed Consent Form (ICF). 2. Aged 18 to 70 years old, male or female. 3. Treatment-naïve, histologically confirmed, resectable Stage II, IIIA, IIIB (N2) NSCLC (AJCC Cancer Staging Manual, version 8). The cTNM stage can be confirmed through positron emission tomography-computed tomography (PET-CT) or pathological biopsy. Resectable stage II NSCLC is defined as radically resectable as assessed by a qualified thoracic surgeon. Resectability is defined as resectable or potentially resectable according to the multidisciplinary consensus statement on the clinical management of subjects with stage III non-small cell lung cancer (version 2019). Resectable NSCLC includes IIIA (N0-1), some N2 with single-station mediastinal lymph node < 2 cm in shortest diameter and some T4 (satellite nodules present in adjacent lobes) N1. Potentially resectable NSCLC includes some stage IIIA and IIIB, including single-station N2 mediastinal lymph nodes with shortest diameter < 3 cm IIIA NSCLC, as well as T3 or T4 centrally located tumors determined as potentially resectable by a surgeon. Pathological puncture verification is strongly recommended for suspicious lesions detected by imaging studies that are presenting as clinically normal, which can lead to TNM stage changes in areas including but not limited to contralateral mediastinal lymph nodes, supraclavicular lymph nodes, solid/subsolid pulmonary nodules, and non-simple ground-glass opacity (GGO). 4. Measurable lesions as defined by the Response Evaluation Criteria in Solid

Tumors (RECIST) version 1.1;

5. Tumor tissue specimens should be provided for pathological evaluation to confirm the diagnosis and for the detection of PD-L1 tumor expression and biomarkers prior to randomization. Tumor tissue specimens can be freshly obtained, while archived samples obtained within 3 months prior to enrollment. Acceptable tumor tissue specimens must contain tumor cells and may be acquired through core needle biopsy, excisional biopsy during bronchoscopy, or surgical resection. Tumor samples acquired through fine-needle aspiration or bronchial brushing are not acceptable.
6. ECOG performance status score 0 or 1;
7. Adequate organ function:
   - Hematology: Absolute neutrophil count (ANC) ≥ 1500/μL; platelet ≥ 100000/μL; hemoglobin ≥ 9.0 g/dL or ≥ 5.6 mmol/L;
   - Renal function: Serum creatinine ≤1.5× upper limit of normal (ULN) or calculated creatinine clearance (CrCl) ≥60 mL/min (using Cockcroft-Gault formula).
   - Hepatic function: Total bilirubin ≤1.5× ULN or direct bilirubin within the normal range for subjects with a total bilirubin level >1.5× ULN; aspartate aminotransferase (AST) and alanine aminotransferase (ALT) ≤2.5× ULN.
   - Endocrine function: Thyroid-stimulating hormone (TSH) within the normal range. Note: subjects may still meet the inclusion criteria if TSH levels are outside the normal range at baseline, as long as T3 and free T4 are within the normal ranges;
   - Coagulation function: International normalized ratio (INR) or prothrombin time (PT), as well as activated partial thromboplastin time (aPTT) ≤1.5× ULN. Exception: For subject receiving anticoagulation therapy, PT or aPTT just needs to be within the proposed therapeutic range of the anticoagulant.
8. Being willing and able to comply with the visits, treatment plan, laboratory examinations, and other study procedures scheduled in the study;
9. Total lung function is able to withstand the proposed pneumonectomy, as evaluated by a surgeon;
10. Women of childbearing potential must undergo a serum pregnancy test within 3

	days prior to the first dose of the investigational product, and the result must be negative. Female subjects of childbearing potential and male subjects whose partners are women of childbearing potential must agree to use highly effective contraceptive methods during the study period and for at least 180 days after the last dose of the study drug.
Exclusion Criteria	Subjects cannot be enrolled in this study if they meet any of the following exclusion criteria: 1. Presence of locally advanced, unresectable, or metastatic disease. Unresectable disease is defined by the following criteria outlined in the multidisciplinary consensus statement on the clinical management of subjects with stage III non-small cell lung cancer[33]: Stage IIIC, Stage IIIB with N3 lymph node involvement, Stage IIIA or IIIB with N2 involvement of single-station mediastinal lymph node where the shortest diameter is ≥3 cm, N2 involvement is characterized by the fusion of multi-station lymph node where the shortest diameter is ≥2 cm on CT, T4 with invasion of the esophagus, heart, aorta, or other adjacent tissues, which in the opinion of the investigator, renders the subject unresectable. 2. Pancoast tumor, or in cases where the tumor histology corresponds to a large cell neuroendocrine carcinoma or sarcomatoid tumor. 3. Participants with known active EGFR-sensitive mutations or ALK translocation. EGFR and ALK mutation status must be identified for the subjects with non-squamous cell carcinoma. 4. Previous systemic anti-tumor therapy, including investigational products, for early-stage NSCLC. 5. History of (non-infectious) pneumonia/interstitial lung disease or ongoing pneumonia/interstitial lung disease that requires steroid treatment. 6. History of active tuberculosis. 7. Active infections requiring systemic treatment. 8. Subjects with any known or suspected autoimmune disease or immunodeficiency, with the following exceptions: patients with a history of hypothyroidism, if hormone replacement therapy is not needed or currently stable while receiving

hormone replacement therapy at physiological dose; subjects with stable type I diabetes whose blood glucose is controlled.

9. Uncontrolled active hepatitis B infection, defined as positive hepatitis B surface antigen [HBsAg] with detectable hepatitis B virus -deoxyribonucleic acid (HBV-DNA) above the ULN at the clinical laboratory of the study center, or active hepatitis C infection, defined as hepatitis C surface antibody seropositive and hepatitis C virus-ribonucleic acid (HCV-RNA) during screening period). Subjects with HBV-DNA <500 IU/mL within 28 days prior to randomization, who received antiviral therapy per local standard of care for at least 14 days and commit to continuous antiviral therapy during the study, are eligible.
10. Known human immunodeficiency virus (HIV) infection (e.g., HIV seropositive).
11. Vaccination with live vaccine within 30 days prior to the first dose, including but not limited to the following: mumps, rubella, measles, varicella/herpes zoster (varicella), yellow fever, rabies, Bacillus Calmette-Guérin (BCG), and typhoid vaccines (inactivated viral vaccines are permitted);
12. Grade ≥2 peripheral neuropathy.
13. Previous use of PD-1/PD-L1 blocking antibodies or drugs targeting another T cell co-signaling receptor (e.g., CTLA-4, OX-40).
14. History of severe allergic reactions to any monoclonal antibodies.
15. History of serious allergic reactions to pemetrexed, paclitaxel or docetaxel, cisplatin, carboplatin or prophylactic medications administered during chemotherapy.
16. Known serious or uncontrolled co-morbid disease, including but not limited to, hemodynamically unstable cardiovascular events within 6 months, symptomatic cerebrovascular events, cirrhosis of the liver with Child-Pugh A classification or higher.
17. History or current evidence of any disease, concomitant medication, or abnormal laboratory results that may confound the safety or efficacy assessment, interfere with the subject's participation throughout the full course of the study, or prevent the subject from receiving the maximum benefit from participating in the study, as judged by the investigators.
18. Presence of any malignancy other than non-small cell lung cancer within 5 years

	prior to the first dose. Exception: subjects with a history of a malignancy with negligible risk of metastasis or death (e.g., expected disease-free survival >5 years) and anticipate a complete cure after definitive treatment (e.g., appropriately treated carcinoma in situ of the cervix, basal or squamous cell skin cancer, or ductal breast carcinoma in situ treated) are eligible.
Efficacy Evaluation	Assessment of pathological response: Resected tumor specimens obtained at the time of definitive surgery must be submitted to the designated central laboratory for blinded assessment of pathological response. In addition, resected tumor specimens will be assessed by the pathologists for surgical margins and pathology response at the study center. Tumor imaging must include contrast-enhanced CT or MRI of the chest and other anatomic sites with suspicious lesions. Subjects who do not receive surgical treatment for causes outside the protocol-defined disease progression will continue attending imaging follow-ups, with the perioperative tumor assessment as baseline, once every 9 weeks until disease progression or the start of new anticancer therapy, whichever comes first. Reset the baseline 30 days after the operation. Radiological evaluation will be performed every 3 months for two years, and then every 6 months afterwards until disease progression or relapse, or initiation of new anti-tumor therapy, whichever occurs first. At the first postoperative radiological evaluation (postoperative baseline), if new lesions are identified in comparison with the preoperative imaging assessment, progressive disease should be recorded. Survival (OS) follow-up: Subjects will undergo regular survival assessments every 3 months after the last dose. Details regarding survival and subsequent anti-tumor treatment will be gathered through telephone follow-up sessions or other means until the subject's death, loss to follow-up, withdrawal of informed consent, or the study's termination by the Sponsor.
Safety Evaluation	Any adverse events occurring in subjects during the study, including clinical symptoms, abnormal vital signs, and abnormality laboratory results, will be recorded in terms of clinical features, severity, time of onset, duration, actions taken and final outcome, and their relationship with the study drug will be assessed and determined. Adverse events will be graded according to the National Cancer Institute

	Common Terminology Criteria for Adverse Events version 5.0 (NCI CTCAE V5.0).
Pharmacokinetic Evaluations	Blood samples will be collected to measure toripalimab concentrations prior to dosing on day 1 of each cycle 1 of the neoadjuvant and adjuvant treatment periods and cycles 1, 5, 9 and 13 of the maintenance treatment period. The study is a double-blind design, and while only the samples obtained from those who received toripalimab will be tested, blood samples will be collected from all subjects to maintain blinding.
Immunogenicity Evaluations	Blood samples will be collected to evaluate the presence of anti-drug antibodies (ADA) prior to dosing on day 1 of each cycle 1 of the neoadjuvant and adjuvant treatment periods and cycles 1, 5, 9 and 13 of the maintenance treatment period. In subjects who are ADA-positive, samples will be further analyzed for ADA titers and evaluated for the presence of neutralizing antibodies. The study is a double-blind design, and while only samples obtained from those who received toripalimab will be tested, blood samples will be collected from all subjects to maintain blinding.
Determination of Sample Size	The sample size is calculated based on investigator-evaluated EFS, and approximately 500 subjects with stage II-III NSCLC meeting enrollment criteria are scheduled to be randomized in this study to the experimental and control arms in a 1:1 ratio, with approximately 400 subjects enrolled in Stage III population. The final analysis of the investigator-evaluated EFS efficacy assessment for the Stage III population will be performed with the expectation of 199 investigator-evaluated EFS events observed in the Stage III population, resulting in an 86% nominal statistical power to detect the superiority of investigator-evaluated EFS compared to the control arm in the Stage III population at a 2-sided 0.05 significance level.

The final analysis of the investigator-evaluated EFS efficacy assessment for the Stage II-III population will be performed with the expectation of 220 investigator-evaluated EFS events observed in the Stage II-III population, resulting in an 89% nominal statistical power to detect the superiority of investigator-evaluated EFS in the Stage II-III population over the control arm at a 2-sided 0.05 significance level (after transmission of the significance level).

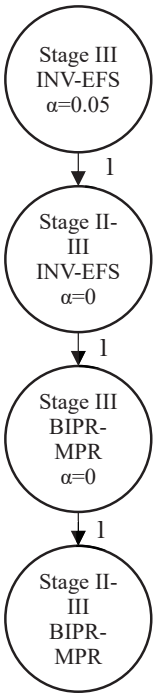
The above sample size will be calculated based on the following assumptions:

- EFS is exponentially distributed.
- The median investigator-evaluated EFS is 14 months in the Stage III population and 40 months in the Stage II population control arm.
- The hazard ratio (HR) between the experimental arm and control arm is 0.65.
- Seventy percent (70%) of the investigator-evaluated EFS information ratio is expected in the Stage III population, and an interim analysis of efficacy for investigator-evaluated EFS in the Stage III population will be performed at approximately 139 events of investigator-evaluated EFS; the interim and final analyses use the α-spending function of the O'Brien-Fleming method (approximated by the Lan-DeMets method) to calculate significance level boundaries.
- All subjects will be enrolled within 28 months, and the Stage III population was randomized between April 2020 and January 2022.
- During the first 12 months of investigator-evaluated EFS event observation, the drop-out rate is 20% in each study group.

The MPR rate assessed by BIPR is 20% in the control arm, and 38% in the experimental arm. With 500 stage II-III subjects, the study provides approximately 99% nominal statistical power to detect the superiority of the experimental arm over the control arm at a two-sided significance level of 0.05 (after transmission of the significance level); with 400 stage III subjects, approximately 98% nominal statistical power is provided to detect the superiority of the experimental arm over the control arm at a two-sided significance level of 0.05 (after transmission of the significance level).

The sample size was calculated using R 4.3.5 and the gsDesign statistical

	software, as well as the EAST 6.5 statistical package.
Statistical Analysis	**Analysis population** The intent-to-treat (ITT) population will be used as the primary analysis population for efficacy analysis in this study and includes all the randomized subjects. The subjects in ITT will be analyzed according to the treatment group randomly allocated in the study. The per-protocol set (PPS) is a subset of the ITT population that includes all randomized subjects without major protocol deviations. Prior to the study database lock, subjects included in PPS will be determined based on protocol deviation information, which will be recorded and stored in one separate document. The PPS will be used for sensitivity analysis of primary efficacy endpoints. The safety analysis set (SS) includes all the subjects who have received any dose of the study drug (including toripalimab or chemotherapy). The subjects in SS will be analyzed according to the treatment they actually receive. The SS will be used for the analysis of safety and biomarkers. The pharmacokinetic analysis set (PKS) includes all subjects who received at least one dose of toripalimab and had PK data collected. The PKS will be used for all pharmacokinetic analyses. **Efficacy analysis** The multiple comparison strategy in this study involves the use of hierarchical testing to statistically test the investigator-evaluated EFS in the Stage III population, investigator-evaluated EFS in the Stage II-III population, BIPR-evaluated MPR in the Stage III population, and BIPR-evaluated MPR rate in the Stage II-III population sequentially at a two-sided significance level of 0.05 to control overall Type I error α level. The figure below illustrates a graphical representation of the multiple comparison strategy described above.



α is the initial significance level

The BIPR-evaluated MPR rate will be analyzed for Stage II-III and Stage III populations, respectively, using the ITT set (Stage II-III) as the primary analysis set. Group comparisons will be performed using a stratified Cochran Mantel Haenszel (CMH) method and 95% CIs (experimental arm minus control arm) will be calculated for the difference in rates between groups, stratified by the same stratification factors as those employed for randomization. Sensitivity analyses of the BIPR-evaluated MPR rate will be performed based on the PPS population for the Stage II-III and Stage III populations, respectively.

The investigator-evaluated EFS will be analyzed for the Stage II-III and Stage III populations, respectively, using the ITT set as the primary analysis set. A stratified log-rank test will be used for the analysis of intergroup differences in EFS, using the same stratification factors as those employed for randomization. The Kaplan-Meier method will be used to estimate the EFS curve and median EFS in each study group. The 95% confidential interval of median EFS will be estimated through the Brookmeyer-Crowley method using log-log function conversion to reach normal approximation. The hazard ratio (HR) of EFS and its 95% confidential

interval will be estimated through the stratified Cox proportional hazard model, using the same stratification factors as those employed for randomization. Sensitivity analyses for investigator-evaluated EFS will be performed based on the PPS population.

The BIPR-and investigator-evaluated pCR rates will be analyzed based on the ITT set for the Stage II-III and Stage III populations, respectively, and the statistical analysis method will be the same as the one used for the BIPR-evaluated MPR rate.

The IRC- and investigator-evaluated DFS will be analyzed for the Stage II-III and Stage III populations, respectively, based on the ITT set, and the statistical analysis method will be the same as the one used for the investigator-evaluated EFS.

A stratified log-rank test will be used for the analysis of intergroup differences in EFS, using the same stratification factors as those employed for randomization. The Kaplan-Meier method will be used to estimate the survival probability rates at 2- and 3- year in each arm; the 95% confidence intervals for 2- and 3-year survival rates will be estimated using the Greenwood formula. The 95% confidence interval for the between-arm difference in OS will be estimated using the normal approximation method.

Safety analysis

All safety data will be summarized based on the safety analysis population.

Medical Dictionary for Regulatory Activities (MedDRA) Version 22.0 or later version will be used to code AEs. For each study treatment group, the number and incidence of events will be tabulated by preferred term and system organ class. Treatment-related or protocol procedural-related AEs, AEs related to radical resection, serious AEs (SAE), immune-related AEs, AEs leading to death, and AEs leading to a subject's discontinuation of study treatment will be tabulated for each group. The common AEs in each group (i.e., the AEs that occurred in 5% or more subjects) will be descriptively summarized.

Descriptive statistics will be used to summarize all laboratory results, physical examinations, vital sign measurements, weights, ECOG PS scores, 12-lead electrocardiogram results, and the number and percentage of subjects with changes from baseline in each study group at each visit. Statistics regarding baseline values and changes from baseline will also be provided.

	Descriptive statistics will be used to summarize the following surgical feasibility results in each study group: percentage of delayed or canceled operations, change of surgical approach and time of operation. The original data will be tabulated. **Pharmacokinetic Analysis** Descriptive statistical analysis will be performed on blood concentrations of toripalimab. Mean trough concentrations over time profile will be plotted. Toripalimab blood concentration data will be included in population pharmacokinetic analyses to identify potential factors affecting toripalimab exposure, such as chemotherapy regimens administered in this study and the development of anti-drug antibodies (ADA). The population pharmacokinetic analyses will be performed under a separate analysis plan and the results will not be reported in the clinical study report (CSR) for this study. **Immunogenicity Study** The incidence of ADA, ADA titers, and neutralizing antibodies in subjects with ADA will be analyzed The effects of ADA on the pharmacokinetics, safety and efficacy will be evaluated in population pharmacokinetic modeling and E-R analyses. The population pharmacokinetic and E-R analyses will be conducted under a separate analysis plan and the results will not be reported in the CSR for this study. **Biomarker Analysis** Potential predictive biomarkers for clinical efficacy (e.g., MPR, EFS, DFS, and OS) from peripheral blood and/or tumor specimens will be explored, and the results will be provided in a tabular summary with descriptive statistics.
Interim analysis and analysis plan	One interim analysis is planned for INV-EFS in Stage III population when 139 (70%) INV-EFS events are observed among Stage III patients. It is expected to occur about 10 months after the randomization of the last subject in Stage III population. The final EFS analysis in Stage III population is planned when 199 INV-EFS events are reached (this final analysis will become descriptive if INV- EFS in the Stage III population has reached statistical significance at the interim analysis). The

	final analysis is expected to occur 12 months after the completion of subject enrollment in the Stage II-III population. The significance level for the interim and final analyses will be calculated based on the actual investigator-evaluated number of EFS events using an O'Brien-Fleming type α-spending function approximated by the Lan-DeMets method. Approximately 220 investigator-evaluated EFS events are expected among the Stage II-III population at the time of the final analysis of INV-EFS in the Stage III population. The INV-EFS in the Stage III and Stage II-III populations, and BIPR-evaluated MPR rates in the Stage III and Stage II-III populations will be performed hierarchically. Actual data cut-off dates will be based on the number of events required and time needed for the completion of necessary clinical data collection. An independent Data Monitoring Committee (IDMC) and an independent Statistical Supporting Group (ISSG) will be set up for this study. For the interim analysis of EFS for the Stage III population, ISSG will generate treatment allocation (unblinded) information and provide to the IDMC, which will make recommendations to the Sponsor based on the analysis. See the IDMC Charter for detailed information about the IDMC's organization, responsibilities and data monitoring program.
End-of-Treatment (EOT)	EOT will occur if any of the following conditions is met: 1. Voluntary withdrawal of consent to participate in the study. 2. Completion of all planned protocol-specified study treatments. 3. Progression of disease that precludes surgery. 4. Post-operative disease progression. 5. Local or distant recurrent metastasis. 6. Pregnancy. 7. Any clinical AEs, abnormal laboratory results, or other medical conditions that, in the investigator's judgment, would alter the risk-benefit assessment for continuation of treatment such that the risks of continued treatment outweigh its benefits. 8. Clinical progression (i.e., clinical deterioration without evidence of

	radiologic disease progression such that the risks of continued treatment outweigh its benefits). 9. Ineligibility, non-compliance or significant protocol deviations after enrollment. 10. Loss to follow-up. 11. Death. 12. Inability to continue treatment in the opinion of the investigator for other reasons (e.g., intercurrent illness). 13. Request for study termination by the Sponsor, investigators or regulatory authorities.
Study Period	Estimated enrollment date of the first subject: April 2020. Estimated enrollment date of the last subject: December 2022. Estimated end date of the study: October 2024.

Study Flow Chart

Study Procedures/ Visits	Screening period (Day -28 to Day -1)			Neoadjuvant treatment period (3 cycles)			Perioperative visits [20]	Surgical Resection (5 week [±7 days] after C3)	Postoperative Safety Visit	Adjuvant treatment (1 cycle after surgery) [24]	Consolidation treatment Period (13 cycles) [23]	End-of-treatment (EOT)	Safety visit	Survival visits [26]
	Day -28 to Day -1	Day -14 to Day -1	Day -3 to Day -1	C1D1	C2D1 (±3d)	C3D1 (±3d)	4 weeks (±7 d) after C3D1		30 days (±7 d) from day of surgery	4 to 8 weeks after surgery	Consolidation treatment starts 3-16 weeks after the end of 1 cycle of adjuvant treatment, 13 cycles (± 3 days) [22]	Withdrawal from study treatment / end of treatment (± 7 days) [25]	30 days (±7 d) after last dose	Once every 3 months for 2 years, once every 6 months (± 7 days) thereafter
Signing of informed consent form [1]	√													
Demographic data	√													
Medical history	√													
Prior anti-cancer treatment	√													
Tumor tissue collection [2]	√													
Mediastinal lymph node evaluation [3]	√													
Radiological imaging evaluation [4]	√						√		√	√	√			√
Biomarker testing in peripheral blood [5]			√				√		√			√		

Study Procedures/ Visits	Screening period (Day -28 to Day -1)			Neoadjuvant treatment period (3 cycles)			Perioperative visits [20]	Surgical Resection (5 week [±7 days] after C3)	Postoperative Safety Visit	Adjuvant treatment (1 cycle after surgery) [24]	Consolidation treatment Period (13 cycles) [23]	End-of-treatment (EOT)	Safety visit	Survival visits [26]
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Inclusion/exclusion criteria check, randomization [6]			√											
Physical examination and body weight [27]		√	√	√	√	√	√		√	√	√	√	√	
Body height			√											
Vital signs and oxygen saturation [7]			√	√	√	√	√		√	√	√	√	√	
ECOG score [27]		√	√	√	√	√	√		√	√	√	√		
12-lead ECG [8]		√		√	√	√	√		√	√	√	√	√	
UCG [9]		√					√		√			√		
Pregnancy test [10]			√									√		
Virus test [11]		√					√					√		
Blood routine [12]		√		√	√	√	√		√	√	√	√	√	

Study Procedures/ Visits	Screening period (Day -28 to Day -1)			Neoadjuvant treatment period (3 cycles)			Perioperative visits [20]	Surgical Resection (5 week [±7 days] after C3)	Postoperative Safety Visit	Adjuvant treatment (1 cycle after surgery) [24]	Consolidation treatment Period (13 cycles) [23]	End-of-treatment (EOT)	Safety visit	Survival visits [26]
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Urine routine [13]		√		√	√	√	√		√	√	√	√	√	
Blood biochemistry [14]		√		√	√	√	√		√	√	√	√	√	
Coagulation [15]		√		√	√	√	√		√	√	√	√	√	
Thyroid function examination [16]		√		√	√	√	√		√	√	√	√	√	
Pulmonary function test		√					√		√			√		
Toripalimab/Placebo				√	√	√				√	√			
Platinum-based doublet chemotherapy [17]				√	√	√				√				
Surgical resection [21]								√						
Concomitant medication/concurrent	√			√	√	√		√	√	√	√	√	√	

Study Procedures/ Visits	Screening period (Day -28 to Day -1)			Neoadjuvant treatment period (3 cycles)			Perioperative visits [20]	Surgical Resection (5 week [±7 days] after C3)	Postoperative Safety Visit	Adjuvant treatment (1 cycle after surgery) [24]	Consolidation treatment Period (13 cycles) [23]	End-of-treatment (EOT)	Safety visit	Survival visits [26]
	Day -28 to Day -1	Day -14 to Day -1	Day -3 to Day -1	C1D1	C2D1 (±3d)	C3D1 (±3d)	4 weeks (±7 d) after C3D1		30 days (±7 d) from day of surgery	4 to 8 weeks after surgery	Consolidation treatment starts 3-16 weeks after the end of 1 cycle of adjuvant treatment, 13 cycles (± 3 days) [22]	Withdrawal from study treatment / end of treatment (± 7 days) [25]	30 days (±7 d) after last dose	Once every 3 months for 2 years, once every 6 months (± 7 days) thereafter
therapy [18]														
Adverse events [19]	√			√	√	√	√	√	√	√	√	√	√	
Survival follow-up and subsequent anti-tumor therapy														√
Pharmacokinetic/immunogenic blood sampling [28]				√	√	√				√	√			

Note: Investigators may perform additional tests at any time. Additional tests should be recorded on the electronic case report form (eCRF) for unscheduled visits.

1. The subject's signature on the written informed consent form (ICF) must be obtained prior to any clinical study procedure, except for tumor radiological imaging, tumor biopsy, and genetic testing performed as part of the standard of care and obtained within the specified time limits prior to the first dose. Subjects with previous screening failure are permitted to be re-screened, provided that they re-sign the ICF and are re-registered to obtain a new subject identifier number at re-screening.
2. The subjects' histologic cancer diagnosis, pathologic subtype, histologic grade, clinical tumor stage, and date of initial diagnosis must be recorded when freshly

biopsied samples are obtained. Archived tumor specimens must be provided for those participants who cannot be safely biopsied. Fifteen formalin-fixed paraffin-embedded (FFPE) unstained sections containing tumor cells must be obtained within 3 months prior to the first dose of study treatment. Otherwise, freshly obtained tumor biopsies must be provided prior to enrollment. For unstained slides, 3 will be used for the determination of preoperative PD-L1 status and the other 12 will be used for biomarker studies (e.g., TMB and WES). Fine-needle biopsies of mediastinal lymph nodes may be provided when a biopsy of the primary tumor is contraindicated or not safely accessible. Histologic confirmation of the presence of tumor cells is required prior to biomarker testing. If the tumor specimen does not provide an evaluable result for PD-L1, the subject does not need to undergo a second biopsy and will be stratified in the group with PD-L1 <1% or non-evaluable PD-L1.

3. Involvement of mediastinal lymph nodes can be confirmed through PET-CT scan or by biopsy obtained by EBUS or mediastinoscopy.
4. Whole-body PET-CT and CT (contrast-enhanced) (thoracoabdominopelvic) will be performed 28 days prior to randomization; MRI/CT (contrast-enhanced) scans of the brain should be performed within 28 days of randomization for subjects with suspected brain metastases; tumor assessments should be performed according to the RECIST1.1 criteria. Subsequent radiological evaluations will include contrast-enhanced CT of the chest and PET-CT/CT or MRI scans of other anatomic sites with suspected recurrence until progressive disease, death, loss to follow-up, or withdrawal of consent. The tumor assessment is divided into preoperative and postoperative phases: the preoperative tumor assessment is performed perioperatively with the screening scan as the baseline; the postoperative tumor assessment (including maintenance period) is performed at 30 days postoperatively as the new baseline, then every 3 natural months for 2 years, and every 6 natural months afterwards. The evaluation frequency will not change regardless of unscheduled examinations or delays in treatment, with a radiological window of ± 7 days.
5. Optional biomarker tests: 8-10 mL peripheral blood will be sent to a central laboratory for plasma ctDNA testing and WES analysis. Detailed information regarding these peripheral blood samples is discussed in Sections 6.1.1, 6.1.2, and 6.1.3.
6. Prior to randomization, the surgeon must confirm that the subject meets the eligibility criteria as resectable or potentially resectable.
7. Vital signs include body temperature, blood pressure, heart rate, respiratory frequency, and resting oxygen saturation. Vital signs will be recorded within 30 minutes prior to the start of infusion and within 60 minutes after the end of infusion on the day of dosing. Additional vital sign measurements will be collected as needed if relevant uncomfortable symptoms occur.
8. 12-lead ECG: Includes measurement of QT, QTc, and P-R interval. Additional ECGs may be performed at any time, as indicated, for chest pain, palpitation, or other cardiac symptoms. If these items are completed within 72 hours prior to the first dose in the screening period, they do not need to be repeated prior to the first dose.
9. Echocardiogram: Includes the left ventricular ejection fraction (LVEF) evaluation and must be performed within 14 days prior to the first dose, at the preoperative, postoperative safety and end-of-treatment visits. Additional unscheduled echocardiograms may be performed as needed for the evaluation of

cardiac function.

10. Women of childbearing age must undergo a serum pregnancy test within 3 days prior to the first dose in the screening period (Note: if the study site cannot obtain a result within 3 days, the test should be obtained within 7 days prior to the first dose) and at the end-of-treatment visit.
11. HIV, HBV and HCV testing will be performed during screening and at the end-of-treatment visit. HBV testing: HBsAg, HBsAb, HBcAb, HBeAg and HBeAb; if positive for HBsAg, the patient must also be tested for HBV-DNA. HCV testing: HCV-Ab; if HCV-Ab seropositive, the patient must also be tested for HCV-RNA.
12. Blood routine: including full blood count with differential. Additional tests may be performed by the investigator if necessary. If completed within 72 hours prior to the first dose in the screening period, no additional test is required prior to the first dose.
13. Urinalysis: Includes the evaluation of white blood cell, red blood cell and urine protein. If urine protein is ≥ 2+ in the screening period, an additional 24-hour urine protein quantification will be required. Subjects with 24-hour urine protein quantification ≥ 1 g at screening are not recommended for enrollment. The 24-hour urine protein quantification will be determined as clinically required during the study; if these items are completed within 72 hours prior to the first dose in the screening period, they do not need to be repeated prior to the first dose.
14. Serum biochemistry: Includes ALT, AST, ALP, serum creatine kinase (CK), serum creatine kinase isoenzyme (CK-MB), TP, ALB, TBIL, DBIL, GLU, BUN, Cr, K+, Na+, Cl-, Ca2+, Mg2+, and P. Additional tests may be performed by the investigator if necessary. If these items are completed within 72 hours prior to the first dose in the screening period, they do not need to be repeated prior to the first dose.
15. Coagulation function: Includes, at a minimum, the international normalized ratio (INR), activated partial thromboplastin time (aPTT), fibrinogen, and thrombin time (TT). If the INR is not available, it can be replaced with prothrombin time (PT). If these items are completed within 72 hours prior to the first dose in the screening period, they do not need to be repeated prior to the first dose.
16. Thyroid function test will be performed at the clinical laboratory at the study center within 72 hours prior to each cycle and includes serum thyroid-stimulating hormone (TSH), free triiodothyronine (FT3), and free thyroxine (FT4). If FT3 and FT4 are not available, T3 and T4 may be obtained as alternatives. If thyroid function tests are performed within 72 hours prior to the first dose, testing does not need to be repeated on C1D1. In subjects with clinically significant changes in thyroid function, appropriate treatment should be initiated and consultation with the department of endocrinology regarding treatment and to rule out hypopituitarism.
17. The time window of administration of pemetrexed, taxanes and carboplatin/cisplatin is +3 days for the first dose and ±3 days for the other doses. Except for radiological examination, all test items need to be completed within 72h prior to administration, and the dose can be continued only after safety evaluation.
18. Concomitant medication/concurrent therapy: All concomitant medications/concurrent therapies administered between 28 days before the first dose of neoadjuvant therapy through 30 days after last dose of study treatment will be documented. Any concomitant medications/concurrent therapies administered

for treatment of for study drug-related AEs should be collected through 90 days after the last study treatment dose. Concomitant medications/concurrent therapies administered for the treatment of a study drug-related SAE or AE of special interest (AESI) will be recorded until resolution of that AE. If a subject initiates a new systemic anti-tumor therapy, only the concomitant medications/concurrent therapies administered for the treatment of study drug-related AEs shall be recorded.

19. The severity grading of AEs will be categorized according to NCI CTCAE V5.0. All AEs occurring between the first dose of the study drug and 30 days after the last dose of the study treatment or initiation of a new anti-tumor treatment, whichever comes first, will be recorded. All SAEs occurring between the first dose of the study treatment and 90 days after the last dose or initiation of a new anti-tumor treatment, whichever comes first, will be recorded. Only the SAEs caused by mandatory protocol-specified interventions (e.g., interventional procedure, biopsy) need to be recorded during screening.
20. All examinations must be completed prior to the operation.
21. Complete resection should be achieved during the operation, which is defined as (1) negative margin, including bronchus, artery, vein, bronchial surroundings, tumor adjacent tissue; (2) at least 6 groups of lymph nodes, including 3 groups in lungs and 3 mediastinal groups; (3) microscopically negative highest lymph node resected; (4) no extranodal involvement.
22. In subjects who are not indicated for radiotherapy, maintenance treatment should be initiated 3 weeks (±3 days) after administration of adjuvant chemotherapy. In subjects who are indicated for radiotherapy, radiotherapy should be initiated within 4 weeks after administration of adjuvant treatment. Maintenance treatment should be initiated 30 days (±3 days) after completion of radiotherapy. An oncology radiologist will determine whether radiotherapy is indicated and determine the appropriate radiotherapy course.
23. Maintenance treatment: Toripalimab/placebo will be given once every 3 weeks for a maximum of 13 cycles. Required safety evaluation, including laboratory tests and recording of AEs, need to be completed in each cycle.
24. The postoperative safety visit and the initiation of adjuvant treatment can be conducted during the same visit. All required assessments for both visits should be performed.
25. Radiological imaging will be performed every 3 months from the date of surgery for 2 years and then every 6 months afterwards. If the EOT visit coincides with an end-of-treatment visit or a safety visit, two visits may be considered as a single visit and should include all of the items indicated on the study flow chart.
26. Subjects who are being followed for survival will undergo this visit, including those who complete/withdraw from the study treatment but have not been withdrawn from the study. Subjects without disease recurrence (local or distant) should undergo scheduled tumor assessments and be followed for survival and subsequent anti-tumor therapy at the study site. Those with disease recurrence will should be followed up for survival and subsequent anti-tumor therapy via telephone.

27. If a physical examination and determination of ECOG performance status score are performed within 72 hours prior to the first dose during screening, they do not need to be repeated prior to the first dose on C1D1.
28. To maintain blinding, samples will be collected from all subjects. A total of 2mL of whole blood will be collected for pharmacokinetic analysis and 3mL of whole blood will be collected for immunogenicity analysis within 2hrs before administration of toripalimab/placebo for each patient in each cycle. The date and time of sample collection (accurate to minute, 24-hour clock) should be accurately recorded each time. Details of the blood sample collection volume and treatment process can be found in Laboratory Manual.

Abbreviations

Abbreviations	English full name
12-Lead ECG	12-Lead electrocardiogram
ADA	Anti-drug antibody
ADI	Actual drug intensity
ADRs	Adverse drug reactions
AE	Adverse event
ALB	Albumin
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
ALK	Anaplastic lymphoma kinase
ANC	Absolute neutrophil count
APTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
AUC	Area under curve
BIPR	Blinded independent pathology review
BP	Blood pressure
BUN	Blood urea nitrogen
BOR	Best overall response
BSC	Best supportive care
Ca	Calcium
CD-ROM	Compact Disc Read-Only Memory
CI	Confidence interval
CYP	Cytochrome oxidase
CNS	Central nervous system
CFDA	China Food and Drug Administration
Cr	Creatinine
CRF	Case report form
CFR	Code of Federal Regulations
CMH	Cochran–Mantel–Haenszel test
Cl-	Blood chlorine
CTCAE	Common Terminology Criteria for Adverse Events
CTLA-4	Cytotoxic T lymphocyte associated antigen 4
ctDNA	Circulating tumor DNA
D	Day
DBIL	Direct bilirubin
DFS	Disease-free survival
DMC	Data monitoring committee
DNA	Deoxyribonucleic acid
EC	Ethics committee
ECG	Electrocardiogram
ECOG PS	Eastern Cooperative Oncology Group Performance Status
EFS	Event-free survival
EGFR	Epidermal growth factor receptor
eCRF	Electronic Case report form
EDC	Electronic data collection
ELISA	Enzyme-linked immunosorbent assay
Fbg	Fibrinogen

Abbreviations	English full name
FDA	Food and Drug Administration
FT3	Free triiodothyronine
FT4	Free thyroxine
FFPE	Formalin fixed paraffin embedded
GCP	Good Clinical Practice
G-CSF	Granulocyte colony stimulating factor
GLU	Blood glucose
GLU-U	Urine glucose
H	Hour
Hb	Hemoglobin
HBV	Hepatitis B virus
HbsAg	Hepatitis B surface antigen
HCV	Hepatitis C virus
HIV	Human immunodeficiency virus
HLA	Human leukocyte antigen
HR	Hazard ratio
IASLC	International Association for the Study of Lung Cancer
IB	Investigator's brochure
IBIL	Indirect bilirubin
ICH	International Conference on Harmonization
IDMC	Independent Data Monitoring Committee
IMAE	Immune-related adverse events
INR	International Normalized Ratio
iSSG	Independent statistical supporting group
IU	International unit
ITT	Intend to treat
IRB	Institutional Review Board
IRC	Independent Review Committee
LLN	Lower limits of normal
LVEF	Left ventricular ejection fraction
LEU	Leukocytes in urine
LDH	Lactate dehydrogenase
LYMPH	Lymphocyte
K+	Serum potassium
KET	Urine acetone bodies
min	Minute
mPFS	Median progression-free survival
MPR	Major pathology response
MRI	Magnetic Resonance Imaging
MedDRA	Medical Dictionary for Regulatory Activities
MOP	Medical Operation Procedure
miRNA	MicroRNA
Na^+	Plasma sodium
NEUT	Neutrophil
NCCN	National Comprehensive Cancer Network
NCI-CTC	National cancer institute Common Terminology Criteria

Abbreviations	English full name
NSCLC	Non-small cell lung cancer
ORR	Objective response rate
OHRP	Office for human research protections
OS	Overall survival
P	Phosphorus
pCR	Pathologic complete response
PD	Progressive disease
PD-1/PD-L1	Programmed death 1/ Programmed death ligand 1
PFS	Progression-Free-Survival
PK/PD	Pharmacokinetics/ Pharmacodynamics
PPS	Per-Protocol Set
PR	Partial response
PT	Prothrombin time
PLT	Blood platelet
PI	Principal investigator
RBC	Red blood cell count
RDI	Relative drug intensity
RECIST	Response evaluation criteria in solid tumors
RNA	Ribonucleic acid
Sec	Second
SAE	Serious adverse event
SAP	Statistical analysis plan
SAS	Statistical analysis system
SD	Stable disease
SOP	Standard Operation Procedure
SS	Safety Analysis Set
TBIL	Total bilirubin
TCM	Traditional Chinese Medicine
TEAE	Treatment-emergent adverse events
TP	Plasma total protein
TLF	Table Listing Figure
TSH	Thyroid-stimulating hormone
TT	Thrombin time
TTOR	Time to objective response
UA	Uric acid
UBIL	Urine bilirubin
ULN	Upper limit of normal
WBC	White blood cell count

1. Background and Scientific Rationale

1.1. Research Background

In 2018, the World Health Organization's International Agency for Research on Cancer issued a Global Cancer Report that presented data on the morbidity and mortality of 36 cancer types from 185 countries and regions. The report showed that there were 18.1 million new cases of cancer and 9.6 million deaths globally in 2018. The disease burden has further exacerbated worldwide. Lung cancer (of which non-small cell lung cancer [NSCLC] accounts for 85% of lung cancer cases) has the highest incidence and is the leading cause of cancer-related deaths (18.4% and 11.6%, respectively). [1] In China, the incidence and mortality of lung cancer is gradually increasing, most likely due to smoking and air pollution. According to data released by the National Cancer Center in 2015, the incidence of lung cancer was 57.26/100,000 while the mortality was 45.87/100,000. [2] In terms of disease stage, localized, locally advanced and advanced stages of disease account for 19%, 24%, and 55% of lung cancer cases, respectively.[3] Radical resection is the main treatment for early and resectable, locally advanced NSCLC, whereas the main treatment for unresectable, locally advanced or metastatic NSCLC is systemic therapy.

1.1.1 Current treatment of Resectable Stage II-III NSCLC

The strategy for treating resectable, stage II-III NSCLC is a multidisciplinary, comprehensive treatment plan involving surgical resection in combination with chemotherapy and radiotherapy. The 5-year survival rate is 25% to 40% for the early (tumor localized to the primary site) and locally advanced (tumor involving adjacent structures or with regional lymph node involvement) stages. [1] The study showed that >50% of patients treated with surgery alone developed recurrence. Perioperative systemic chemotherapy is one of the methods used to reduce disease recurrence, but systemic chemotherapy with surgery only improves the 5-year survival rate by 5% compared with surgery alone. The time to recurrence is relatively short and the median DFS is only 12 months in stage III patients. Currently, the efficacy of neoadjuvant/postoperative adjuvant treatment is still limited and the 5-year survival rates range between 25%-40%. [4]

1.1.2 Neoadjuvant Treatment for Resectable, Stage II-III NSCLC

1.1.2.1 Neoadjuvant Chemotherapy Combined with Surgery Versus Surgery Alone

Previous studies showed that similar to postoperative adjuvant chemotherapy, neoadjuvant chemotherapy followed by surgical resection significantly prolonged disease-free survival (DFS), progression-free survival (PFS), OS, and the time to distant metastasis in NSCLC subjects when compared with surgery alone. [5,6,7] The CHEST study (Scagliotti et al.) demonstrated that 3 cycles of neoadjuvant chemotherapy with cisplatin and gemcitabine, followed by surgical resection, prolonged PFS and OS as compared with surgery alone. A subgroup analysis showed that the survival advantage for neoadjuvant chemotherapy was mainly in subjects with locally advanced (stage IIB/III) disease. In addition, the National Comprehensive Cancer Network (NCCN) noted that for selected patients with stage II-III-N2 NSCLC, with a good response to neoadjuvant chemotherapy, the possibility of long survival benefits or even complete cures might be achieved following neoadjuvant chemotherapy and surgical resection. [8,9,10,11,12]

In terms of safety, a retrospective analysis showed that neoadjuvant treatment prior to surgery did not increase the risks of surgery or the incidence of postoperative complications. [24,25] Regarding whether delayed surgery due to prior neoadjuvant treatment affected survival, research shows (Pless et al.) that there was no difference in survival rates when the interval between neoadjuvant treatment and surgery was within 6 weeks. [13]

1.1.2.2 Neoadjuvant Chemotherapy Combined with Surgery Versus Postoperative Adjuvant Chemotherapy

Postoperative adjuvant chemotherapy has been shown to prolong survival in NSCLC patients. However, poor physical performance after surgical resection often hinders the implementation of adjuvant chemotherapy. The tolerability of preoperative and postoperative systematic treatment was compared in the Phase III clinical study NATCH. This study enrolled a total of 624 subjects with stage I, II, and partial stage III NSCLC, and the results showed that 90% of the subjects completed systemic treatment in the preoperative group, while only 60% of subjects completed it in the postoperative group.

The main reasons for failure of completion were mainly serious postoperative complications and failure to recover from surgical stress. [12,13]

The effect of neoadjuvant chemotherapy versus adjuvant chemotherapy on survival was compared in NSCLC patients in the CSLC0501 study. A total of 214 patients with resectable stage IB-III NSCLC were enrolled and randomized into the neoadjuvant or adjuvant chemotherapy group. 100% and 87.4% of the patients completed chemotherapy in the 2 groups, respectively. The primary endpoint, 3-year DFS, were 43% and 56% and were not statistically significant. There were no statistically significant differences observed in the secondary endpoints of 3-year OS rate and 5-year OS rate. [14] Results from a meta-analysis also showed that neoadjuvant and adjuvant chemotherapy had similar impacts on survival [15]. However, given the advantage of neoadjuvant chemotherapy in terms of the accessibility and tolerability, neoadjuvant chemotherapy was shown to be a safe and feasible strategy. [22,23]

1.2. Scientific Rationale

Tumor immunotherapy mobilizes cytotoxic T cells to kill tumor cells, and is one effective and safe way to treat cancer. In recent years, an improved understanding of the molecular mechanism of tumor-host response control has led to significant progress in the identification of signaling pathways that regulate checkpoints associated with the anti-tumor immune response. Programmed death-1 (PD-1) is a protein receptor expressed on the surface of T cells. Discovered in 1992, PD-1 belongs to the CD28 family. PD-1 has 2 ligands: PD-L1 and PD-L2. PD-L1 is mainly expressed on T cells, B cells, macrophages and dendritic cells, and its expression is up-regulated upon stimulation by inflammatory cytokines. Meanwhile, PD-L1 is also expressed on tumor-infiltrating lymphocytes and tumor cells. Tumor cells transmit negative regulatory signals through PD-L1 molecules that bind to PD-1 on T lymphocytes, leading to immunosuppression of tumor antigen-specific T cells and tumor cell escaping from immune surveillance and killing.

PD-1/PD-L1pathway blockers represent the new class of tumor immunotherapeutic drugs. These drugs can regulate the anti-tumor activity of T lymphocytes by blocking the PD-1/PD-L1 signaling pathway and enhancing the patient's own immune response against tumors. Traditional cytotoxic anti-tumor treatment, such as chemotherapy, can promote T

cell activation as chemotherapy-induced tumor cell death results in the release of tumor antigens, thereby enhancing the patient's immune response to tumor cells.

Currently, PD-1/PD-L1 pathway blockers, have achieved breakthrough progresses in the clinical management of various cancers. In the field of lung cancer treatment, KEYNOTE-010, 024, 189, and 407 studies have demonstrated the efficacy of pembrolizumab [16] as monotherapy or combined therapy for advanced, epidermal growth factor receptor (EGFR) wild type NSCLC. [18,19,20,21] However, despite the demonstration of effectiveness, the adoption of adjuvant treatment for NSCLC in clinical practice has been limited, possibly due to the long follow-up period needed to conduct and complete trials evaluating adjuvant treatment. Neoadjuvant treatment may provide a similar survival benefit to adjuvant treatment, and evaluations of its effectiveness based on pathologic response can be achieved more rapidly. [28]

Early exploration of neoadjuvant immunotherapy has yielded positive results. One Phase II study showed that neoadjuvant immunotherapy was safe and tolerable, either as monotherapy or combined with chemotherapy, with a majority of subjects demonstrated a favorable pathologic response (the study endpoint). (See Table 1 for details)

Table 1: Early Clinical Trials Exploring Neoadjuvant Immunotherapy

Trial name	Therapeutic regimen	Study Treatment	Sample size	Efficacy
Checkmate159	IO mono	Nivolumab 3 mg/kg administered 28 and 14 days prior to surgery	21	MPR: 45% (9/20)
NEOSTAR	IO mono	Arm A: Nivolumab 3 mg/kg Q2W for 3 doses prior to surgery; followed by adjuvant treatment	14	MPR: 28% (4/14)
	IO + IO	Arm B: Nivolumab 3 mg/kg and ipilimumab 1 mg/kg Q2W for 3 doses prior to surgery, followed by adjuvant treatment	12	MPR: 33% (4/12)
LCMC3	IO	Atezolizumab 1200 mg, Q3w, 2 cycles; Postoperative adjuvant: Atezolizumab 1200 mg for 1 year	47	MPR: 22% (10/45) PR: 7% (3/45)
NADIM	IO + chemo	Nivolumab 360 mg + carboplatin 6 + paclitaxel 200 mg/m2 Q3w for 3 cycles; Postoperative adjuvant Nivolumab for 1 year, 240 mg, Q2w for the first 4	30	MPR: 80% RR:70%

		months, 480 mg, Q4w for the last 8 months		
NCT02716038	IO + chemo	Atezolizumab + carboplatin and albumin-bound paclitaxel) for 2 cycles, radiological evaluation, with additional 2 neoadjuvant cycles for those with response or proceeding directly to surgery for those without a response	14	MPR: 50% (7/14)

Based on the significant unmet need as evidenced by poor 5-year survival rates and promising clinical data with other drugs in this class, this Phase III clinical study is designed to evaluate the efficacy and safety of adding toripalimab to perioperative chemotherapy for patients with resectable Stage II-III NSCLC.

1.3. Potential Risks and Benefits

1.3.1. Known Potential Risks

Any investigational product or therapy may have severe or unpredictable toxicity.

The overall adverse reaction profiles for PD-1 blocking antibodies administered as monotherapy are superior to that for traditional chemotherapy or targeted therapy. The adverse reaction profile of PD-1 antibodies when combined with chemotherapy is primarily characterized by an increased incidence of chemotherapy-induced adverse reactions, including weakness, nausea, rash, and alopecia, as compared with chemotherapy alone, but is generally well tolerated. In addition, immune-related adverse reactions may occur. The most common are immune-related interstitial pneumonitis, rash, and thyroiditis, while immune-related leukoderma, colitis, nephritis, hepatitis, uveitis, adrenal insufficiency, and neuropathy generally occur at a low incidence (≤1%). The majority of these immune-related adverse reactions are mild and manageable, but a small percentage are serious or life-threatening. Under the established procedure for toxicity management, the majority of immune-mediated adverse events can be sufficiently controlled.

For all therapeutic proteins, including monoclonal antibodies, other risks include infusion reactions, which are mainly characterized by chills, rigors, and cyanosis of the face and extremities, followed by fever, nausea, vomiting, headache, dizziness, and irritability.

In severe cases, patients may also experience delirium, coma, hypotensive shock and respiratory failure. Infusion reactions may be caused by contamination of the drug with pyrogens or other impurities, low temperature of the diluted solution, high drug concentrations in the diluted solution and rapid infusion rates. As a fully-humanized monoclonal antibody, toripalimab has a low incidence of infusion reactions, which are generally mild in severity.

The chemotherapy regimens used in this study are platinum-based doublets recommended in the NCCN guidelines with well-established adverse reaction spectrum. In clinical studies, the most common adverse reactions (incidence ≥20%) for platinum-based doublet chemotherapy regimens include weakness, nausea, decreased appetite, and hematologic toxicity.

Subjects participating in this study also need to undergo an invasive tumor tissue biopsy, which is an essential step to determine the PD-L1 tumor expression level. Such invasive biopsies can generally be successfully performed; however, anaphylactic shock due to anesthesia, local pain, local infection, local hemorrhage, and local nerve damage are potential complications that cannot always be predicted. When necessary, either due to complications or inadequate tissue biopsy, a second biopsy may be needed to fully characterize the pathology and conduct all necessary tests.

Relevant medical examinations during the study may also bring risks to subjects. Frequent radiological examination may expose subjects to low-dose radiations. However, a periodic radiological examination is essential for the determination of disease status.

1.3.2. Known Possible Benefits

As described above, lung cancer has the highest incidence and is the highest cause of cancer mortality. Approximately 30% of lung cancer at initial diagnosis are localized, intermediate to late stage III, with a 5-year survival rate of less than 40% with the existing standard of care. The standard of care for the initial treatment of resectable NSCLC is surgery combined with systemic neoadjuvant and/or adjuvant chemotherapy; however, as compared with the 5-year survival rates of 25% to 35% for surgery alone, the addition of

adjuvant and neoadjuvant chemotherapy provides only a 5% absolute increase in the 5-year survival rates. Based on the data summarized above, the addition of immunotherapy to neoadjuvant chemotherapy for the treatment of resectable stage II-IIIA NSCLC is safe, feasible, and may further improve the benefits of neoadjuvant chemotherapy. At the 2021 AACR annual meeting, Checkmate-816, a phase III study reported the addition of nivolumab to neoadjuvant chemotherapy significantly increased the pathological complete response (pCR) rate compared with neoadjuvant chemotherapy alone in resectable stage IB-IIIA NSCLC, 24% vs. 2%.

Table 2 Registration studies of neoadjuvant immunotherapy in NSCLC

Trial name	Study Population	Sample size	Study Treatment			Primary Endpoint(s)	Secondary Endpoint(s)
			Treatment group	Preoperative neoadjuvant therapy	Postoperative adjuvant therapy		
Checkmate 816 (NCT02998528)	Patients with early, resectable stage IB-III NSCLC, EGFR/ALK WT	350	Study group 1	Nivolumab Q2W × 3 cycles, Ipi 1 mg/kg, first cycle	Diagnosis and treatment practice	pCR EFS	OS, MPR
			Study group 2	Nivolumab + platinum-based doublet drug chemotherapy, Q3W × 3 weeks	Diagnosis and treatment practice		
			Control arm	Platinum-based doublet drug chemotherapy, Q3W × 3 weeks	Diagnosis and treatment practice		
KEYNOTE-671 (NCT03425643)	Patients with stage IIB-IIIA NSCLC	786	Study group	Pembrolizumab + platinum-based doublet drug chemotherapy, Q3W × 4 cycles	Pembrolizumab Q3W, 13 cycles	EFS OS	MPR, pCR, procedural complications, PRO
			Control arm	Placebo +	Placebo, 13 cycles		

				platinum-based doublet drug chemotherapy, Q3W × 4 cycles			
Impower 030 (NCT03456063)	Patients with resectable, stage II, IIA, or selected stage IIIB NSCLC, EGFR/ALK WT	374	Study group	Atezolizumab + platinum-based doublet drug chemotherapy, Q3W × 4 cycles	Atezolizumab Q3W, 16 cycles	MPR EFS	EFS、OS、pCR、MRP、DFS、HRQoL
			Control arm	Placebo + platinum-based doublet drug chemotherapy, Q3W × 4 cycles	Clinical diagnosis and treatment specifications/supportive care evaluated by investigators		
AEGEAN (NCT03800134)	IIA-partial IIIB	300	Study group	Durvalumab + platinum-based doublet drug chemotherapy, Q3W × 3 cycles	-	MPR	OS、EFS、DFS、pCR
			Control arm	Placebo + platinum-based doublet drug chemotherapy, Q3W × 3 cycles	-		
BGB-A317-315	Resectable, stage II-IIIA	380	Study group	Tislelizumab (200 mg) ± platinum-based doublet drug, Q3W × 3-4 cycles	Tislelizumab (400 mg), Q6W, 8 cycles	MPR EFS	pCR, OS, DFS, HRQoL
			Control arm	Placebo ± platinum-based doublet drug, Q3W × 3-4 cycles	Placebo, Q6W, 8 cycles		
SHR-1316	II-IIIB	456	Study group	SHR-1316 ±	SHR-1316, Q3W, 16	MPR	pCR, OS,

(NCT04316364)				platinum-based doublet drug, Q3W × 3 cycles	cycles	EFS	ORR, DFS
			Control arm	Placebo ± platinum-based doublet drug, Q3W × 3 cycles	Placebo, Q3W, 16 cycles		
Checkmate 77t (NCT04025879)	II-IIIB	452	Study group	Nivolumab ± platinum-based doublet drug, Q3W × 4 cycles	Nivolumab	EFS	OS, pCR, MPR, SAE
			Control arm	Placebo ± platinum-based doublet drug, Q3W × 4 cycles	Placebo		

2. Study Objectives and Endpoints

2.1. Study Objectives

Primary Study Objectives

- To compare the event-free survival (EFS), as evaluated by investigators, of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable, stage III NSCLC.
- To compare the major pathologic response (MPR) rates, as evaluated by a central blinded independent pathologic review (BIPR) committee, of neoadjuvant toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable, stage III NSCLC.
- To compare the EFS, as evaluated by investigators, of toripalimab combined with platinum-based doublet drug chemotherapy versus placebo combined with platinum-based doublet drug chemotherapy for subjects with resectable, stage II-III NSCLC.
- To compare the MPR rates evaluated by a BIPR in neoadjuvant toripalimab combined with platinum-based doublet drug chemotherapy versus placebo combined with platinum-based doublet drug chemotherapy for subjects with resectable, stage II-III

NSCLC.

Secondary objectives

Efficacy:

To compare the BIPR- and investigator-evaluated pathologic complete response (pCR) rates, IRC-evaluated EFS, disease-free survival (DFS), 2- and 3-year overall survival (OS) rates, and OS of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage III and Stage II-III NSCLC.

Safety:

To evaluate the safety, tolerability, feasibility, and incidence of preoperative and postoperative complications of toripalimab combined with platinum-based doublet chemotherapy in subjects with resectable Stage III and Stage II-III NSCLC.

Pharmacokinetics: To evaluate the pharmacokinetic properties of toripalimab.

Immunogenicity: To evaluate the immunogenicity of toripalimab and its effect on pharmacokinetics, safety and efficacy.

Exploratory Study Objective:

To explore potential predictive biomarkers for clinical efficacy in tumor tissue specimens and peripheral blood samples (including PD-L1 expression in tissue specimens, tumor mutation burden [TMB], whole exome sequencing [WES], and ctDNA levels).

2.2. Study Endpoints

Primary Study Endpoints

- EFS as evaluated by investigators in Stage III NSCLC
- MPR rate as evaluated by BIPR in Stage III NSCLC
- EFS as evaluated by investigators in Stage II-III NSCLC
- MPR rate as evaluated by BIPR in Stage II-III NSCLC

Secondary Study Endpoints

- pCR rate as evaluated by BIPR and investigators in Stage III NSCLC
- EFS as evaluated by IRC in Stage III NSCLC
- pCR rate as evaluated by BIPR and investigators in Stage II-III NSCLC
- EFS as evaluated by IRC in Stage II-III NSCLC
- DFS as evaluated by IRC and investigators in Stage III NSCLC
- DFS as evaluated by IRC and investigators in Stage II-III NSCLC

- OS in Stage III NSCLC, and in Stage II-III NSCLC
- 2- and 3-year OS rates in Stage III NSCLC, and in Stage II-III NSCLC
- Safety: Adverse events (AEs), abnormal laboratory results, and treatment-related serious adverse events (SAEs) graded according to NCI CTCAE v5.0; feasibility of surgical resection, percentage of surgery delays or cancellations, change in type of surgery, and duration of surgery
- Pharmacokinetics: Toripalimab blood concentrations
- Immunogenicity: Incidence, titer, and, if required, neutralizing antibodies of anti-drug antibodies (ADAs).

Exploratory Study Endpoints

Exploratory analysis of potential biomarkers related to clinical efficacy (including PD-L1 expression in tissue specimens, TMB, WES, and change of ctDNA in peripheral blood samples).

3. Study Design

3.1. Overall Design

This is a randomized, double-blind, placebo-controlled, multicenter phase III clinical study to evaluate the efficacy and safety of toripalimab combined with platinum-based doublet chemotherapy versus placebo combined with platinum-based doublet chemotherapy in subjects with resectable stage II-III non-small cell lung cancer (NSCLC).

500 subjects are planned to be enrolled in this study and randomized at a 1:1 ratio to receive study treatment in the experimental arm or control arm, stratified by the following factors:

- Disease stage: II vs. IIIA vs. IIIB
- PD-L1 status: PD-L1 expression ≥ 1% vs. PD-L1 expression < 1% or non-evaluable
- Planned surgical type: pneumonectomy vs. lobectomy
- Pathological subtype: non-squamous cell carcinoma vs. squamous cell carcinoma

Screening will be initiated upon the signing of the informed consent form (ICF). Eligible subjects will be randomized. Prior to randomization, tumor tissue must be provided, either as a fresh biopsy or as paraffin-embedded archival tumor blocks or slides obtained ≤3 months prior to randomization for detection of PD-L1 tumor status. Neoadjuvant treatment

should be initiated within 3 days after randomization.

Experimental arm:

Neoadjuvant treatment period: Toripalimab and platinum-based chemotherapy for 3 cycles

Adjuvant period: Toripalimab and platinum-based chemotherapy for 1 cycle

Maintenance treatment period: Toripalimab for a maximum of 13 cycles.

Control arm:

Neoadjuvant treatment period: Placebo and platinum-based chemotherapy for 3 cycles

Adjuvant period: Placebo and platinum-based chemotherapy for 1 cycle

Maintenance treatment period: Placebo for a maximum of 13 cycles.

Platinum-based doublet chemotherapy regimens were selected by the investigator according to institutional practices. Squamous cell carcinoma: taxanes + platinum; non-squamous cell carcinoma: pemetrexed + platinum.

• Toripalimab /placebo: A fixed dose of 240mg once every 3 weeks, on Day 1 of each cycle, intravenously administered over 30 ±10 min [over at least 60 min for the first dose] for a total of 17 doses.

• Pemetrexed combined with platinum (non-squamous histology only): Once every 3 weeks, intravenously administered on Day 1 of each cycle for 4 cycles (3 cycles before surgery and 1 cycle after surgery). Pemetrexed 500 mg/m^2 IV over 10 minutes or according to institutional standards; cisplatin 75 mg/m^2 IV over 60 minutes or carboplatin AUC = 5 IV over 30 minutes, after completion of pemetrexed infusion, or in accordance with institutional standards.

• Docetaxel or paclitaxel combined with platinum (squamous histology only): Once every 3 weeks, for a total of 4 cycles (3 cycles before surgery and 1 cycle after surgery). Docetaxel 60 mg/m^2 to 75 mg/m^2 (60 minutes IV) or paclitaxel 175 mg/m^2 IV (30 minutes IV) will be administered intravenously on Day 1 of each cycle or according to institutional standards; cisplatin 75 mg/m^2 (60 minutes IV) or carboplatin AUC = 5 (30 minutes IV) will be administered intravenously after completion of docetaxel or paclitaxel infusion on Day 1, or according to institutional standards.

The investigator is allowed to adjust the combination treatment cycle according to the patient's clinical condition. In the absence of significant intolerability issues, combination therapy should be administered for 4 cycles.

The study treatment will continue until the completion of 13 cycles of maintenance treatment, intolerable toxicity, progression of disease, withdrawal of consent, judged by

investigators that the subject needs to be withdrawn from the treatment, or meeting other criteria of drug discontinuation specified in the protocol.

The tumor status will be evaluated using RECIST 1.1 during screening (as baseline), prior to operation and 30 days after operation, then once every 3 natural months (±7 days) afterward, until radiologically confirmed recurrence or progression of disease.

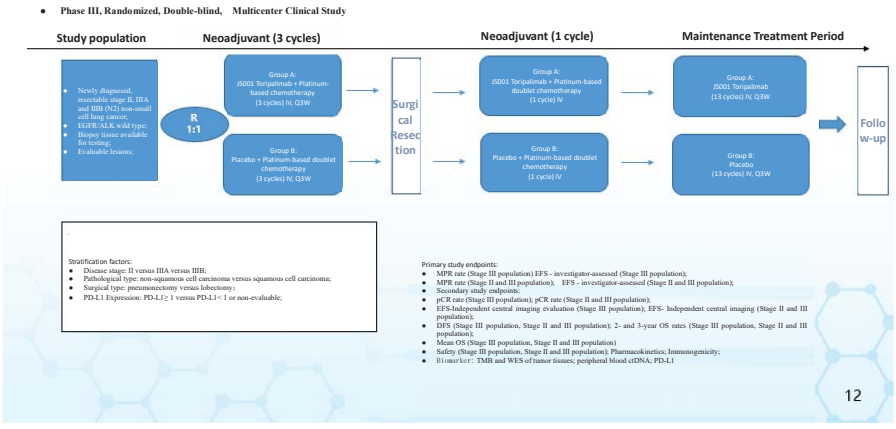
The preoperative safety visit will be performed 4 weeks ±7 days after administration in the 3rd cycle and 30 days after operation.

The post-operative safety visit should occur 30 days after surgical resection.

Safety will be evaluated on an ongoing basis during the study. The severity evaluation and grading will be performed using NCI CTCAE v5.0.

An end-of-treatment visit should occur within 3 days of the decision to permanently terminate study treatment. At this visit, subjects should undergo a complete physical examination, including vital signs, laboratory testing, radiologic tumor imaging, etc. During the survival follow-up period, subjects who do not experience disease recurrence should return to the study site to undergo the scheduled radiological tumor imaging and provide information on subsequent anti-tumor treatment and survival. For those who experience disease recurrence, follow-up visits will be conducted via telephone for the collection of survival and subsequent anti-tumor therapy.

Figure1: Diagram of Study Design



3.2. Rationale for Selection of MPR as Primary Endpoint

Pathological response after systemic neoadjuvant therapy has been shown to be an effective prognostic factor for survival. [16] In 2014, the US Food and Drug

Administration issued a new guideline accepting pCR as the primary study endpoint for evaluating neoadjuvant treatment of early breast cancer, thereby supporting accelerated approval for this indication. In early-stage resectable NSCLC, the median pCR rate is 4% following neoadjuvant therapy, so it is difficult to determine whether pCR is a prognostic factor for survival. Subsequently, it was observed that the MPR rate, as evaluated by a BIPR, may be a strong predictor for survival. MPR is defined as the presence of a viable tumor involving ≤10% of the tumor bed. [17] Therefore, in the neoadjuvant setting for stage III NSCLC, the use of MPR as the primary endpoint is appropriate.

3.3. Rationale for Dose Selection of Toripalimab

The data on the samples in each dose group of toripalimab (dose levels 0.3, 1, 3, and 10 mg/kg once every 2 weeks and 240 mg fixed dose) from 3 Phase I dose-escalation clinical studies (CT1, CT2, and CT3) were summarized for the pooled analysis of pharmacokinetics (PK). The study results showed a generally linear PK after administration of 1, 3, and 10 mg/kg toripalimab IV infusion in subjects, and a non-linear PK profile was observed when the dose was increased to 10 mg/kg in some of these Phase I studies. Toripalimab shows an obvious “long-acting” characteristic commonly observed in antibody drugs, and the plasma drug concentration reaches a steady state after the continuous administration of the toripalimab IV infusion for consecutive 3 to 4 doses.

The steady-state lowest plasma drug concentration ranged from 20 to 40 μg/mL at 3 mg/kg once every 2 weeks. The in vitro study showed that PD-1 receptor on the surface of T cells could be saturated when the concentration of toripalimab was >20 nM or 3 μg/mL, and the concentration was usually maintained at 25 μg/mL in peripheral blood, in consideration of the restricted entry of antibody macromolecules in the tumor microenvironment, so as to ensure the complete occupancy of PD-1 in the lymphocytes in the tumor microenvironment. Each Phase I study showed that the complete occupancy of the PD-1 receptor could be maintained throughout the treatment in different dose groups (0.3, 1, 3, and 10 mg/kg, once every 2 weeks). In accordance with the above PK results and receptor occupancy data, the dose selected for the CT4 Phase II pivotal study was 3 mg/kg once every 2 weeks.

The 360 mg fixed dose once every 3 weeks was explored in 2 early studies (CT5 and CT7). The preliminary PK results from 9 subjects showed that compared with 3 mg/kg once

every 2 weeks (from 11 subjects in CT1), the steady-state peak concentration was increased to 153%, the steady-state trough concentration was increased to 133%, and AUC from time zero to Day 85 was increased by 138%. The PK model predicted a similar trough concentration (~32 μg/mL) and exposure level for toripalimab dosage regimens of 240 mg every 3 weeks and 3 mg/kg every 2 weeks.

Based on the comprehensive analysis of PK and receptor occupancy for toripalimab and the results from the Phase II pivotal melanoma study, the toripalimab dosage regimen of 240 mg as a fixed dose every 3 weeks was supported as the recommended dose for Phase III clinical studies.

3.4. Methods to Minimize Deviation

3.4.1. Enrollment/Randomization/Blinding Procedures

Randomization is the primary method of ensuring comparability between the study groups. All personnel involved in the study should do everything possible to ensure the strict implementation of randomization. The generation, review, quality assurance, testing, and verification of the randomization protocol will strictly follow the Sponsor's relevant standard operating procedure (SOP).

Block randomization will be used in this study. Treatment allocation will be generated by nonblinded statisticians prior to the initiation of enrollment using SAS software. Treatment allocation will be carried out using an interactive network response system (IxRS).

Eligible subjects will be screened and randomized at a 1:1 ratio to either the experimental arm or control arm according to the following stratification factors:

- Disease stage: II vs. IIIA vs. IIIB
- PD-L1 status: PD-L1 expression ≥1% versus PD-L1<1% or non-evaluable
- Planned surgical type: Pneumonectomy versus lobectomy
- Histopathologic subtype: Squamous cell carcinoma versus non-squamous cell carcinoma

3.4.2. Unblinding Treatment Assignment

A double-blind study design is used in this study, where subjects and investigators are not aware of the study treatment allocated to subjects. During the study, the Sponsor, all subjects, staff at the study site, consultants, and their representatives will remain blind to treatment allocation.

Planned interim analyses for this study will be presented in 10.5 Interim Analysis . Subject enrollment will continue during the interim analysis. The results, including treatment group (non-blind) information, will be generated by an independent statistical support team (iSSG) and provided to the independent data monitoring committee (IDMC). If necessary, limited personnel from the Sponsor may have unblinded access to these results, including treatment group information, so as to provide necessary information to the IDMC or implement IDMC recommendations. In this case, details regarding individuals who received access to unblinded information will be carefully recorded. Prior to the completion and unblinding of this study, the iSSG should not be involved in the protocol amendment, selection of statistical methods, statistical analysis plan, protocol deviation and data cleaning after the interim analysis, to reduce bias.

Prior to the conclusion of the study, all study sites and investigators will remain blinded to treatment assignment. The treatment allocation information will be restricted to select statisticians and statistical programmers responsible for preparing datasets or executing the interim analysis for the IDMC.

In case urgent unblinding is needed for subject management in a medical emergency (e.g., when a serious adverse event [SAE] occurs and knowledge of treatment assignment is necessary to provide effective management of the SAE), the investigators or the designated representatives should contact the Sponsor’s medical monitor or pharmacovigilance personnel to discuss the subject’s condition prior to unblinding of treatment assignment by the IxRS; unblinding of treatment assignment will occur only upon mutual agreement of investigators or designated representatives of the Sponsor. Under extremely critical circumstances, such as the subject’s life is endangered or permanent damage will occur without appropriate management based on knowledge of the treatment,

urgent unblinding can be requested by the investigator or their designated representatives by contacting the IxRS. In such cases, the Sponsor should be informed by investigators or designated representatives immediately after necessary measures are taken. Urgent unblinding can be performed through IxRS in this system.

When a subject experiences unequivocal disease progression and knowledge of the treatment assignment is needed to decide the most appropriate subsequent treatment, the investigator communicates with the Sponsor on the principle of ensuring the best benefit to the subject and refers to the Site Unblinding Process.

If the investigator desires information regarding the study drug for any other reason, they should contact the medical monitor and provide explanations for any premature unblinding of a subject's treatment assignment.

Once the patient is unblinded, the study treatment will be permanently terminated. Note: Discontinuation of study treatment is not the same as withdrawal or termination of the study, as specified in 6.3 Withdrawal From the Study or Termination of Study Treatment.

4. Study Population

4.1. Inclusion Criteria

Subjects must meet all of the following inclusion criteria to be eligible for enrollment in this study:

1. Have sufficient understanding of this study and voluntarily sign the Informed Consent Form (ICF).

2. Aged 18 to 70 years old, male or female.

3.Treatment-naïve, histologically confirmed, resectable, Stage IIIA, IIIB (N2) NSCLC (AJCC Cancer Staging Manual, version 8). The clinical tumor, node, and metastasis (cTNM) stage can be confirmed through positron emission tomography-computed tomography (PET-CT) or by pathological biopsy. Resectability is defined as resectable or potentially resectable according to the multidisciplinary consensus statement on the clinical management of subjects with stage III non-small cell lung cancer.

Resectable NSCLC includes any of the following cTNM stages:

- Stage II is defined as radically resectable, as assessed by a qualified thoracic

surgeon

- Stage IIIA (N0-N1), some N2 with single-station mediastinal lymph node < 2 cm in shortest diameter, and some T4 (satellite nodules present in adjacent lobes) N1

Potentially resectable NSCLC includes the following cTNM stages:

- Stage IIIA and IIIB based on ipsilateral mediastinal nodes (N2) less than 3 cm in shortest diameter
- Stage III with T3 or T4 centrally located tumors determined to be potentially resectable by a surgeon

Pathological puncture verification is strongly recommended for suspicious lesions detected by imaging studies that are presenting as clinically normal, which can lead to TNM stage changes in areas including but not limited to contralateral mediastinal lymph nodes, supraclavicular lymph nodes, solid/subsolid pulmonary nodule(s), and non-simple ground glass opacity (GGO).

4. Measurable lesions as defined by the Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

5. Tumor tissue specimens should be provided for pathological evaluation to confirm the diagnosis and for the detection of PD-L1 tumor expression and biomarkers prior to randomization. Tumor tissue specimens can be freshly obtained, while archived samples obtained within 3 months prior to enrollment. Acceptable tumor tissue specimens must contain tumor cells and be acquired through core needle biopsy, excisional biopsy during bronchoscopy, or surgical resection. Tumor samples acquired through fine-needle aspiration or bronchial brushing are not acceptable.

6. ECOG performance status score of 0 or 1.

7. Adequate organ function:

- Hematology: Absolute neutrophil count (ANC) ≥1500/μL; platelet ≥100000/μL; hemoglobin ≥9.0 g/dL.
- Renal function: Serum creatinine ≤1.5× upper limit of normal (ULN) or calculated creatinine clearance (CrCl) ≥60 mL/min (using Cockcroft-Gault formula).
- Hepatic function: Total bilirubin ≤1.5× ULN or direct bilirubin within the normal range for subjects with a total bilirubin level >1.5× ULN; aspartate

aminotransferase (AST) and alanine aminotransferase (ALT) ≤2.5× ULN.

- Endocrine function: Thyroid-stimulating hormone (TSH) within the normal range. Note: subjects may still meet the inclusion criteria if the TSH levels are outside the normal range at baseline, as long as T3 and free T4 are within the normal ranges.
- Coagulation function: International normalized ratio (INR) or prothrombin time (PT), as well as activated partial thromboplastin time (aPTT) ≤1.5× ULN.

8. Being willing and able to comply with the visits, treatment plan, laboratory examinations, and other study procedures scheduled in the study.

9. Total lung function is able to withstand the proposed lung resection procedure as assessed by a surgeon.

10. Women of childbearing potential must undergo a serum pregnancy test within 3 days prior to the first dose of the investigational product, and the result must be negative. Female subjects of childbearing potential and male subjects whose partners are women of childbearing potential must agree to use highly effective contraceptive methods during the study period and for at least 180 days after the last dose of the study drug.

4.2. Exclusion Criteria

Subjects cannot be enrolled in this study if they meet any of the following exclusion criteria:

1. Presence of locally advanced, unresectable, or metastatic disease. Unresectable disease is defined by the following criteria outlined in the multidisciplinary consensus statement on the clinical management of subjects with stage III non-small cell lung cancer: Stage IIIC, Stage IIIB with N3 lymph node involvement, Stage IIIA or IIIB with N2 involvement of single-station mediastinal lymph node where the shortest diameter is ≥3 cm, N2 involvement characterized by the fusion of multi-station lymph node where the shortest diameter is ≥2 cm on CT, T4 with invasion of the esophagus, heart, aorta, or other adjacent tissues, which in the opinion of the investigator, renders the subject unresectable.
2. Pancoast tumor, or in cases where the tumor histology corresponds to a large cell neuroendocrine carcinoma or sarcomatoid tumor.

3. Participants with known active EGFR-sensitive mutations or ALK translocation. EGFR and ALK mutation status must be identified for the subjects with non-squamous cell carcinoma.
4. Previous systemic anti-tumor therapy, including investigational products, for early-stage NSCLC.
5. History of (non-infectious) pneumonia/interstitial lung disease or ongoing pneumonia/interstitial lung disease that requires steroid treatment.
6. History of active tuberculosis.
7. Active infections requiring systemic treatment.
8. Subjects with any known or suspected autoimmune disease or immunodeficiency, with the following exceptions: patients with a history of hypothyroidism, if hormone replacement therapy is not needed or currently stable while receiving hormone replacement therapy at a physiological dose; subjects with stable type I diabetes whose blood glucose is controlled.
9. Uncontrolled active hepatitis B infection, defined as positive hepatitis B surface antigen [HBsAg] with detectable hepatitis B virus-deoxyribonucleic acid (HBV-DNA) above the ULN at the clinical laboratory of the study center, or active hepatitis C infection, defined as hepatitis C surface antibody seropositive and hepatitis C virus-ribonucleic acid (HCV-RNA) during screening period. Subjects with HBV-DNA <500 IU/mL within 28 days prior to randomization, who received antiviral therapy per local standard of care for at least 14 days and commit to take antiviral therapy continuously during the study, are eligible.
10. Known human immunodeficiency virus (HIV) infection (e.g., HIV seropositive).
11. Vaccination with a live vaccine within 30 days prior to the first dose, including but not limited to the following: mumps, rubella, measles, varicella/herpes zoster (varicella), yellow fever, rabies, Bacillus Calmette-Guérin (BCG), and typhoid vaccines (inactivated viral vaccines are permitted).
12. Grade ≥2 peripheral neuropathy.
13. Previous use of PD-1/PD-L1 blocking antibodies or drugs targeting another T cell co-signaling receptor (e.g., CTLA-4, OX-40).
14. History of severe allergic reactions to any monoclonal antibodies.
15. History of serious allergic reactions to pemetrexed, paclitaxel or docetaxel, cisplatin, carboplatin, or prophylactic medications administered during chemotherapy.

16. Known serious or uncontrolled co-morbid disease, including but not limited to, hemodynamically unstable cardiovascular events within 6 months, symptomatic cerebrovascular events, cirrhosis of the liver with Child-Pugh A classification or higher.
17. History or current evidence of any disease, concomitant medication, or abnormal laboratory results that may confound the safety or efficacy assessment, interfere with the subject's participation throughout the full course of the study, or prevent the subject from receiving the maximum benefit from participating in the study, as judged by the investigators;
18. Presence of any malignancy other than non-small cell lung cancer within 5 years prior to the first dose. Exception: subjects with a history of a malignancy with negligible risk of metastasis or death (e.g., expected disease-free survival >5 years) and anticipate a complete cure after definitive treatment (e.g., appropriately treated carcinoma in situ of the cervix, basal or squamous cell skin cancer, or ductal breast carcinoma in situ treated) are eligible.

4.3. Randomization

Subjects who do not meet eligibility criteria for the study are not allowed to be randomized into the study. If an ineligible subject is randomized and receives treatment, or the subject no longer meets eligible criteria after enrollment but before the first dose of study treatment, the Sponsor, medical monitor and investigators must discuss whether the subject will continue or be withdrawn from the study and any decisions reached must be documented.

Subjects who previously failed screening can be re-screened, provided that they re-sign the ICF and are registered under a new subject identification number for rescreening.

The treatment will be initiated within 3 days of randomization.

4.4. Requirements of Lifestyle

Toripalimab should not be administered to pregnant women. Its effect on pregnant and lactating women is unclear; however, a number of non-clinical studies have shown that the PD-1/PD-L1 signaling pathway is essential for the maintenance of pregnancy through maternal immunity to fetal tissue tolerance and embryo-fetal survival, and thus adverse effects are expected for pregnancy. No developmental or reproductive toxicity

study of toripalimab has been carried out. Based on the critical role of the PD-L1/PD1 pathway for fetal tissue tolerance through maternal immunity to maintain pregnancy, toripalimab' s effects on human fertility or fetal development are unknown, but adverse effects are expected.

In accordance with the mechanism of action of the chemotherapeutic agent, chemotherapy may damage the fetus in pregnant women. Subjects should be informed of the potential risks of chemotherapy to the fetus if they receive it during pregnancy or become pregnant during chemotherapy. Female subjects with childbearing potential should be informed to avoid pregnancy. Male subjects should be advised to practice highly effective contraception during the treatment and for 6 months thereafter. Contraceptive measures or abstinence are recommended. In addition, due to the possibility of irreversible infertility caused by chemotherapy, male subjects are advised to seek counseling on sperm preservation before treatment starts.

Women of childbearing age must undergo a serum pregnancy test within 3 days prior to the first dose in the screening period, and the results must be negative.

Subjects in both groups—female subjects of childbearing potential and male subjects whose partners are women of childbearing potential, should be sterilized, or must agree to use highly effective contraceptive methods during the study and for 180 days after the last dose of the study drug.

The investigators or their designated personnel, in consultation with the subjects, will select a suitable contraceptive method for the subjects and their partners from the highly effective options listed below. The investigators will confirm that the subject is aware of the correct usage of the selected contraceptive methods and remind them of the need for contraception during scheduled study visits. In addition, the investigator will ensure that the subject is aware that if pregnancy is suspected or confirmed, the investigators must be notified immediately.

A "highly effective contraceptive method" is defined as a method used alone or in combination with other methods continuously and properly, with an annual failure rate of <1%. Highly effective contraceptive methods include the following:

1. Hormone contraceptive methods that inhibit ovulation (e.g., oral, insertion, injection, implantation, and percutaneous), used throughout the treatment period.
2. Intra-uterine contraceptive device.
3. Male/female condoms used concurrently with spermicide (i.e., foam, gel, film, cream, or suppository).
4. Males sterilized through vasectomy.
5. Ligation, occlusion, or resection of bilateral fallopian tubes.

5. Study Treatment

5.1. Investigational product

5.1.1. Obtaining Study Drug

The investigational product (toripalimab/placebo) will be provided by the Sponsor, uniformly packaged and of an appropriate quality (per certificate of analysis).

The concomitant anti-tumor drugs and prophylactic medications for nausea or allergic reaction AEs are non-investigational drugs and therefore not provided by the Sponsor. These drugs are marketed products that can be purchased by the study center and should be stored in accordance with the package insert or summary of product characteristics.

5.1.2. Basic Information

Study drug 1 (investigational product): toripalimab/placebo

Table 1: Toripalimab (JS001) and placebo

Investigational product/control drug	Manufacturer	Supply concentration and formulation
Toripalimab	Suzhou Zhonghe Biosciences Co., Ltd.	Sterile liquid containing 240 mg toripalimab. The nominal concentration is 40 mg/mL, containing 20 mM sodium citrate, 2.5% (w/v) mannitol, 50 mM NaCl and 0.02% (w/v) polysorbate 80. The pH is 6.0. Each bottle contains 6 mL of effective volume.
Placebo	Suzhou Zhonghe Biosciences Co.,	Sterile liquid containing 20 mM sodium citrate, 2.5% mannitol, 50 mM NaCl, 0.02% polysorbate 80. The pH

	Ltd.	is 6.0. Each bottle contains 6 mL of effective volume.

Study drug 2: pemetrexed disodium for injection

Dosage form: Sterile lyophilized powder for injection

Active ingredients: Pemetrexed disodium

Storage conditions: Store according to current instructions for use.

Study drug 3: paclitaxel injection

Dosage form: A colorless to slightly yellow viscous liquid

Active ingredients: 5β, 20-epoxy-1, 2α, 4, 7β, 10β, 13α-Hexahydroxy taxone-11-ricor-9-keto-4, 10diacetate-2-benzoate-13-(2R, 3 S)-benzoyl-3-phenylisoserine

Storage conditions: protected from light in an airtight container, below 25°C

Study drug 4: docetaxel injection

Dosage form: Sterile aqueous injection

Active ingredient: Docetaxel

Storage conditions: 2℃ to 8℃, away from light in a sealed container

Study drug 5: carboplatin for injection

Dosage form: Sterile aqueous injection

Active ingredients: Cis-I, I-cyclobutane dicarboxylate diamine platinum

Storage conditions: Store in a cool (no more than 20°C) container, away from light

Study drug 6: cisplatin for injection

Dosage form: Sterile aqueous injection

Ingredients: (cis) Diaminodichloroplatinum, sodium chloride, polyethylene glycol 400.

Storage conditions: protected from light in a sealed container

5.1.3. Study Drug Appearance, Packaging, and Labeling

The labeling and packaging of each investigational product shall be based on the actual packaging.

The labeling of study drug will be carried out in accordance with the relevant regulations and guidelines of Good Clinical Practice (GCP). The content of label includes

but is not limited to: drug name, drug number, packaging specifications, production batch number, term of validity, instruction for use, storage condition, Sponsor's name, and marked with "for clinical study use only".

5.1.4. Storage and Stability of the Drug

Investigators or their authorized representatives (e.g., pharmacists) will ensure all the study drugs are stored in a controlled, secure area that meets the storage condition, and the storage complies with the requirements of applicable regulations.

The investigational product should be stored in its original container and in accordance with the drug label. If the storage conditions on the label are inconsistent with other information on the investigational product (e.g., Pharmacy Manual), the storage conditions on the label shall prevail.

The study center must record daily the highest and lowest temperatures of all storage areas (e.g., freezing, refrigerated or room temperature). The recording cycle should start from the receipt of the investigational product and continue until the completion of the study. Even with a continuous monitoring system, the study center should also have a record log to ensure the correct storage temperature. The temperature monitoring device and storage device (e.g., freezer) should be checked regularly to ensure their proper functioning.

Any deviation from the conditions on the product label should be reported promptly. The study center should take active measures to return the investigational product to the appropriate storage conditions as soon as possible, and all temperature deviations and measures taken will be reported to the Sponsor.

5.1.5. Management, Distribution, and Disposition

The management, distribution, and disposition of the investigational products for the study will be performed by designated personnels from the study site's GCP pharmacy. Investigators must make sure that all investigational products are only used for the subjects who participate in the study and the dosage and administration method are in accordance with the study protocol. The remaining investigational product "toripalimab" will be

returned to the Sponsor. Expired or remaining drug solutions will be destroyed directly according to the standards for medical waste. The investigational product shall not be transferred to any non-clinical study participant.

The investigational products are preserved as specified in the label. The drug receipt form must be signed in duplicate when the drug arrives at the study center, one copy for the clinical study center and the other for the Sponsor. If the remaining drugs and empty boxes need to be recovered at the end of the study, both parties shall sign the drug recovery sheet. The distribution and recovery of each drug shall be promptly recorded on a dedicated record sheet.

Monitors are responsible for monitoring the supply, use, and storage of the investigational products and the treatment of the remaining products.

Used investigational products can be recovered and destroyed by the Sponsor; alternatively, the study center can be commissioned to destroy them. Before commissioning, the monitor needs to verify the study center's destruction process, relevant package inserts are present, and a relevant certificate of destruction is provided after the drugs have been destroyed.

5.2. Drug Preparation

5.2.1. Drug Preparation

The investigational product should be prepared according to the Pharmacy Manual and the package insert for approved products, by qualified or experienced investigators such as doctors, pharmacists, nurses, or medical assistants (as permitted by the operation guidance of the country or research institute).

The investigational products contain cytotoxic drugs. Only qualified personnel who are familiar with how to minimize improper exposure to themselves and their environment can be allowed to prepare, distribute, and safely discard investigational products.

Preparation of toripalimab/placebo:

Please refer to the Pharmacy Manual for the mixing method, concentration (preparation), and **usage of the solution**. Since this product does not contain any antimicrobial preservatives or bacteriostatic agents, care must be taken to ensure that the prepared liquid is sterile.

Toripalimab dose: 240 mg fixed dose

Preparation of pemetrexed disodium injection:

Drug preparation is based on the drug package insert approved by CFDA or the clinical routine operating practice.

Pemetrexed dose: 500 mg/m^2

The formula for calculating body surface area (BSA): BSA (m^2) = 0.20247 x height $(m)^{0.725}$ x weight $(kg)^{0.425}$.

If there is a difference of more than 10% between the body weight on administration day and the weight used for calculating the previous dose, the dose must be recalculated.

Preparation of paclitaxel injection:

Drug preparation is based on the drug package insert approved by CFDA or the clinical routine operating practice.

Paclitaxel dose: 175 mg/m^2

The formula for calculating body surface area (BSA): BSA (m^2) = 0.20247 x height $(m)^{0.725}$ x weight $(kg)^{0.425}$.

If there is a difference of more than 10% between the body weight on administration day and the weight used for calculating the previous dose, the dose must be recalculated.

Preparation of docetaxel injection:

Drug preparation is based on the drug package insert approved by CFDA or the clinical routine operating practice.

Docetaxel dose: 60 mg/m^2 or 75 mg/m^2

The formula for calculating body surface area (BSA): BSA (m^2) = 0.20247 x height $(m)^{0.725}$ x weight $(kg)^{0.425}$.

If there is a difference of more than 10% between the body weight on administration day and the weight used for calculating the previous dose, the dose must be recalculated.

Preparation of carboplatin injection:

Drug preparation is based on the drug package insert approved by CFDA or the clinical routine operating practice. Once diluted, the product should be used up within 8 hours. Direct sunlight should be avoided during infusion and storage.

The dose of the carboplatin injection should be calculated according to the AUC and endogenous creatinine clearance. The value of AUC (mg/mL x min) is 5.

Carboplatin is administered according to the Calvert formula of AUC:

Carboplatin dose (mg) = the defined AUC (mg/mL x min) × [creatinine clearance (mL/min) +25].

Cockcroft formula is used to calculate the endogenous creatinine clearance, as follows:

If serum creatinine in units of mg/dL:

$$\text{Creatinine clearance rate in males (mL/min)} = \frac{(140 - \text{age}) \times (\text{body weight})\ a}{72 \times \text{serum creatinine}}$$

$$\text{Creatinine clearance rate in females (mL/min)} = \frac{0.85 \times (140 - \text{age}) \times (\text{body weight})\ a}{72 \times \text{serum creatinine}}$$

Serum creatinine concentration (μmol/L):

$$\text{Creatinine clearance rate in males (mL/min)} = \frac{(140 - \text{age}) \times (\text{body weight})\ a}{0.818 \times \text{serum creatinine}}$$

$$\text{Creatinine clearance rate in females (mL/min)} = \frac{0.85 \times (140 - \text{age}) \times (\text{body weight})\ a}{0.818 \times \text{serum creatinine}}$$

Age, “a”, is expressed in years and body weight is expressed in kg.

If the serum creatinine concentration is below the lower limit of the normal value, calculate creatinine clearance using the lower limit of the normal value to avoid toxicity.

Preparation of cisplatin injection:

Drug preparation is based on the drug package insert approved by CFDA or the clinical routine operating practice.

Administered dose: The administration dose will be calculated based on BSA, 75 mg/m^2 IV infusion, once every 3 weeks, on Day 1 of each cycle.

The formula for calculating body surface area (BSA): BSA (m^2) = 0.20247 x height (m)$^{0.725}$ x weight (kg)$^{0.425}$.

A sufficient hydration treatment must be performed 2-16 hours before and within at least 6 hours after drug administration. This product needs to be administered through

intravenous infusion after being diluted with normal saline or a 5% glucose/sodium chloride solution.

If there is a difference of more than 10% between the body weight on administration day and the weight used for calculating the previous dose, the dose must be recalculated.

5.2.2. Investigation Products Accountability

The study site will complete the preparation of the investigational product and record according to the study drug manual. Toripalimab/placebo is prepared for infusion in the whole bottle; meanwhile, it should be ensured that the administration principle of chemotherapeutic drugs should follow the principle of sufficient amount. For possible liquid preparation error in clinical practice, the difference between the planned dose and the actual dose should be controlled within ± 5%. The documentation system of clinical study center should include all relevant or required information on drug preparation and administration.

5.3. Usage

5.3.1. Treatment Plan

Experimental arm:

Neoadjuvant treatment period: Toripalimab and platinum-based doublet chemotherapy for 3 cycles;

Adjuvant treatment period: Toripalimab and platinum-based doublet chemotherapy for 1 cycle;

Maintenance treatment period: Toripalimab monotherapy for 13 cycles.

Control Arm:

Neoadjuvant treatment period: Placebo and platinum-based doublet chemotherapy for 3 cycles;

Adjuvant treatment period: Placebo and platinum-based doublet chemotherapy for 1 cycle;

Maintenance treatment period: Placebo for 13 cycles

The selection of the platinum drug (cisplatin or carboplatin) will be determined by the investigator.

Each treatment cycle is 21 days (±3 days). All drugs in the treatment regimen will be administered on Day 1 of each cycle. Subjects must complete all protocol-specified safety evaluations within 72 hours prior to dosing on Day 1 of each treatment cycle as well as radiological tumor imaging and blood collection for biomarker testing, as per protocol. It is recommended that subjects are hospitalized for a 72-hour observation period after the first dose, when feasible. The order of administration and dosage of the study drugs are shown in Table 2.

Order of drug administration: On Day 1 of each cycle, drugs will be administered sequentially. The toripalimab/placebo IV infusion will be administered first, followed by the pemetrexed, paclitaxel, docetaxel, and platinum IV infusion after an interval of at least 30 mins, and the dosing interval between chemotherapeutic drugs may be in accordance with local standard of care.

Toripalimab/placebo will be administered for at least 60 mins for the first dose. If the first dose is tolerated without an infusion-related AE, subsequent doses of toripalimab may be infused over a period of 30 mins (±10 mins). If an infusion reaction occurred during the most recent previous infusion, the intravenous infusion of toripalimab should last for over 60 mins.

Table 3: List of Administration and Dosage of Investigational Product

Name of investigational	Dose/route of administration	Order of administration	Day of Dosing and Interval between
Toripalimab/Placebo	240 mg/IV infusion over 60 min Q3W	1	D1
Pemetrexed	500 mg/m^2/IV infusion, ≥10 min, Q3W	2	D1, interval ≥ 30 min
Paclitaxel	175mg/m^2/IV infusion, Q3W	2	D1, interval ≥ 30min
Docetaxel	60-75mg/m^2/IV infusion, Q3W	2	D1, interval ≥ 30 min
Carboplatin	AUC 5, IV infusion, ≥30 min, Q3W	3	D1, dose interval in accordance with local standard of care
Cisplatin	75mg/m^2, IV infusion, Q3W	3	D1, dose interval in accordance with local standard of care

5.3.2. General Recommendations for Treatment Plan Adjustments

If an adverse reaction is related to some of the drugs in the combination regimen, in the judgment of the investigator, only the dose of the drugs causing toxicity will be adjusted to include dose interruption, delay, or termination. If it is not possible to attribute an adverse event to specific drugs, dose adjustments should apply to all drugs in the regimen.

If one or more of the chemotherapy drugs is not tolerable, the dose can be reduced twice. If a chemotherapeutic drug is not tolerated after two dose reductions, that drug(s) will be permanently discontinued. Principles for dose adjustment: The first dose adjustment will be 75% of the initial dose, and the second dose adjustment will be 50% of the initial dose. Once reduced, the dose of the chemotherapeutic drug cannot be increased in subsequent cycles.

If treatment is interrupted/delayed due to treatment-related AEs caused by only some of the drugs in the regimen, the remaining drugs will be administered as scheduled. When the toxicity subsides, all drug treatments will be carried out as scheduled. The investigator

should seek to avoid disrupting the combination treatment regimen and strive to give all planned study treatments within the specified time window of each administration cycle.

In situations where dose adjustment is not specified in the protocol, investigators need to consider the benefit/risk ratio of the subjects before making a decision on dose adjustment. It is recommended to suspend the administration and reduce the dose as specified in the protocol. If the investigator has other considerations for dose adjustment, the investigator should record the reasons for comprehensive consideration in the original medical record and communicate with the Sponsor. Once the dose is reduced, it cannot be readjusted unless some drugs are discontinued.

5.3.2.1. Study Drug (Toripalimab/Placebo) Treatment Adjustment

Increases or decreases in the dose of toripalimab/placebo is not permitted, only dose delay is permitted.

The window for administration of each toripalimab dose is calculated based on the date of the first dose. Generally, delay in administration of toripalimab/placebo should not exceed 3 days, but can be more than 3 days with documentation of the reason for the delay. If delayed for more than 3 days, the next dose of toripalimab/placebo should be based on the actual dosing date in the previous cycle with 3 days' time window. In the combination treatment period, if delayed for more than 7 days, it is recommended not to compensate for the missed dose in this cycle. Instead, the drug will be continued in the next cycle, and the time window will be calculated based on the date of the first dose. The maximum permissible timeframe for toripalimab dose delay/suspension due to adverse events is 12 weeks.

If the treatment is interrupted due to an AE, it should recommence at the original dose once the AE has been resolved. Adjustments to the toripalimab treatment regimen due to immune-related adverse reactions is detailed in the Table below. Toripalimab must be suspended or resumed if the investigator believes that the operations listed in the Table cannot be performed or encounters circumstances not listed in the Table during the clinical

operation. In this case, investigators must take into account the benefit/risk ratio of the subject, and the decision should be discussed and agreed upon with the Sponsor.

Subjects are allowed to interrupt the dose for other medical events unrelated to the study treatment, surgery, or accidents (holidays) and should resume study treatment within 3 weeks after dose interruption unless the investigators and the Sponsor have other resolutions upon discussion. The reason for treatment interruption should be recorded in the electronic case report form (eCRF).

Table 4: Recommended Treatment Adjustment (Toripalimab)

Immune-related adverse reactions	Severity (NCI CTC v5.0)	Treatment adjustment
Pneumonia	Grade 2	Dose interruption, until recovery to Grades 0-1
	Grades 3-4 or recurrent Grade 2	Permanently discontinued
Diarrhea and Colitis	Grades 2-3	Dose interruption, until recovery to Grades 0-1
	Grade 4	Permanently discontinued
Hepatitis	Grade 2, AST or ALT at 3-5 × ULN or total bilirubin at 1.5-3 × ULN	Dose interruption, until recovery to Grades 0-1
	Grades 3-4, AST or ALT > 5 × ULN, or total bilirubin > 3 × ULN	Permanently discontinued
Nephritis	Grades 2-3 elevated blood creatinine	Dose interruption, until recovery to Grades 0-1
	Grade 4 elevated blood creatinine	Permanently discontinued
Endocrine disorders	Grades 2-3 symptomatic thyroid function decreased Grades 2-3 hyperthyroidism Grades 2-3 hypophysitis, Grade 2 adrenal insufficiency Grade 3 hyperglycemia or type I diabetes	Dose interruption, until recovery to Grades 0-1
	Grade 4 thyroid function decreased, Grade 4 hyperthyroidism,	Permanently discontinued

	Grade 4 hypophysitis, Grades 3-4 adrenal insufficiency, Grade 4 hyperglycemia or type I diabetes	
Cutaneous adverse reaction	Grade 3 rash	Dose interruption, until recovery to Grades 0-1
	Grade 4 skin rash, Stevens-Johnson syndrome, or toxic epidermal necrolysis	Permanently discontinued
Thrombopenia	Grade 3	Dose interruption, until recovery to Grades 0-1
	Grade 4	Permanently discontinued
Others	Grades 2-3 increased blood amylase and lipase Grades 2 pancreatitis Grade 2 myocarditis [a] Grades 2-3 other immune-associated AEs (first occurrence)	Dose interruption, until recovery to Grades 0-1
	Grade 4 increased blood amylase and lipase Grades 3-4 pancreatitis Grades 3-4 myocarditis Grades 3-4 encephalitis Grade 4 other immune-associated AEs (first occurrence)	Permanently discontinued
Recurrent or persistent adverse reaction	Recurrent Grades 3-4 (except endocrine disease) Grades 2-3 adverse reaction not improved to Grades 0-1 within 3 months after the last dose (except endocrine disease) Corticosteroid unable to be reduced to ≤10 mg/day prednisone equivalent dose within 3 months after the last dose	Permanently discontinued
Infusion-related Reaction	Grade 2	Reduction of infusion rate or dose interruption, dose resumption can be considered and closely observed once the symptoms have been alleviated

	Grades 3-4	The product must be discontinued immediately, and symptomatic treatment will be administered

a: The safety remains unclear regarding whether this product can be resumed after myocarditis improves to Grade 0-1 after treatment.

5.3.2.2. Adjustment of the Dose of Platinum-based Chemotherapy

Adjustment of chemotherapy dosing includes dose reductions and delays. The window for the administration of each chemotherapy dose is calculated from the date of the first dose. Generally, delay in administration of chemotherapy should not exceed 3 days, but can be more than 3 days with documentation of the reason for the delay. If delayed for more than 3 days, the next dose of chemotherapy should be based on the actual dosing date in the previous cycle with 3 days' time window. The maximum permissible timeframe for chemotherapy dose delay/suspension due to adverse events is 6 weeks. Dose adjustment will be carried out according to the package insert for the drug and clinical diagnostic and therapeutic specification, and the recommended dose adjustment is specified in the Table below. If investigators, taking into consideration the benefit/risk ratio of the subjects during the clinical operation, believe that the operations listed in the following Table cannot be carried out or encounter a situation not listed in the Table, the decision to suspend or resume the drug should be made upon discussion with the Sponsor.

Subjects are allowed to interrupt the dose for other medical events unrelated to the study treatment, surgery, or accidents (holidays). Subjects should resume study treatment within 3 weeks after dose interruption unless the investigators and the Sponsor have other resolutions upon discussion. The reason for treatment interruption should be recorded in the electronic case report form (eCRF).

Table 5: Dose Adjustment for Hematologic Toxicity of Platinum-based Doublet Chemotherapy

Toxicity	Pemetrexed	Cisplatin	Carboplatin
Neutrophil count decreased			

Grade 4	One dose level reduced, and G-CSF prevention considered in subsequent cycles	One dose level reduced, and G-CSF prevention considered in subsequent cycles	One dose level reduced, and G-CSF prevention considered in subsequent cycles
Reduced platelet count			
Grade 3	Reduce by 1 dose level	Reduce by 1 dose level	Reduce by 1 dose level
Grade 4	Reduce by 1 dose level	Reduce by 1 dose level	Reduce by 1 dose level
Anemia			
Grade 2	Reduce by 1 dose level	Reduce by 1 dose level	Reduce by 1 dose level
Grade 3	Reduce by 1 dose level	Reduce by 1 dose level	Reduce by 1 dose level
Grade 4	Drugs suspended	Drugs suspended	Drugs suspended

Table 6: Dose Adjustment for Non-hematologic Toxicity of Platinum-based Doublet Chemotherapy

Toxicity	Pemetrexed	Cisplatin	Carboplatin
Grade ≥3 febrile neutropenia	Reduce by 1 dose level	Reduce by 1 dose level	Reduce by 1 dose level
Grade ≥3 diarrhea	Reduce by 1 dose level	No change	No change

Grade ≥3 allergic reaction	Discontinuation	Discontinuation	Discontinuation
Grade 2 neuropathy	No change	Reduce by 1 dose level	No change
Grade ≥3 neuropathy	Discontinuation	Discontinuation	Discontinuation
Creatinine clearance<50 mL/min	Creatinine clearance<45 mL/min, drug discontinuation	Discontinuation	Creatinine clearance<20 mL/min, drug discontinuation
Other Grade ≥3 toxicities (not including fatigue, joint pain, and muscle pain)	Adjustment as appropriate	Adjustment as appropriate	Adjustment as appropriate

During treatment with taxanes, the treatment should be delayed and subsequent doses should be adjusted in accordance with the principle on dose reduction for the subjects with relevant hematological events (neutropenia, neutrophil <500/mm3 for 7 days) or other Grade 3/4 nonhematological toxicity (except Grade ≥3 peripheral neuropathy). Treatment with taxanes should be discontinued directly once Grade 3 and above peripheral neuropathy occurs.

If the investigator has additional considerations regarding chemotherapy drug dose adjustment, communication with the Sponsor is required.

5.3.3. Prophylactic Medication

Toripalimab/placebo should be administered without prophylaxis for adverse reactions.

Pemetrexed:

Vitamin Supplementation: In order to reduce pemetrexed-induced toxicity, instruct subjects to take low-dose folic acid preparations or folic acid-containing compound vitamins orally every day. Preventive supplementation with vitamins can be carried out in accordance with the package insert for the pemetrexed or folic acid preparation. As stated

in the package insert for pemetrexed, folic acid (350 to 1000 μg) must be taken orally, once a day for at least 5 days in the first 7 days prior to the first dose of pemetrexed and throughout the treatment as well as 21 days after the last dose of pemetrexed. In the week prior to the first dose of pemetrexed, subjects must also receive vitamin B12 1000 μg intramuscularly, and then once every 3 cycles thereafter. Subsequent doses of vitamin B12 should be administered on the same day as pemetrexed. In clinical trials, doses of folic acid typically range from 350 to 1000 μg, while the dose of vitamin B12 is 1000 μg. The most frequently used oral dose of folic acid is 400 μg in clinical studies.

Taxanes:

Subjects receiving docetaxel will be given prophylactic corticosteroids as follows: dexamethasone 8 mg orally twice a day on the day prior to, on the day of, and one day after the administration of docetaxel. This protocol-required premedication is not considered a prohibited concomitant medication nor a protocol deviation.

Subjects receiving paclitaxel will be given prophylactic corticosteroids as follows: dexamethasone 20 mg orally twice per day on the day prior to and on the day of the administration of paclitaxel. This protocol-required premedication is not considered a prohibited concomitant medication nor a protocol deviation.

Carboplatin: Hydrate as clinically indicated.

Cisplatin: Adequate hydration, in accordance with the package insert, initiated 2 to 16 hours before and continuing for at least 6 hours after the administration of cisplatin.

Prophylactic medication may refer to the instructions for use or clinical practice.

5.4. Concomitant Therapies

Concomitant therapy is any therapy other than protocol-specified therapy administered to or taken by the subject for the prevention or treatment of concomitant medical conditions.

In accordance with GCP requirements, all the concomitant medications, blood products, and non-drug interventions (e.g., biopsy) received by subjects during the screening phase (28 days prior to the first dose) through 30 days after the last dose of study treatment should be recorded in the eCRF.

Dose adjustments for chemotherapy for subjects in the experimental and control arms should consider the contraindications and precautions specified in the latest version of the package insert for that drug and local standards of care.

5.4.1. Permitted Concomitant Medications

Concomitant medications/therapies permitted throughout the study:

- Subjects are permitted to receive supportive care for disease-related symptoms.
- Use of local, intra-ocular, intra-articular, intranasal, and inhaled corticosteroids (very little systemic absorption) is permitted. Physiologic steroid replacement (≤10 mg/day prednisone) for adrenal insufficiency is also permitted.
- Subjects are permitted to receive a short-course (up to 3 weeks) of corticosteroid for the prevention of allergic reactions (e.g., allergy to contrast agent dye) or treatment of non-autoimmune disease (e.g., delayed-type hypersensitivity caused by contact allergens).

5.4.2. Prohibited Concomitant Medication

In the screening, treatment, and, for subjects without disease progression, follow-up periods of the study, subjects are not permitted to receive any of the following:

- Systemic anti-tumor chemotherapy or biologic therapy not specified in this clinical protocol.
- Immunotherapy that is not specified in this clinical protocol.
- Investigational products other than toripalimab/placebo, pemetrexed**,** docetaxel, paclitaxel cisplatin and carboplatin.
- Vaccination within 4 weeks prior to and during the administration of the study treatment with live vaccines, including but not limited to measles, mumps, rubella, varicella, yellow fever, rabies, Bacille-Calmette-Guerin, and typhoid (oral) vaccines, is not permitted. **Inactivated** vaccines for the prevention of pneumonia, influenza, or COVID-19 are permitted. Other inactivated viral vaccines may be permitted after discussion with and agreement from the Sponsor.
- The use of any corticosteroids is not permitted other than treatment of study drug-induced, immune-related AE, as physiologic replacement for treatment of

endocrinopathy, or as permitted for an individual subject may be obtained after discussion with the Sponsor.

- The use of Traditional Chinese Medicine (TCM) preparations approved by the CFDA for treating cancer (including but not limited to Xi Huang Wan, Jinlong Jiaonang, Kanglaite Injection, Aidi Injection, Kangai Injection, and Xiao'aiping Pian) is not permitted. Refer to the relevant classification and regulations specified in the Expert Consensus on Use of Palliative Chinese Patent Medicine in Oncology (2013 version) for TCM preparations approved for treating cancer.

5.4.3. Supportive Care

During the study treatment period, all subjects should be given the best supportive care. The administration of palliative and supportive care for disease-related symptoms will be based on the investigator's judgment and relevant guidelines (e.g., the American Society of Clinical Oncology Guidelines, the Standards for the Diagnosis and Treatment of Primary Lung Cancer [2015 version] in China).

During the study treatment period, all subjects should be given the best supportive care.

Subjects who are stable on physiologic hormone replacement for endocrine insufficiency, such as those with a history of autoimmune-mediated hypothyroidism who are receiving a stable dose of thyroid hormone replacement, are eligible for this study.

5.4.4. Immunomodulators and Corticosteroids

Immunomodulatory drugs (i.e., immunostimulants or immunosuppressants) are not permitted except for the management of immune-related AEs.

Long-term and systemic use of corticosteroids for palliative or supportive treatment purposes is not permitted. Corticosteroids at the prescribed dose, as indicated in the instructions, are permitted for a cumulative period of ≤1 week as prophylactic medication for chemotherapy or contrast media allergy. Systemic corticosteroid treatment in individual subjects may be permitted after discussion with the Sponsor. Short-term (less than 21 days) use of systemic corticosteroids at therapeutic doses is permitted for treatment of non-autoimmune disease (e.g., delayed allergic reaction due to contact allergens).

Non-systemic corticosteroids (topical, inhaled, locally injected (e.g., joint), or eye drops) are permitted. Systemic corticosteroids (e.g., prednisone or equivalent ≤ 10 mg/day) at physiologic replacement doses (e.g., adrenal replacement steroid dose) are permitted.

5.4.5. Anti-inflammatory Drugs

If there are no known or predictable drug interactions with chemotherapeutic drugs and the drug is not listed in Section 5.4.2, non-steroidal anti-inflammatory drugs (NSAIDs) may be administered to treat pain.

5.4.6. Surgical Procedures Not Specified in the Protocol

Any surgical procedures other than those specified in the protocol should only be performed as needed during the study treatment. These procedures should be scheduled to minimize the potential influence of the study treatment on wound healing and the risk of hemorrhage, with investigational product dosing at least one week prior to surgery if possible. The timing of the administration after surgery is dependent on the clinical evaluation of wound healing and postoperative recovery.

6. Study Procedures

6.1. Study Procedures

6.1.1. Screening Period of Study Visits

Prior to conducting any protocol-required screening tests or clinical study procedures designated in the study, a written ICF on participation in this study must be obtained except for the tumor radiology, tumor tissue biopsy, and gene test. The ICF for the subjects who are not subsequently enrolled and enrolled subjects will be maintained at the study site.

All protocol-specified screening evaluations must be completed, and results must be reviewed to confirm that the subject meets all of the inclusion criteria before randomization. Investigators will keep the screening log to record information for all screened subjects and confirm each subject's eligibility, or when applicable, record the reason for screening failure.

Screening Period

Tumor imaging: If the protocol-required radiological imaging studies are performed at the study site prior to the signing of the ICF and within 28 days prior to the first dose as part of routine medical care at the clinical study site, there is no need to repeat radiologic imaging studies during screening. Radiological imaging of all known or suspected tumor sites should be performed. PET-CT is an essential parameter for the determination of cTNM staging.

Fifteen FFPE unstained tumor sections, obtained within 3 months prior to the first dose of the study drug or as a fresh biopsy during screening, must be provided prior to enrollment. Three slides will be used for the detection of preoperative PD-L1 status and the other 12 of which (optional) will be used for the detection of TMB and WES in tumor tissue. If the tumor specimen does not provide an evaluable result for PD-L1, the subject does not need to undergo a second biopsy and the subject will be stratified in the group with PD-L1 <1% of non-evaluable PD-L1.

Tumor specimens must be evaluated for EGFR and ALK status in subjects with non-squamous cell carcinoma. If test results are not available when the ICF is signed, the testing of available tumor specimens should be conducted at the local study site. If the testing cannot be conducted at the local study site, it must be performed at a central laboratory. If testing is required to determine EGFR and ALK status, at least 18 additional slides (this number may be changed by the testing unit based on the quality of the slides provided) must be provided prior to enrollment.

About 8-10ml of peripheral venous blood will be collected and sent to the central laboratory for biomarker detection of changes in plasma ctDNA content (optional) and the whole exome sequencing of blood cells (optional). From each sample, 8 ml of peripheral blood plasma will be used for ctDNA content determination (optional), and 2 ml of peripheral blood will be used for whole exome sequencing after centrifugation to obtain blood cells (optional). Details of blood collection and processing of these biomarker samples can be found in the Laboratory Manual.

Physical examination: A complete physical examination will be performed at screening.

Concomitant medication/concurrent therapy: All drugs administered within 28 days prior to the first dose of the investigational product must be recorded in the case report

form, including the generic name, administration method, reason for drug use, and start and end date will be detailed in the case report form (CRF).

AE: All SAEs caused by the mandatory protocol-specified interventions (e.g., interventional procedure, biopsy) from signing the ICF to the first dose of the investigational product will be recorded in the CRF.

See the **Study Flow Chart** for specific study procedures during the screening phase.

6.1.2. Neoadjuvant Treatment Period and Preoperative Visit

Eligible subjects who pass screening will enter the preoperative neoadjuvant treatment period and start the study treatment within 3 days after randomization.

The visit in the preoperative treatment period includes visits for the 3 cycles of preoperative neoadjuvant treatment and the perioperative visit.

Precautions for 3 Cycles of Preoperative Neoadjuvant Treatment:

The window period is ±3 days for all the cycles of therapy, except for the window period of +3 days for C1D1.

Pre-dose examination on C1D1: Hematology tests (i.e., complete blood cell count [CBC] with differential), urinalysis, serum chemistry, and coagulation function tests performed within 72 hours prior to the first dose in screening period, do not need to be repeated prior to study drug administration on C1D1.

Physical examination: Targeted specialized physical examinations will be performed during treatment.

Thyroid function test: Thyroid functions will be evaluated within 72 hours prior to administration in each cycle. In the case of clinically significant changes in thyroid function, request a consultation from the department of endocrinology regarding treatment and to rule out hypopituitarism.

Tumor assessment: The same radiological examination method should be used at baseline and subsequent evaluations, and evaluated by the same investigator whenever possible. Subjects who are suspected of having disease progression should undergo unscheduled tumor evaluations as needed.

Perioperative visit: Performed 4 weeks ± 7 days after completing the third cycle of neoadjuvant preoperative treatment. This visit includes perioperative safety and surgical feasibility assessments. The safety analysis involves evaluating preoperative

complications, while the surgical feasibility analysis involves evaluating the planning of the surgical type, whether to delay the operation, and surgical difficulty.

Peripheral blood samples of 8 mL will be collected for ctDNA content determination at the perioperative visit. Details of the blood collection and treatment process can be found in the **Laboratory Manual**.

Study drug treatment: Study procedures specified in the protocol must be done prior to the dose for each cycle. Dosing can only occur after a review of the safety evaluations and justification of continued treatment, as needed. If dose interruption or adjustment occurs due to hematological toxicity, the complete blood cell count needs to be repeated every week until recovery to normal. In the case of neutrophil counts $\leq 1.0\times10^9$/L or platelet counts $\leq 50\times10^9$/L, the frequency of repeated examination needs to be increased (once every 2 to 3 days). If ALT or AST increases by 3 or 2 times from the abnormal baseline value during the trial, the frequency of laboratory testing should be increased to once or twice per week (recommended).

Pharmacokinetic/immunogenicity sample collection: Serum samples will be collected prior to the administration of toripalimab/placebo in Cycles 1, 2, and 3 and sent to the central laboratory for plasma concentration/immunogenicity testing. Details of the blood collection and treatment process can be found in the Laboratory Manual.

See the **Study Flow Chart** for the specific test items and procedures required for each visit.

AE evaluations and concomitant medications/concurrent therapies will be recorded at each visit.

6.1.3. Operation Period

The preoperative visit should occur within 4 weeks (±7 days) and the surgery should be performed within 5 weeks (±7 days) after administration of the third cycle of neoadjuvant therapy (C3D1). This visit requires the collection of relevant parameters for surgical difficulty analysis, including the actual surgical type performed, duration of surgery, intraoperative blood loss, intraoperative fluid intake and output, whether conversion occurred, surgical approach, and completeness of resection.

Details regarding the handling and shipping of the postoperative specimens can be found in the **Laboratory Manual**.

Postoperative complications and the information on postoperative recovery parameters should also be recorded, including the drainage type, daily volume of drainage, time to removal of drainage tube(s), volume drained at removal of drainage tube, postoperative daily fluid intake and output volumes, and length of hospitalization.

6.1.4. Postoperative Safety Visit

Subjects who undergo complete surgical resection should complete this visit 30 (±7) days after surgery.

This visit should include a full physical examination, radiological tumor imaging, laboratory examination, and recording of AEs. See the **Study Flow Chart** for details of specific test items.

During this visit, 8 mL of peripheral blood will be collected for ctDNA content determination (optional).

6.1.5. Adjuvant Treatment Period

This visit should be completed 4 to 8 weeks after surgery.

This visit should include a full physical examination, radiological tumor imaging (if the safety visit took place after operation, the subject can be exempted from this visit; otherwise, it must be carried out and used as the postoperative baseline), laboratory examination, documentation of drug usage, and recording of AEs. See the **Study Flow Chart** for details of specific test items.

Study drug treatment: Study procedures specified in the protocol must be done prior to the first dose for each cycle. Dosing can only occur after a review of the safety evaluations and justification of continued treatment, as needed. If dose interruption or adjustment occurs due to hematological toxicity, the complete blood cell count needs to be repeated every week until recovery to normal. In the case of neutrophil counts $\leq 1.0\times10^9$/L or platelet counts $\leq 50\times10^9$/L, the frequency of repeated examination needs to be increased (once every 2 to 3 days). If ALT or AST increases by 3 or 2 times from the abnormal baseline value during the trial, the frequency of laboratory testing should be increased to once or twice per week (recommended). For subjects who have completed 4 cycles of neoadjuvant treatment before surgery, no administration will be performed in this cycle.

Pharmacokinetic/immunogenicity sample collection: Serum samples will be collected prior to the administration of toripalimab/placebo and sent to the central laboratory for

plasma concentration/immunogenicity testing. Details of the blood collection and treatment process can be found in the **Laboratory Manual**.

6.1.6. Maintenance Treatment Period

Subjects not indicated for radiotherapy will start the maintenance treatment period 3 weeks (±3 days) after the end of adjuvant treatment (postoperatively 1 cycle); those with an indication for radiotherapy will start radiotherapy within 4 weeks after the end of adjuvant treatment (postoperatively one cycle) and start maintenance treatment within 30 days (±3 days) after the end of radiotherapy.

Toripalimab/placebo will be given once every 3 weeks for a total of 13 cycles during the maintenance treatment period. Complete laboratory tests and recording of adverse events should be completed on the day of administration (± 3 days). See the **Study Flow Chart** for details of specific items.

Study drug treatment: Study procedures specified in the protocol must be done prior to the first dose for each cycle. Dosing can only occur after a review of the safety evaluations and justification of continued treatment, as needed. If dose interruption or adjustment occurs due to hematological toxicity, the complete blood cell count needs to be repeated every week until recovery to normal. In the case of neutrophil counts $\leq 1.0\times10^9$/L or platelet counts $\leq 50\times10^9$/L, the frequency of repeated examination needs to be increased (once every 2 to 3 days). If ALT or AST increases by 3 or 2 times from the abnormal baseline value during the trial, the frequency of laboratory testing should be increased to once or twice per week (recommended).

Pharmacokinetic and immunogenicity sample collection: Serum samples will be collected before the administration of toripalimab/placebo in Cycles 1, 5, 9, and 13 and sent to the central laboratory for plasma concentration/immunogenicity testing. Details of the blood collection and treatment process can be found in the **Laboratory Manual**.

6.1.7. Follow-up Visit Period

Safety visit:

The safety visit will be conducted at the study site at 30 days (±7 days) after the last dose of the study drug, and the visit will include hematology, blood biochemistry and other relevant examinations. See Study Flow Chart for the details.

Additional unscheduled safety visits can be performed for the purpose of tracking AE outcomes when necessary.

Survival Visit:

The survival visit will be performed once every 3 months (± 7 days) for 2 years after the last dose and every 6 months thereafter, and the visit will include a tumor follow-up assessment, survival assessment, and subsequent anti-tumor therapy. Survival visits should stop once the subject experiences progressive disease, dies, is lost to follow-up, and withdraws consent. Follow-up sessions will be conducted by telephone until the subject dies, is lost to follow-up, withdraws informed consent, or the Sponsor terminates the study.

See the **Study Flow Chart** for details.

6.1.8. End-of-Treatment Visit

Subjects who complete or decide to withdraw from study treatment are required to complete this visit with a window of ± 7 days.

This visit should include a complete physical examination, laboratory tests, and collection of AEs. Tumor imaging is not performed at this visit; instead, imaging information will be based on the most recent radiological evaluation every 3 months for 2 years, followed by routine radiological evaluations every 6 months thereafter. If a subject decides to withdraw on the day of maintenance treatment administration, the two visits may be considered as a single visit and should include all of the items indicated on the **Study Flow Chart**.

During this visit, 8 ml peripheral blood will be collected for ctDNA content determination.

6.1.9. Unscheduled Visits

Unscheduled visits for the assessment of potential AEs occurring during the study may take place at any time. The following information should be collected at unscheduled visits:

- AEs
- Concomitant medications/treatments

All relevant examinations will be performed (including radiological examinations, if available).

6.2. Study Tests

6.2.1. Laboratory Examination

- Hematology: Complete blood cell count (CBC) with differential to include white blood cell count, absolute neutrophil count, lymphocyte, monocyte, basophil, eosinophil, red blood cell count, hemoglobin, and platelet count.
- Serum biochemistry: Alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), serum creatine kinase (CK), serum creatine kinase isoenzyme (CK-MB), total protein (TP), albumin (ALB), total bilirubin (TBIL), direct bilirubin (DBIL), serum urea nitrogen (BUN) or urea (Urea), serum creatinine (Cr), serum electrolytes (K+, Na+, Cl-, Ca2+, Mg2+, and P), and blood glucose (GLU).
- Urinalysis: Includes white blood cells, red blood cells, and urine protein. If urine protein is ≥ 2+ at screening, an additional assessment of 24-hour urine protein quantification is required.
- Thyroid function: Serum thyroid-stimulating hormone (TSH); free triiodothyronine (FT3); free thyroxine (FT4); if FT3 and FT4 are unavailable, T3 and T4 can be used instead.
- Coagulation function: Includes international normalized ratio (INR), activated partial thromboplastin time (APTT), fibrinogen (FIB), and thrombin time (TT); if INR is unavailable, prothrombin time (PT) shall be used instead.
- Virology tests: Includes tests for HIV, HBV, and HCV as follows: HBV testing: HBsAg, HBsAb, HBcAb, HBeAg and HBeAb; if HBsAg is positive, further testing of HBV-DNA is necessary. HCV testing: HCV-Ab; if HCV-Ab is seropositive, further testing of HCV-RNA is also necessary.
- Pregnancy test: Serum pregnancy test.
- 12-lead electrocardiogram (ECG): Includes measurements of QT, QTc, and P-R interval.
- Echocardiogram: Includes left ventricular ejection fraction (LVEF).
- Pulmonary function: Includes forced vital capacity, forced expiratory volume in the first second, maximum ventilation volume, and the percentage of the actual measurement of carbon monoxide dispersion relative to the expected

measurement. If the pulmonary function is abnormal, a blood gas analysis must also be performed to evaluate pulmonary function; if the pulmonary function is not abnormal, oxygen saturation should be determined by digital pulse oximetry.

6.2.2. Radiologic Tumor Imaging

High-resolution CT plus oral/intravenous contrast or contrast-enhanced magnetic resonance imaging (MRI) are the preferred methods for radiologic tumor imaging. If the subject is known to be allergic to the contrast agents, radiologic imaging with contrast should be performed only after prophylaxis for contrast allergies in accordance with local standards of care. If administration of the contrast agent is contraindicated, non-contrast-enhanced scans should be performed.

Screening (baseline) evaluation, including contrast-enhanced CT scans of the thoracoabdominopelvic region, PET-CT, and contrast-enhanced MRI/CT scans of the brain (when suspected), should be performed within 28 days prior to administration. Subsequent tumor radiology assessment should include scans of the thorax and all suspected disease sites using the same imaging modality as at baseline. The frequency of radiologic tumor monitoring can be increased, as clinically indicated.

If a subject suffers general clinical deterioration without evidence of radiologic disease progression, such that the risks of continued treatment outweigh its benefits, the reason for treatment discontinuation should be reported as “clinical progression.” Every effort must be made to document radiologic disease progression after the discontinuation of study treatment in these subjects. Evaluation should also be performed when disease progression (e.g., worsening of symptoms) is suspected or when the subject withdraws from treatment.

6.2.3. Pharmacokinetic/Immunogenicity Sample Collection

Serum samples for PK and ADA should be collected within 2 hours prior to administration on Day 1 of Cycles 1, 2, and 3 in the neoadjuvant treatment period, on Day 1 of Cycle 1 in the adjuvant treatment period, and on Day 1 of Cycles 1, 5, 9 and 13 of the maintenance treatment period.

Central experiments will test samples for toripalimab concentration/ADA/neutralizing antibodies using a validated method. Only samples from subjects treated with toripalimab need to be tested.

6.2.4. Collection of Tumor Tissues/Biomarker Test in Peripheral Blood

Fifteen white formalin-fixed paraffin-embedded (FFPE) sections from subjects diagnosed with Stage II and Stage III NSCLC will be provided before enrollment, either archived within 3 months before the first dose or freshly obtained. Among them, 3 (must) will be used for the determination of PD-L1 status, and the other 12 (optional) will be used for the detection of optional biomarker studies (e.g., TMB and WES). In addition, subjects with non-squamous cell carcinoma who are unable to be tested for EGFR/ALK at the study site may be sent to the central laboratory for testing and will provide an additional 18 sections (the number of sections may change due to actual testing unit requirements and the quality of the sections provided). Formalin-fixed paraffin-embedded tissue blocks are preferred over glass slides, and newly obtained biopsies are preferred over archival tissue. The criteria for fine needle aspiration, cytology smear for pleural effusion, or punch biopsy are insufficient for reviewing biomarker testing. Requirements for tumor tissue sampling, sample preservation, transportation, and analysis are detailed in the Laboratory Manual.

8 ml peripheral blood (optional) samples will be collected for exploratory biomarker analysis (detection of changes in ctDNA content) at four time points: screening phase, perioperative period, postoperative 30 days and withdrawal from the study. In addition, 2 ml of peripheral blood (optional) will be collected during the screening phase and centrifuged to obtain blood cells for whole exome sequencing (negative control).

Surgically resected tumors, including lymph nodes, must be collected on the day of surgery, processed within 72 hours at the designated head of the site pathology department, sectioned, and then sent to BIPR for pathologic review.

6.2.5. Vital Signs, Physical Examination, and ECOG Performance Status Score

Vital signs include body temperature, blood pressure, pulse, respiratory rate, resting oxygen saturation, and weight. Vital signs will be collected on the day of dosing, within 30 minutes prior to the start of infusion and 60 minutes after the end. Vital signs will be collected additionally at any time if relevant uncomfortable symptoms occur.

Physical examination: Full physical examination will be performed at screening and at the end-of-treatment visit, and will evaluate the head, eye, ear, nose, throat, neck, heart, chest (including lung), abdomen, extremities, skin, lymph nodes, nervous system, and general condition. A targeted physical examination will be performed at other visits by the investigator.

ECOG score: See Annex I.

6.3. Withdrawal From the Study or Termination of Study Treatment

Subjects may withdraw from the study at any time or be withdrawn from the study by the investigator or the Sponsor for any of the following reasons: subject safety, inability to comply with the study, or study termination by the investigator or Sponsor.

6.3.1. Criteria for Withdrawal from the Study

Reasons for subjects to withdraw from the study may include:

1. Withdrawal of informed consent for participation in the study.
2. Subjects with any clinical AE, abnormal laboratory results, or comorbidities for whom investigators consider continuous participation in the study as not meeting their best interest.
3. Other conditions that investigators consider as necessary to withdraw the subject from the study, such as the loss of the ability to freely express their will due to custody or isolation.
4. Loss to follow-up;
5. Death;
6. Request for study termination by the Sponsor, investigators, or regulatory authorities.

6.3.2. Criteria for Termination of Study Treatment

The criteria for termination of study treatment are as follows:

Subject must discontinue study-drug treatment if any of the following occurs:

1. Subjects' voluntary withdrawal from the study or withdrawal of informed consent to participate in the study.
2. Completion of all planned protocol-specified study treatments.
3. Progression of disease that precludes surgery.
4. Post-operative disease progression.

5. Local or distant recurrent metastasis.
6. Pregnancy.
7. Any clinical AEs, abnormal laboratory results, or other medical conditions that will preclude the benefit of continued treatment of the subject.
8. General deterioration of health status, leading to inability to continue participating in the study.
9. Ineligibility, non-compliance or significant protocol deviations after enrollment.
10. Loss to follow-up.
11. Death.
12. Inability to continue treatment in the opinion of the investigator for other reasons (e.g., intercurrent illness).
13. Request for study termination by the Sponsor, investigators, or regulatory authorities.

6.3.3. Procedures to be Followed when Subjects Withdraw from the Study or Permanently Discontinue Study Treatment

If a subject fails to return for scheduled visits, investigators should make every effort to contact the subject. In any case, the final state of a subject should be recorded to the greatest extent possible. The investigator should inquire the reasons for withdrawal, and the subjects will be required to attend a final visit. Every effort must be made to complete the efficacy and safety evaluations specified in the protocol at the time of withdrawal from the study, complete safety follow-up visits, and record AEs and outcomes fully. Investigators can advise or provide new or alternative therapeutic methods to subjects according to their actual condition.

Upon permanent termination of the study treatment, subjects will be monitored for survival in accordance with the protocol until death, loss to follow-up, withdrawal of informed consent to remain on study, or termination of the study by the Sponsor.

If a subject refuses further visits, his/her survival status should continue to be monitored until the subject withdraws consent to disclose further information or be contacted. In this case, no further study evaluations or additional data collection should be performed. If a subject withdraws informed consent to remain in the study, the Sponsor can retain and analyze all data obtained prior to withdrawal from the study.

6.4. Premature Termination or Suspension of the Study

The study may be terminated prematurely or suspended for various reasons. This may be the decision of a regulatory agency, change in the IRB/EC's opinion, efficacy or safety issues of the study drug, or according to the Sponsor's judgment. The Sponsor reserves the right to stop the research and development of toripalimab at any time for other reasons. The party who makes the decision to terminate or suspend the study will send a written notice, record the reason for study termination or suspension, and communicate them to investigators, the Sponsor, and regulatory authorities.

The reasons for premature termination or interruption of the study may include:

- Unexpected, significant, or unacceptable risks to the subjects.
- Study treatment is ineffective.
- Available efficacy results supporting the premature termination of the study.
- Major protocol deviations during the implementation of the study.
- Slow accrual that makes it extremely unlikely to complete the study
- Incomplete or missing data.
- Study results no longer informative.

Once the aforementioned drug safety, protocol compliance, and data quality issues causing study suspension are resolved and agreed to by the Sponsor, ethics committee (EC), and regulatory authority (National Medical Products Administration [NMPA] in China), the study may resume.

6.5. End of Study Visit and Subsequent Treatments

The end of this study is defined as the date of the last subject’s visit (LPLV) or the date of the last data point collection required for statistical analysis, whichever occurs first. The last data point required for statistical analysis will be described in detail in the statistical analysis plan. The Sponsor is entitled to terminate the study at any time due to specific reasons (e.g., major safety issues, force majeure, etc.). The duration of this study is estimated to be about 53 months from the randomization of the first subject to the end of this study.

The Sponsor reserves the right to discontinue the provision of the study drug in any of the following circumstances: a) the health authorities reject the application for marketing; b) the study is terminated due to safety issues; c) subjects are able to obtain study drugs through government or privately sponsored medical plans; or d) alternative drugs are available on the local market.

7. Study Evaluations

7.1. Efficacy Evaluation

MPR rate is defined as the percentage of subjects with a viable tumor involving ≤10% of the tumor bed. All surgically resected tumors, including tumor tissue margin and lymph node samples, must be processed within 72 hours. They will be subsequently submitted to the BIPR for pathological analysis. Regional lymph nodes, including mediastinal lymph nodes, hilar lymph nodes, and intrapulmonary lymph nodes, should be resected if feasible. For right-sided tumors, mediastinal lymph node sampling should include groups 4, 7, and 9; for left-sided tumors, mediastinal lymph node sampling should include groups 5/6, 7, and 9. The sampling of hilar lymph nodes should include groups 10 and 11. The surgeon at the study site may follow his/her standard practice for mediastinal lymph node resection; however, group 7 mediastinal lymph nodes must be resected. The entire tumor specimen should be inspected macroscopically. For pathologic evaluation, the tumor specimen should be sectioned at a thickness of 0.5cm across the longest diameter to acquire the complete cross-section. Carcinoma in situ should not be included in the estimation of residual active tumor.

Event-free survival (EFS), as assessed by the IRC and investigator, is defined as the time from randomization to the first documented disease progression under RECIST 1.1 criteria resulting in inoperability, local progression resulting in inoperability and inability to undergo tumor resection, and postoperative disease progression, local recurrence, or distant metastasis or death for any reason under the RECIST 1.1 criteria, whichever occurs first. Disease progression is considered to have occurred at the first postoperative radiological evaluation (baseline assessment at 30 days after surgery) when apparent new lesions are identified in comparison with the preoperative radiographic evaluation. EFS will be evaluated separately by the investigator at each study site and by the IRC.

Pathologic complete response (pCR) rate, as assessed by the BIPR and investigator, is defined as the percentage of subjects with no residual tumor in the lungs and lymph nodes. The investigator at each study site and the BIPR will evaluate the pCR using the same method as the one for MPR.

Disease-free survival (DFS), as assessed by the IRC and the investigator, is defined as the time from the date of surgery until the first documented local or distant disease, or death of any cause, whichever comes first.

Survival (OS) follow-up: Subjects will be monitored for survival every 3 months after the last dose, and information regarding survival and subsequent anti-tumor treatments will be obtained through telephone follow-up sessions or other means until death, loss to follow-up, withdrawal of informed consent, or termination of the study by the Sponsor.

7.2. Safety Evaluation

7.2.1. Safety Parameters

The safety parameters encompass the following:

- The incidence and severity of study treatment (any study drug)-related AEs, including serious AEs, abnormal laboratory examination, unexpected safety events, and AEs of special interest.
- The incidence and severity of study drug (toripalimab/placebo)-related AEs, including serious AEs, abnormal laboratory examination, unexpected safety events, and AEs of special interest.
- AEs related to surgery, including serious AEs, abnormal laboratory results, and parameters related to surgical feasibility and difficulty such as preoperative state, intraoperative findings, and postoperative recovery.

7.2.2. Safety Evaluation Criteria

The evaluation of an AE includes type, incidence, severity (in accordance with NCI CTCAE version 5.0), start and end date, classification as special interest, immune-related, or surgical complication, correlation with the study treatment, and impact on the outcome.

7.3. Pharmacokinetic Evaluation

Toripalimab concentrations will be measured and tested in all toripalimab-treated subjects using a validated method at a central laboratory.

7.4. Immunogenicity Evaluation

The presence of anti-drug antibodies (ADAs) will be detected using a validated method at a central laboratory. If the sample is ADA-positive, the ADA titer will be determined and the sample will be assessed for the presence of neutralizing antibodies.

7.5. Biomarker Evaluation

The exploratory objective of this study is to explore the correlation between potential biomarkers (including PD-L1, TMB, WES expression in tumor tissue, and changes in ctDNA in peripheral blood samples) and disease status, resistance mechanisms, and response to toripalimab. It includes tumor tissue biomarker detection and peripheral blood biomarker detection.

PD-L1 tumor expression is a stratification factor for randomization and, therefore, a mandatory test in this study.

On the other hand, the detection of the changes in ctDNA and TMB/WES in tumor tissue is optional since these measurements are affected by the number of tissue sections.

7.5.1. Biomarker Evaluation Variable

The specimen will be analyzed using immunohistochemistry to determine the expression status of PD-L1. In addition, some immunohistochemical analyses may also be completed to determine the abundance of other protein markers associated with the progression of NSCLC disease.

The tumor tissue specimen will be used for the detection of TMB and WES and to explore potential prognostic factors (optional).

Ten (10) mL of peripheral venous blood will be collected at designated time points and analyzed at a central laboratory for optional biomarker tests. Analyses will assess

changes in plasma ctDNA assay and WES in circulating peripheral blood mononuclear cells. The control for WES is the analysis of the pre-treatment tumor tissue specimen.

7.5.2. Biomarker Evaluation Method

An FFPE tumor tissue block or unstained sections of tumor specimen from an archived tumor specimen obtained within 3 months prior to the first dose of the study drug or from a fresh biopsy obtained during screening) will be used for biomarker evaluation.

Tumor tissue testing for biomarkers will be carried out in the central laboratory determined by the Sponsor. Tumor tissues will be tested for PD-L1 status and optional TMB and/or WES determination.

Subjects may choose to have a peripheral blood sample collected for ctDNA testing. Subjects for WES testing will also have 2ml of peripheral blood collected as a negative control.

The assay methodology for the collection, processing, shipping and handling of biomarker specimens is fully described in the Laboratory Manual for Special Testing. Detailed flow charts are shown in the **Laboratory Manual**.

8. Adverse Events

All AEs/SAEs occurring during the study period must be recorded on the AE/SAE page of the eCRF, regardless of their relationship to the study drug.

8.1. Definition

8.1.1. Adverse Event Definition

An AE is defined as an undesirable medical event that occurs in the subject who receives a drug in a clinical study, and it does not necessarily have a causal relationship with the treatment. AEs can be any unfavorable and unexpected symptoms, signs, abnormal laboratory results, or diseases, such as the following conditions:

1) Aggravation of original (i.e., prior to enrollment of clinical study) medical condition/disease (including the exacerbation of symptoms, signs, and abnormal laboratory results);

2) Any new AEs: Any undesirable medical condition newly occurred (including symptom, sign, newly diagnosed disease);

3) Clinically significant abnormal laboratory test value or result that is not caused by concomitant disease.

The study staff should record any adverse events occurring in the subjects in detail, including: adverse events and descriptions of all relevant symptoms, time of onset, severity, reasons for adverse events, correlations with the investigational product, duration, actions taken, and final outcome.

The names of adverse events should be recorded as the names of diagnoses or diseases in the CRF. If the name of the adverse event can not be determined or the investigator determines that the diagnostic name or disease name can not be used instead, the clinical condition or symptom will be recorded on CRF as the name of the adverse event.

8.1.2. Serious Adverse Event Definition

An SAE is defined as an undesirable event that meets any of the following criteria:

- Is fatal (i.e., the adverse event actually causes or leads to death)
- Is life-threatening (i.e., the adverse event, in the view of the investigator, places subjects at immediate risk of death)
- Requiring hospitalization or prolongation of existing hospitalization, except as noted below
- Leading to permanent or significant disability/incapability (i.e., the adverse event causing substantial damage to the subject's normal activities)
- Congenital deformity or birth defect
- **Other important medical events** (defined as events that, although not immediately life-threatening, resulting in death or hospitalization, may jeopardize the subject or require intervention [e.g., medication or surgery] to prevent serious consequences listed in the definition above based on appropriate medical and scientific judgment). Examples of such events include, but are not

limited to, allergic bronchospasm requiring intensive treatment in an emergency room or at home, hematologic cachexia or convulsions not requiring hospitalization, underlying drug-induced liver injury, suspected transmission of a pathogen (e.g., pathogenic or nonpathogenic pathogen) via study drug, pregnancy, secondary neoplasm, etc.

Adverse events resulting in hospitalization or prolonged hospitalization in a clinical study should be considered SAEs. Hospitalization does not include the following: rehabilitation facility, nursing home routine emergency room admission, or same-day surgery (e.g., outpatient/same-day/ ambulatory surgery).

Hospitalization or prolongation of hospitalization not associated with worsening of an AE is not an SAE.

For example:

- Hospitalization for the treatment of a preexisting condition without the development of new AEs or worsening of the preexisting conditions (e.g., for work-up of persistent pre-treatment laboratory test abnormality).
- Administrative admission (e.g., for annual physical examination).
- Hospitalization specified in the protocol during the clinical trial (e.g., according to the protocol).
- Optional admission not associated with the aggravation of clinical adverse events (e.g., elective surgery).
- The scheduled treatment or surgery shall be recorded in the whole protocol and/or the subject's baseline data.
- Hospitalization to receive blood products.
- Hospitalization for medical reimbursement.
- Hospitalization or prolonged hospitalization solely due to disease progression of the primary tumor is not considered an SAE.

Invasive (e.g., surgery) and non-invasive procedures for the diagnosis and treatment of a disease or medical condition should not be reported as SAEs; however, the

underlying disease, event or condition leading to these procedures should be reported as an SAE. For example, acute appendicitis occurring during the SAE reporting period should be reported as an SAE and the procedure (appendectomy) should be recorded as the treatment for that AE.

8.1.3. Unexpected Safety Event Definition

In order to determine the requirements for reporting a single AE case, the Sponsor will use the following documents as references to evaluate the expectedness of these events.

- Toripalimab Investigator's Brochure
- Post-marketing package insert for pemetrexed, paclitaxel, docetaxel, carboplatin, and cisplatin

If the nature, severity, specificity, and outcome of the event are inconsistent with those described in the IB and/or post-marketing package insert, the event is an unexpected event.

8.1.4. Adverse Events of Special Interest (AESI)

Adverse events of special interest in this study include:

- Suspected immune-related myocarditis: Elevated myocardial enzymes, with change in ECG or clinical symptom.
- Events in accordance with Hy's law, including ALT or AST >3× baseline during treatment, accompanied by total bilirubin >2× ULN (where the direct bilirubin ≥35%); or ALT or AST >3× baseline during treatment, accompanied by clinical jaundice; and no cholestasis or hyperbilirubinemia due to other reasons.

8.1.5. Immune-related Adverse Events

The mechanism of action of toripalimab will lead to the activation and proliferation of T cells, and may cause immune-related AEs during the conduction of the clinical study.

The optimal strategy and experience for monitoring of immune-related toxicity are accumulating; therefore, the signs and symptoms of irAE should be monitored carefully in

the subject receiving toripalimab. In case of no other alternative etiology (e.g., infection or progression of disease), the signs or symptoms of all the inflammation-related events, such as myocarditis, enterocolitis, pneumonia, dermatitis, hepatitis and endocrine disease, need to be considered for the possibility that they are immune related.

8.2. Reporting

8.2.1. Adverse Events

Following informed consent and prior to the start of study medication, only SAEs caused by mandatory protocol-specified interventions (e.g., interventional procedure (e.g., tissue biopsy), discontinuation of study medication) will be reported.

All AEs should be recorded from the first dose to 30 days after the last dose or the start of a new anti-tumor therapy, whichever comes first. Any AE associated with the study drug should be followed from the start to the outcome. All SAEs will be recorded from the first dose until 90 days after the last dose or the start of a new anti-tumor therapy, whichever occurs first. Only the SAEs caused by mandatory protocol-specified interventions (e.g., interventional procedure, biopsy) need to be recorded during the screening period.

At each visit, investigators must inquire whether the subject had an AE since the last visit. All AEs (including SAEs and AESIs) should be followed by investigators until they fully resolve, return to baseline level or Grade ≤ 1, the AE stabilizes (e.g., stroke with residual permanent deficit), an alternative explanation is identified for the AE, the subject is lost to follow-up, or the subject dies. The follow-up information should be provided in a timely manner and as specified by the protocol.

8.2.2. Serious Adverse Events

SAE information should be collected from the time the subject signs the informed consent form until 90 days after the last dose or before starting a new anti-tumor therapy. During the safety follow-up period, SAEs occurring in subjects who start taking other anticancer drugs are to be reported. SAEs occurring more than 90 days after the last dose do not need to be reported unless they are suspected to be related to the study drug. Deaths

for any reason within 90 days after the last dose should be reported as SAEs. If other criteria for an SAE are not met, the second malignancy should be captured as a “medically important event”).

Except for serious adverse events that are not required to be reported immediately as specified in the protocol or other documents (e.g., investigator's brochure), investigators should immediately report all serious adverse events to the Sponsor in writing and then provide a detailed and written follow-up report in a timely manner. The identification code of the subject in the clinical trial should be indicated in the SAE report and follow-up report rather than the subject's real name, citizenship number, address, and other personal identifying information. Adverse events and laboratory abnormalities specified in the protocol that are important for safety evaluation should be reported to the Sponsor according to the requirements and timelines of the protocol.

For fatal events, the investigator should provide the Sponsor and the ethics committee with other information, such as autopsy reports and medical discharge reports.

The investigator shall promptly sign and read the relevant clinical trial safety information provided by the Sponsor, consider whether the subject’s treatment should be adjusted accordingly, communicate with the subject as soon as possible if necessary, and report the suspected and unexpected serious adverse reactions provided by the Sponsor to the Ethics Committee.

The detailed documentation of SAE should include symptoms, severity, the correlation with the investigational product, time of onset, management time, measures taken, follow-up duration and method, and outcome. If the investigator determines that a serious adverse event is not related to investigational therapy but potentially related to the study condition (e.g., termination of the original treatment or comorbidity during the trial), the relationship should be described in detail on the narrative page of the SAE report form.

If there is a change in the severity of an ongoing serious AE or its relationship with the study drug, a follow-up report should be submitted immediately.

If the investigator considers that a previously reported SAE is misreported, then correction, withdrawal, or de-escalation instructions may be given in the follow-up report and reported according to the SAE reporting procedure.

Any AE/SAE/AESI should be followed up until the end of the event, such as resolution of the event, stabilization of symptoms and signs, resolution to baseline or grade ≤ 1, loss to follow-up, and death.

8.2.3. Adverse Events of Special Interest

AEs of special interest specified in the clinical study protocol, regardless of their seriousness, must be reported by investigators to the Sponsor immediately (i.e., within 24 hours of becoming aware of the event).

8.2.4. Laboratory Test Abnormality

Blood/urine/other samples for laboratory tests will be collected according to the study flow chart and tested at the central laboratory.

The following abnormal laboratory test results should be recorded on the non-serious AE page of the corresponding CRF or SAE report form (written report form or electronic report form):

- All laboratory findings that are clinically significant or meet SAE definitions
- All abnormal laboratory test results that require the subject to discontinue or interrupt the study drug
- All abnormal laboratory test results that require the subject to receive specific symptomatic treatment.

Wherever possible, the investigator should use the clinical rather than the laboratory term (e.g., anemia rather than low hemoglobin value) for reporting.

Note that CTCAE grade 4 laboratory abnormalities alone should not be considered life-threatening unless the associated clinical signs/symptoms or diagnoses meet the criteria for a life-threatening SAE.

8.2.5. Pregnancy

If a female subject is pregnant during the clinical study, she will be withdrawn from the group and the investigator should report it to the Sponsor within 24 hours of becoming aware of the pregnancy. the Sponsor's Clinical Trial Pregnancy Report/Follow-Up Form must also be filled out.

If the male subject's partner becomes pregnant during the clinical study, the subject may continue to participate in the clinical trial and the investigator should report it to the Sponsor within 24 hours of becoming aware of the pregnancy. The Junshi Clinical Trial Pregnancy Report / Follow-Up Form should also be filled out.

The investigator should follow up on the pregnancy until 1 month after delivery, and details should be reported to the Sponsor.

If the outcome of pregnancy is stillbirth, spontaneous abortion, or fetal malformations, it is considered an SAE and needs to be reported per the time limit requirements.

If a subject concurrently has an SAE during pregnancy, the NMPA report form for SAE also needs to be filled in and the SAE must be reported according to the SAE reporting procedure.

8.2.6. Death

All deaths occurring during the study treatment or safety follow-up periods of this protocol must be reported as follows:

When death is not due to (or not clearly due to) disease progression, the AE causing the death must be reported as an SAE within 24 hours. The description of the death should be recorded in the eCRF Death form. The SAE report should include an assessment of the contribution, if any, of disease progression to the subject's death and indicate the primary and secondary causes of death.

Unexplained deaths should always be reported as SAEs. The description of the death should be recorded in the eCRF Death form. An autopsy may be helpful in assessing the

cause of death; if an autopsy was performed, a copy of the results should be sent to the Sponsor.

Deaths occurring after the last protocol-specified safety visit should be recorded on the eCRF Death page. If the death is attributed to delayed toxicity of the study drug, the death should also be reported as an SAE.

8.2.7. Second Malignancy

A second malignancy (i.e., not the primary NSCLC for which the patient was enrolled in this protocol) identified after enrollment in the study should be reported as an SAE. If other criteria for an SAE are not met, the second malignancy should be captured as a “medically important event”.

8.2.8. Overdose

Overdose refers to accidental or intentional dosing at an excessive and medically meaningful dose.

Overdose and medication errors are not considered AEs, but AEs may be caused by an overdose or medication errors.

Overdoses and dosing errors must be recorded in the eCRF dosing module.

All AEs related to overdose or medication errors should be documented in the eCRF. Overdoses and medication errors that result in SAEs should be reported to the Sponsor in accordance with SAE reporting requirements.

8.3. Criteria for Grading Severity of Adverse Events

Refer to NCI CTCAE version 5.0 for the classification of the severity of adverse drug reactions.

For AEs that are not listed in the table from NCI CTCAE version 5.0, refer to the criteria in the following table:

Table 7: Criteria for Grading the Severity of AEs Not Listed in NCI CTCAE Version 5.0

Grade	Criteria

1	Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated.
2	Moderate; minimal, local or noninvasive intervention indicated; limiting age-appropriate instrumental ADL*.
3	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL**. Self-care refers to bathing, dressing, undressing, eating, using the toilet, and taking medications rather than being bedridden
4	Life-threatening consequences; urgent intervention indicated.
5	Death related to AE

8.4. Criteria for Determining the Correlation Between AEs and Investigational Product

The investigators should determine whether the AE has a certain relationship with the study drug and make corresponding “related” or “unrelated” judgments based on their understanding of subjects and the pre- and post-event conditions and evaluations of any potential reason. Assessment should be performed in accordance with the following guidelines:

- Temporal relationship between the occurrence of the event and the start of the study drug
- The course of events, in particular, the impact of dose reduction, discontinuation of study drug, or rechallenge of study drug (if applicable), should be considered
- The events are known to be related to the study drug or similar therapy
- The events are known to be related to the study diseases
- Presence of risk factors or use of concomitant medications known to increase the incidence of the event
- Presence of non-therapeutic related factors known to be associated with the occurrence of the event

8.5. Recommended Treatment for Adverse Events

8.5.1. Recommended Management of Adverse Events from the Study Drug

The AEs induced by immuno-oncology (I-O) drugs vary from the AEs seen with other anti-tumor drugs. Early identification and management of the immune-related AEs (irAEs) is required to reduce the incidence of serious toxicities. Toripalimab belongs to this class of drugs, so it is necessary to early identify and deal with the adverse events caused by toripalimab and reduce the incidence of serious toxicities. For specific measures, please refer to Section 5.6 Management of Immune-related Adverse Events (irAEs) and Other Special Adverse Events in the Investigator's Brochure.

8.5.2. Infusion-related Reaction

As toripalimab is a fully-humanized monoclonal antibody, the possibility of infusion reaction is low, and no preventive medication is needed prior to infusion. The pemetrexed injection may lead to infusion reactions and needs preventive medication according to the latest package insert. Once an infusion reaction occurs, the infusion should be slowed down or interrupted based on the condition while clinical supportive treatment is given, and preventive medication should be given prior to the subsequent doses. Symptoms and signs of acute infusion reaction (including cytokine release syndrome, angioedema, allergic shock, and allergic reaction, refer to NCI CTCAE version 5.0 for the terms and criteria) usually appear during or not long after drug infusion and disappear within 24 hours after completion of infusion. The symptoms and signs include allergic reaction/hypersensitivity (including drug-induced fever), cough, chills, shivering/trembling, dizziness, headache, fatigue (weakness, somnolence), rash/desquamation, pruritus/itching, joint pain, muscle pain, hypo- or hypertension, nausea, vomiting, sweating (perspiration), tachycardia, tumor pain, urticaria (Rubella), dyspnea (shortness of breath) or bronchospasm. All the Grade 3 or 4 infusion reactions should be reported to the Sponsor within 24 hours after awareness of the event, and reported as SAE in the case that they meet the SAE criteria.

Response to allergic reactions should be based on the medical practice and guidelines of the research site. The following are the recommendations for the treatment of infusion reactions for reference.

For the management of delayed hypersensitivity (e.g., pruritus occurred one week after the end of infusion), symptomatic treatment should be given (e.g., oral antihistamine or

corticosteroids). The specific recommendations for the treatment of infusion reactions can be seen in the table below.

Table 8: Recommendations for the Treatment of Infusion Reactions

CTCAE Grade	Clinical symptoms	Clinical management	Action to be Taken for Toripalimab/Placebo
1 (Mild)	Mild transient reaction;	Observation and monitoring until recovery. Prophylactic medication for infusion-related reactions is recommended prior to administration of the next dose, e.g., diphenhydramine 50 mg or equivalent and/or acetaminophen 325-1000 mg, administered at least 30 min prior to infusion	Continue infusion
2 (Moderate)	Moderate reaction requiring treatment or dose interruption that can be rapidly relieved after symptomatic treatment (e.g., antihistamines, nonsteroidal anti-inflammatory drugs, anesthetics, bronchodilators, or intravenous fluid support)	Normal saline IV infusion, diphenhydramine 50 mg IV or equivalent and/or acetaminophen 325-1000 mg; Observation and monitoring until recovery. Corticosteroids or bronchodilators can be considered if clinically required. Prophylactic medication is recommended prior to the next infusion: Pre-infusion prophylaxis is recommended: Diphenhydramine 50 mg, or equivalent and/or acetaminophen 325-1000 mg, administered at least 30 min prior to administration of toripalimab Cortisol (equivalent to 25 mg hydrocortisone), if necessary.	Suspend infusion. After the resolution of symptoms, resume at 50% of the initial infusion rate. If there is no complication within 30 minutes, the infusion rate can be increased to the original infusion rate. Observe closely. If the symptoms recur, terminate the infusion.

≥3(Severe or Life-Threatening)	Grade 3: Serious reaction, no rapid relief after treatment and/or dose interruption; or recurrence after remission; sequela occurred requiring hospitalization. Grade 4: Life-threatening	Infusion of toripalimab must be discontinued immediately. Start normal saline IV infusion. Bronchodilator, 1:1000 epinephrine solution 0.2-1 mg subcutaneously, or 0.1-0.25 mg 1:10000 epinephrine solution slowly IV, and/or diphenhydramine 50 mg plus methylprednisolone 100 mg or equivalent intravenously, when necessary, are recommended. Comply with the study site's guidelines for allergic reactions. Observation and monitoring until recovery.	Permanently discontinue toripalimab.

9. Independent Radiological and Independent Central Pathological Evaluations

Radiological tumor imaging will be evaluated by radiologists at each study site and by the IRC for this study.

Radiologists at each study site will determine tumor response status according to RECIST version 1.1.

The IRC assessment will be performed by two radiologists who are blinded to treatment assignment. If the results are not concordant between these two experts, the definitive assessment will be made by a third radiologist. See IRC Charter for the details.

The evaluation of pathologic outcomes (MPR, pCR) will be conducted by pathologists at each study site and by the BIPR for this study.

The pathologist will evaluate according to the pathological assessment criteria. The central BIPR will be performed by two pathologists who are blinded to treatment assignment. If the pathologic results are not concordant between these two pathologists, the definitive assessment will be made by a third pathologist. See the BIPR manual (BIPR Charter) for the details.

10. Data Analysis/Statistical Methods

The statistical analysis will be performed strictly according to ICH E9 and the Biostatical Guideline for Clinical Trials issued by NMPA.

A detailed summary of the statistical analysis plan and methods for this study can be found in the statistical analysis plan (SAP). If there are any changes to the study protocol that are judged by the Sponsor or the principal investigator (PI) to have an important effect on the SAP, the SAP will be revised to align with the study protocol. The SAP may also be revised, but if the revised SAP involves major and/or critical factors for the protocol, such as the definition of the endpoint or its analysis, the revision will be reflected in a revised version of the protocol. Tables, listings, and figures (TLF) generated by statistical analysis of study data will be reported in a standardized format.

10.1. Hypothesis Testing/Estimation

In accordance with the primary objectives of this study, the primary efficacy hypotheses (alternative hypotheses) for this study are described below.

H1: In subjects with resectable, stage III NSCLC, EFS evaluated by investigators shows superiority in toripalimab combined with platinum-based doublet chemotherapy (treatment arm) compared to placebo combined with platinum-based doublet chemotherapy (control arm).

H2: In subjects with resectable, stage III NSCLC, the MPR rate evaluated by BIPR shows superiority in neoadjuvant toripalimab combined with platinum-based doublet chemotherapy (treatment arm) compared to placebo combined with platinum-based doublet chemotherapy (control arm).

H3: In subjects with resectable, stage II-III NSCLC, EFS evaluated by investigators shows superiority in toripalimab combined with platinum-based doublet chemotherapy (treatment arm) compared to placebo combined with platinum-based doublet chemotherapy (control arm).

H4: In subjects with resectable, stage II-III NSCLC, the MPR rate evaluated by BIPR shows superiority in neoadjuvant toripalimab combined with platinum-based doublet

chemotherapy (treatment arm) compared to placebo combined with platinum-based doublet chemotherapy (control arm).

See 10.5 Interim Analysis for a detailed explanation regarding the definition of interim analysis, and see 10.6 Multiple Comparisons for details regarding the strategies for multiple comparisons.

10.2. Study Endpoints

10.2.1. Efficacy Endpoint

See 7.1 Efficacy Evaluation for the definitions of the primary study endpoint of BIPR-evaluated MPR rate and investigator-evaluated EFS, the secondary study endpoint of BIPR-evaluated and investigator-evaluated pCR, IRC-evaluated EFS, IRC-evaluated and investigator-evaluated DFS.

OS is defined as the time from randomization to death from any cause. Subjects will be censored at the time of their last contact (last known alive date) if they do not die during the study.

10.2.2. Safety Endpoints

The measurements and definitions of safety endpoints are detailed in 8 Adverse Events.

10.3. Analysis Set

The intent-to-treat (ITT) population will be used as the primary analysis population for efficacy analysis in this study and includes all randomized subjects. The subjects in ITT will be analyzed according to the treatment group randomly allocated in the study.

The per-protocol set (PPS) is a subset of the ITT population that includes all randomized subjects without major protocol deviations. Prior to the study database lock, the subjects included in PPS will be determined based on protocol deviation information, recorded, and stored in one separate document. The PPS will be used for sensitivity analysis of primary efficacy endpoints.

The safety analysis set (SS) includes all the subjects who have received any dose of the study drug (including toripalimab or chemotherapy). The subjects in SS will be analyzed

according to the treatment they actually receive. The SS will be used for the analysis of safety and biomarkers.

The pharmacokinetic analysis set (PKS) includes all subjects who received at least one dose of toripalimab and for whom there is PK data. The PKS will be used for all pharmacokinetic analyses.

10.4. Statistical Analysis Method

In general, the primary efficacy variables will be evaluated through statistical testing and statistical inference, and the nominal P value will be calculated to show the correlation between treatment and efficacy for other efficacy variables without control of multiple comparisons. The descriptive statistics by method include number, mean, standard deviation, median, minimum, and maximum for measurement data. For enumeration data, the descriptive statistics will consist of the number, case-time, and percentage. The Kaplan-Meier method will be used to estimate the median survival for time-event data, and a survival curve will be plotted. When necessary, the overall 95% confidential interval of median time should also be estimated. Unless otherwise noted, the significance level is two-sided at 0.05.

The SAS 9.4 software package will be used to process and analyze the statistical analysis content designed in this study.

10.4.1. Statistical Method for Efficacy Analysis

10.4.1.1. MPR Rate Evaluated by BIPR

Using the ITT population as the population for primary analysis and the PPS set as the population for sensitivity analysis, a comparison of the BIPR-evaluated MPR rate will be performed. The stratified Cochran–Mantel–Haenszel (CMH) test will be used for between-arm comparisons, and the 95% confidence interval of the difference in BIPR-evaluated MPR rates between arms will be calculated. The stratification factors in the CMH method are the same as those used for randomization.

10.4.1.2. EFS Evaluated by Investigators

The investigator-evaluated EFS will be analyzed using the ITT population as the population for primary analysis and the PPS population as the population for sensitivity

analysis. A stratified log-rank test will be used for the analysis of the intergroup difference in EFS, using the same stratification factors as those for randomization. The Kaplan-Meier method will be used to estimate the EFS curve and the median EFS in each study group. The 95% confidential interval of median EFS will be estimated through the Brookmeyer-Crowley method using log-log function conversion to reach normal approximation. The hazard ratio (HR) of EFS and its 95% confidential interval will be estimated using the stratified Cox proportional hazard model, using the same stratification factors as those for randomization.

Subjects who do not experience an EFS event will be censored for EFS on the date of their last valid tumor evaluation. Subjects who do not undergo a post-operative tumor evaluation will be censored for EFS on the date of their surgery. Subjects who do not experience an EFS who but start alternative anticancer therapy will be censored for EFS on the date of the last radiologic tumor evaluation prior to the start of subsequent anticancer therapy. Details regarding the censoring rules will be provided in a separate statistical analysis plan.

10.4.1.3. pCR Rate Evaluated by BIPR and Study Site Investigator

The pCR rate will be analyzed in the ITT population, using the same statistical analysis method as the one used for the MPR rate.

10.4.1.4. Disease-free Survival Evaluated by IRC and Investigators

DFS will be analyzed in the ITT population, using the same statistical analysis method as the one used for EFS.

Subjects who do not undergo surgical resection will not be excluded from the DFS analysis. Subjects who do not experience a DFS event will be censored for DFS on the date of the last radiological tumor imaging evaluation. Subjects who do not experience a DFS event but start alternative anticancer therapy will be censored for DFS on the date of the last radiological tumor imaging evaluation prior to the start of alternative anticancer therapy.

10.4.1.5. Overall Survival (OS)

OS will be analyzed in the ITT population. A stratified log-rank test will be adopted for analyzing the difference in OS between arms, using the same stratification factors as those used for randomization factors. The Kaplan-Meier method will be used to estimate the survival probability rates at 2 and 3 years in each arm. the 95% confidence intervals for 2- and 3-year survival rates will be estimated using the Greenwood formula. The 95% confidence interval for the difference in OS between arms will be estimated using the normal approximation method. The analysis methods for the OS HR and its 95% confidence interval will be the same as those used for EFS analysis. Subjects who are still alive at the time of the analysis will be censored for OS on the latest known survival date.

10.4.2. Safety Analysis

All safety data will be summarized based on the SS.

Medical Dictionary for Regulatory Activities (MedDRA) Version 22.0 or later versions will be used to code AE. For each study treatment group, the number and incidence of events will be tabulated by preferred term and system organ class. Treatment-related or protocol procedure-related AEs, AEs related to radical resection, SAEs, immune-related AEs, AEs leading to death, and AEs leading to a subject's discontinuation of study treatment will be tabulated for each arm. Descriptive summaries will be provided for the most common AEs in each arm, i.e., occurring in ≥5% of subjects. A treatment-emergent adverse event (TEAE) is defined as any new AE occurring from the first dose in the neoadjuvant treatment period up to 90 days after the last dose or the start of a new anti-tumor therapy, whichever comes first. Any pre-existing AE that is exacerbated from baseline is also considered a TEAE. Non-TEAEs will be presented in the form of a list.

Descriptive statistics will be used to summarize all of the laboratory results, physical examination findings, vital sign measurements, weight measurements, ECOG performance status, 12-lead ECG findings by number, and percentages of subjects with abnormal results or changes from baseline in each arm at each visit. Baseline results and changes from baseline will be presented in the form of a list.

Descriptive statistics will be used to summarize surgical feasibility in each study group, including the percentage of delayed or canceled operations, change in surgical approach, time of operation, and original data, all of which will be tabulated.

Descriptive statistics will be used to summarize the intraoperative findings and postoperative recovery in subjects who undergo surgery in each study arm.

10.4.3. Pharmacokinetic Analysis

Descriptive statistical analysis will be performed to calculate toripalimab pharmacokinetics using trough blood concentrations.

Toripalimab blood concentration data may be included in updated population pharmacokinetic analyses to examine potential factors affecting toripalimab exposure, such as chemotherapy regimens administered in this study and the development of anti-drug antibodies (ADA). If population pharmacokinetic analyses are conducted, the analyses will be performed under a separate analysis plan and the results will not be reported in the clinical study report (CSR).

10.4.4. Immunogenicity Evaluation

The incidence of ADA, ADA titers, and the incidence of neutralizing antibodies in subjects with ADA will be analyzed. Additionally, the potential effects of immunogenicity on pharmacokinetics, safety and efficacy will be explored.

The effects of ADA on the pharmacokinetics, safety, and efficacy will be explored in population pharmacokinetic modeling and E-R analyses if required. Population pharmacokinetic and E-R analyses will be conducted under a separate analysis plan and the results will not be reported in the CSR.

10.4.5. Biomarker Analysis

Potential predictive biomarkers will be explored for efficacy (e.g., predicting effects on MPR, EFS, DFS, and OS) in peripheral blood and tumor specimens, and the results will be provided in a tabular summary with descriptive statistics.

10.4.6. Subject Disposition

Subjects who failed screening (those who have signed informed consent but do not receive any treatment) will not be included in any analyses, but the reasons for screening failure will be summarized and reported in the form of a list. Those who have failed previous screening are permitted to be re-screened in this study.

The number and percentage of subjects who complete the study protocol, terminate the study prematurely, and are included in or excluded from each analysis population will be summarized and analyzed.

Reasons for withdrawal from the study will be summarized and tabulated. The contents of the table need to include the date of the first and last dose and the reasons and date of withdrawal.

10.4.7. Demographic and Baseline Characteristics

Descriptive statistics characterizing the ITT population will be provided for demographic characteristics (e.g., sex, age, height, and weight), medical history, smoking history, and alcohol use in the ITT population and by treatment assignment (as randomized).

10.5. Interim Analysis and Analysis Plan

A 70% investigator-evaluated EFS information ratio (approximately 139 investigator-evaluated EFS events) is expected in the Stage III population, and an interim analysis to assess the efficacy of investigator-evaluated EFS in the Stage III population is planned to occur 10 months after the randomization of the last subject in the Stage III population.

In the Stage III population, a final analysis will be performed to assess the efficacy of investigator-evaluated EFS in the Stage III population at the time when 199 investigator-evaluated EFS events are observed (this analysis is descriptive if investigator-evaluated EFS in the Stage III population has reached statistical significance at the interim analysis). This is expected to occur 12 months after the last subject enrollment in the Stage II-III population. The significance level for the interim and final analyses will be calculated based on the actual investigator-evaluated number of EFS events using an O'Brien-Fleming type α-spending function approximated by the Lan-DeMets method.

Approximately 220 investigator-evaluated EFS events are expected to be observed in the Stage II-III population at the final analysis of investigator-evaluated EFS in the Stage III population. Once that occurs, a final analysis assessing the efficacy of investigator-evaluated EFS in the Stage II-III population will be performed, and a final analysis of the

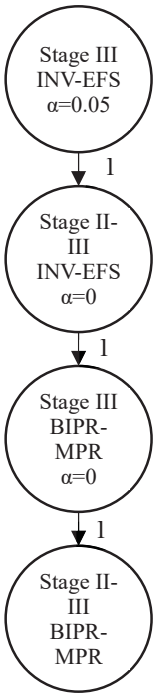
efficacy of BIPR-evaluated MPR in the Stage III and Stage II-III populations will be performed.

The above time point estimates are based on investigator-evaluated rates of EFS occurrence in a blinded fashion as of April 2022, and data cut-off dates for the final analysis may differ between the Stage III and Stage II-III populations, with actual data cut-off dates based on the number of events required and/or completion of necessary clinical data collection.

An independent Data Monitoring Committee (IDMC) and an independent Statistical Supporting Group (ISSG) will be constituted for this study. For the interim analysis of EFS for the Stage III population, which includes treatment allocation (unblinded) information, ISSG will be generated and provided to the IDMC, which will make recommendations to the Sponsor based on the analysis. See the IDMC Charter for detailed information about the IDMC's organization, responsibilities, and data monitoring program.

10.6. Multiple Comparisons

The multiple comparison strategy adopted in this study will use hierarchical testing to statistically test the investigator-evaluated EFS in the Stage III population, investigator-evaluated EFS in the Stage II-III population, BIPR-evaluated MPR in the Stage III population, and BIPR-evaluated MPR rate in the Stage II-III population sequentially at the two-sided significance level of 0.05 to control the overall Type I error α level. The graphical method presents the multiple comparison strategy described above, as shown in the figure below.



α is the significance level initially

Figure2: Diagram of overall type I error control with hierarchical testing

10.7. Sample Size and Power

The sample size is calculated based on investigator-evaluated EFS, and approximately 500 subjects with stage II-III NSCLC meeting enrollment criteria will be randomized to the test and control arms in a 1:1 ratio, with approximately 400 subjects enrolled in the Stage III population.

The final analysis of the investigator-evaluated EFS efficacy assessment for the Stage III population will be performed with the expectation of 199 investigator-evaluated EFS events observed in the Stage III population, resulting in 86% power to detect the superiority of investigator-evaluated EFS compared to the control arm in the Stage III population at a 2-sided significance level of 0.05.

The final analysis of the investigator-evaluated EFS efficacy assessment for the Stage II-III population will be performed with the expectation of 220 investigator-evaluated EFS events observed in the Stage II-III population, resulting in an 89% nominal statistical power to detect the superiority of investigator-evaluated EFS in the Stage II-III population over the

control arm at a 2-sided significance level of 0.05 (after transmission of the significance level).

The above sample size will be calculated based on the following assumptions:

- EFS is exponentially distributed.
- The median investigator-evaluated EFS is 14 months in the Stage III population and 40 months in the Stage II population control arm
- The hazard ratio (HR) between the experimental arm and control arm is 0.65
- Seventy percent (70%) of the investigator-evaluated EFS information ratio is expected in the Stage III population, and an interim analysis assessing the efficacy of investigator-evaluated EFS in the Stage III population will be performed at approximately 139 investigator-evaluated EFS. The interim and final analyses use the α-spending function of the O'Brien-Fleming method (approximated by the Lan-DeMets method) to calculate significance level boundaries.
- All subjects were enrolled within 28 months, and the Stage III population was randomized between April 2020 and January 2022.
- During the first 12 months of investigator-evaluated EFS event observation, the drop-out rate was 20% in each study group.

The MPR rate assessed by BIPR is 20% in the control arm, and 38% in the experimental arm. With 500 stage II-III subjects, the study provides approximately 99% nominal statistical power to detect the superiority of the experimental arm compared to the control arm at a two-sided significance level of 0.05 (after transmission of the significance level); with 400 stage III subjects, the study provides approximately 98% nominal statistical power to detect the superiority of the experimental arm compared to the control arm at a two-sided significance level of 0.05 (after transmission of the significance level).

The sample size will be calculated using R 4.3.5 and the gsDesign statistical software, as well as the EAST 6.5 statistical package.

10.8. Subgroup Analysis

The analysis of investigator-evaluated EFS in the Stage III NSCLC subgroup will be used as the confirmatory analysis for BIPR-MPR rates in the Stage III subgroup. The study endpoints, primary analysis methods, and multiple comparison strategy for Type I error control are described above.

Exploratory subgroup analyses will be described and specified in the SAP.

10.9. Drug Compliance and Exposure

This study will analyze drug use in the study group and control arm based on the SS population, and the analysis will include drug compliance, actual number of treatment cycles, duration of drug exposure, actual cumulative dose, actual dose intensity (ADI) and relative dose intensity (RDI). The detailed definition will be further described and specified in the separate SAP.

11. Data Management

11.1. Data Entry

In this study, subjects' data shall be entered into electronic case report forms (eCRF) and transferred to a Sponsor-designated electronic data system (EDS) to incorporate this clinical data with data from other sources.

The clinical data shall be managed in accordance with applicable Clinical Data Interchange Standards Consortium (CDISC) and data cleaning procedures to ensure the integrity of the data. For example, incorrect and inconsistent data are removed. AEs will be encoded by the Medical Dictionary for Regulatory Affairs (MedDRA), and the concomitant medication name will be encoded by the World Health Drug Dictionary (WHO Drug). The eCRF shall be kept by the Sponsor, and a copy shall be mailed to the investigator as the Investigator's Copy. The staff at the study site are responsible for completing the eCRFs.

For all subjects who signed an ICF and are considered eligible during screening, the investigator or authorized personnel shall record carefully complete all items in the eCRF.

An item without any record shall be filled with UK/NA/ND as appropriate. All data in the eCRF must be verified against original source data/documents. The investigator must attach the original laboratory results or a copy of the results to the subject's investigational medical records.

The investigator shall evaluate all abnormal laboratory tests and indicate whether the result is clinically significant or not clinically significant. The investigator shall fill out the eCRF in strict accordance with the case report form completion guidelines.

11.2. Database Lock

The data may be locked when the following conditions are met:

All data have been entered into the database;

All queries have been resolved;

The criteria for each analysis population have been finalized.

Once locked, data files cannot be changed.

11.3. Data Monitoring and Audit

Authorized representatives (e.g., inspectors, auditors) from the Sponsor must be allowed to visit all study center sites on a regular basis to evaluate the quality, completeness, and reliability of the study data. They will review the study records on-site, compare these records directly with the original documents, discuss the implementation of the study with the investigators, and confirm whether the study facility meets GCP requirements.

The scope of monitoring includes whether the protocol is followed; whether all CRFs are filled out correctly and completely; whether the CRFs are consistent with the original documents, including but not limited to study medical records and laboratory test reports; and whether data is wrong or missing. Monitors need to check the contents of the eCRF against the original document line-by-line to ensure consistency between the data in the eCRF and the original document. This process is also known as source data verification.

In addition, inspectors in government departments may also evaluate the study conduct and data. The staff must be allowed to inspect study facilities and access eCRFs,

original documents, and other study documents. The audit report must be kept confidential. If any government department plans to conduct an inspection, the investigators must immediately notify the Sponsor and promptly forward the inspection report to the Sponsor.

11.4. Data Archiving

After completion of the study, the subject's eCRF will be archived in PDF format, generated directly from the EDC and saved on a CD-ROM. A copy of the CD-ROM will be provided to the Sponsor and each institution for archival storage to facilitate audits and/or inspections by regulatory authorities.

The study data shall be stored and managed in accordance with the requirements of the GCP. The required clinical study documents (trial master file) shall be kept for 2 years after the approval of the investigational product for marketing or 5 years after the termination of the clinical study, whichever is earlier.

11.5. Protocol Deviation

A protocol deviation refers to any practice that does not comply with the clinical study protocol, GCP, SOP, or Medical Operating Practices (MOP) for this study. Failure to comply with any of these documents by the subject, investigator, or other study staff may result in protocol deviations. In the event of a protocol deviation, if Corrective and Preventive Action (CAPA) is required, the study center should prepare the appropriate corrective actions and implement them immediately.

The study site is responsible for maintaining continuous vigilance in order to discover and report protocol deviations in a timely manner. All deviations must be documented in the original study file, reported to the Sponsor, and submitted to the local IRB in accordance with local ethics regulations. The PI or study staff at the study center are responsible for understanding and complying with local ethics requirements. Additional details on the handling of protocol deviations can be found in the MOP.

12. Source Documents and Access to Source Data/File

In accordance with the International Council for Harmonisation (ICH) E6 requirements regarding the protection of personal information, relevant regulations, and study institutions, each study center must properly keep the treatment and scientific research records related to this study. As one part of the studies funded or involved by the Sponsor, each study center should allow the Sponsor's authorized representatives and regulatory health authorities to inspect clinical records for the purpose of quality review and audit and the evaluation of safety, study progress, and validity of the data. This section should also describe who has access to these records.

The original (source) data include all the information necessary to evaluate the clinical study conduct and results, including the original record of clinical findings, observations, or other records. Examples of these original documents and data records include but are not limited to hospital records, clinical and office software, laboratory records, memoranda, subject diaries or assessment lists, pharmacy dispensation records, consultation records, data recorded on automated instruments, verified accurate and complete copies or transcripts, microfilm, photographs, microfilm strip or disk, and X-rays and subject's documents and records stored at the pharmacy, clinical laboratory and imaging departments participating in the study.

The investigators are responsible for ensuring that the source data are accurate, clear, contemporaneous, and traceable to the subject, regardless of whether the data is written by hand or entered electronically. In routine clinical study activities, if the source data is created (first entered), modified, maintained, archived, retrieved, or electronically transmitted through a computer system (and/or any other type of electronic device), such systems must comply with all applicable laws and regulations governing the use of electronic records and/or electronic signatures. Such systems may include but are not limited to electronic medical record/health records (EMR/HER), AE tracking/reporting, evaluations required in the protocol, and/or drug counting records.

When written records obtained through such systems are used in place of the electronic format for specified activities, they should be accompanied by certified copies. A certified copy is a dated and signed duplicate of the verified original information, which contains all essential attributes and information present in the original.

13. Quality Assurance and Quality Control

In order to ensure the quality of the study, the clinical study plan is discussed and established by the Sponsor and investigators together prior to the study's initiation. GCP training is performed for all the relevant study personnel involved. Training topics include but are not limited to the following: GCP, reporting of AEs, study details and procedures, completion of electronic CRFs, completion and retention of study records, informed consent, and inclusion of women of childbearing potential.

Each study site must manage the investigational product in accordance with the Pharmacy Manual, and this includes activities such as the receipt, storage, distribution, drug accountability, and destruction of the IP, as applicable.

In accordance with the GCP guidelines, necessary procedures must be taken in the study design and implementation to ensure the accuracy, consistency, completeness, and credibility of all data collected. All results, including abnormal findings observed in the clinical study, should be recorded promptly and audited to ensure the reliability of the data. Each instrument, device, reagent, and standard used in protocol-specified tests in the clinical study should conform to specifications.

The investigators must enter the required information into the eCRF. The study monitors must ensure that eCRFs are complete and accurate and instruct personnel at the study center for any necessary modifications and supplementations.

Relevant regulatory authorities (e.g., NMPA), Institutional Review Board (IRB)/Independent Ethics Committee (IEC), the Sponsor's monitors and/or auditors may conduct a systematic review of the study-related activities and documents. The audit should aim to assess the study's adherence to the protocol, SOP, and requirements of relevant laws and regulations, and ensure that the study data are promptly, accurately and

completely recorded. The audit should be performed by personnel who are not directly involved in the clinical study.

14. Applicable Laws and Human Subject Protection

14.1. Applicable Laws and Regulations

In accordance with the requirements of relevant health authority regulations, the Sponsor must apply to the State Food and Drug Administration of the People's Republic of China and receive approval for a new drug clinical study prior to initiating the clinical study. The IND/CTA number assigned by the NMPA for toripalimab is 2015L05752.

The relevant legal authorities for the design of this protocol are:

1) Measures for the Administration of Drug Registration

2) Good Clinical Practice in China

4) Ethical principles and consensus based on international ethical guidelines, including the Declaration of Helsinki and the international ethical guidelines established by the Council for International Organizations of Medical Sciences (CIOMS)

5) ICH Guidelines

6) Other applicable laws and regulations

14.2. Ethical Considerations

This study will be conducted in accordance with the Declaration of Helsinki, Good Clinical Practice (GCP) guidelines established by the China Food and Drug Administration (CFDA) and relevant regulations, and Good Clinical Practice (GCP) guidelines developed by the International Conference on Harmonisation (ICH).

The study will be conducted in accordance with the protocol. The approval of and agreements regarding the protocol, all amendments, and the ICF by the IRB/IEC must be obtained prior to the initiation of the study.

The Sponsor and investigator should not amend this study protocol without mutual agreement. Investigators should only amend or deviate from the currently approved version of the study protocol prior to approval by the IRB/IEC to eliminate direct and

immediate injury to subjects. All deviations from the protocol and reasons for deviations must be recorded. The investigator should submit a list of protocol deviations, the reasons for each deviation, as well as a proposed amendment to the protocol to the IRB/EC for review as soon as possible.

During the clinical study, any amendment to the study protocol should be submitted to the IRB/IEC prior to implementation. A corresponding amendment should also be made to other study documents when necessary and be submitted and/or approved as required by the IRB/IEC. Investigators are responsible for the periodic submission of study reports to the IRB/IEC in accordance with relevant IRB/IEC procedures. Additionally, the investigator is responsible for informing the IRB/IEC when the study has ended.

All cases that may seriously violate the specifications must be reported to the Sponsor immediately. Serious violations of the specifications refer to violations of GCP conditions and principles associated with the study or violations of the study protocol, which may have a significant impact on the safety or physical or mental health of the study subjects or the scientific value of the study.

The personnel involved in the implementation of this study must have the appropriate qualifications, including educational background, training, and experience.

14.3. Institutional Review Board/Independent Ethics Committee

Before initiation of the study, the investigators must obtain a signed and dated written approval/agreement from the IRB/IEC for the study protocol, ICFs, subject recruitment materials (e.g., advertising), and all other written materials to be provided to subjects. The investigators or the Sponsor will also submit to the IRB/IEC a copy of the IB, a product package insert, and any updates to be provided to the subject.

The investigators or the Sponsor must submit reports, updates, and other materials (such as expedited safety reports, revisions, and management letters) to the IRB/IEC in accordance with the requirements of the corresponding regulatory health authority or the agency’s procedures.

14.4. Informed consent

14.4.1. Informed Consent Form and Other Written Information Required of Subjects

The Sponsor will provide the investigators with a sample ICF that contains all of the elements required by the ICH, GCP, and regulatory authorities. The sample ICF will be consistent with the ethical principles established in the Declaration of Helsinki.

The sample ICF will describe the investigational product and study procedures in detail and give a sufficient explanation of the study's potential risks to the subjects. Prior to administering the investigational product, a signed and dated written consent on the ICF must be obtained.

All amendments to the ICF must also be approved by the IRB/IEC. The IRB/IEC will decide whether subjects need to sign the new version of the ICF if they have already signed a previous version.

The ICF shall also state that the subject's personal information and identity will be kept confidential but that the Sponsor's medical representative and regulatory health authorities can review the subject information as needed.

14.4.2. Informed Consent

The informed consent process starts before seeking an individual's agreement for participation in the clinical study and continues throughout the study. The risks and possible benefits of participation in the study will be discussed thoroughly with the subjects and their families. Subjects will be asked to read and review the ICF approved by the IEC. Investigators will explain the clinical study to subjects and answer any questions. Subjects will have the opportunity to carefully review the written informed consent and ask questions before signing. Subjects will have the opportunity to discuss the study with their representatives and fully consider it before agreeing to participate in the study. Only after the subject has signed the ICF will they be able to enter the study. Throughout the course of the study, subjects can withdraw informed consent for study participation at any time. The subject will keep one copy of the ICF. Even if subjects refuse to participate in

this study, their rights will be fully protected, with a particular emphasis on ensuring that the quality of their medical care will remain unaffected.

14.5. Confidentiality of Subject Information

The confidentiality of a subject's information will be strictly upheld by investigators, study personnel involved, and the Sponsor and their representatives. Subject confidentiality covers biological samples and genetic test results as well as clinical data. Additionally, the study protocol, related documents, relevant data, and all other information obtained as part of the study will be kept strictly confidential. Without prior written approval by the Sponsor, no study-specific data or other information can be revealed to an unauthorized third party.

The study monitors, other authorized representatives of the Sponsor, the IRB, and representatives of pharmaceutical companies that supply the investigational products can monitor or audit documents and records kept confidential by investigators. This includes but is not limited to the subject's medica and pharmacy/medication records. The study center should have access to these records.

The subject's contact information will be stored securely at each study center and provided to appropriate study personnel for internal use during the study. At the end of the study, all the records will continue to be stored in a secure place in accordance with the time limit specified by the local IRB and applicable regulations.

The data collected for statistical analysis and the scientific report will be uploaded and stored in the EDC (Medidata RAVE) and should not include the subject's contact or identification information. Instead, each subject will be given a unique study identification number that will be associated with all study data for that subject. Study data entry and the study management system used by study personnel at each clinical study center will be kept confidential and password-protected.

After obtaining the consent of the subject and the approval of the local IRB, de-identified biological samples (in this study, this includes venous blood samples and tumor tissue paraffin-embedded blocks or sections sent to the central laboratory for biomarker

testing) will be kept in the central laboratory. Data from analyses of biological samples can be shared only with the Sponsor and study centers. These samples can only be used to detect biomarkers, and the subject's privacy will be protected throughout the process.

Subjects have the right to withdraw their consent for the use of their biological specimens or data at any time. However, if the study is completed, the subject will not be able to withdraw consent to use any data or test results derived from any biological specimen collected prior to the request for withdrawal from the study.

The central laboratory may be accessed for study data and/or samples when the study is completed.

15. Publication of Study Results

Data obtained in this clinical study must be kept strictly confidential. All materials and results are the property of the Sponsor.

After completion of the study or at an earlier time point, if the Sponsor determines that the study data is sufficient for analysis, investigators can prepare the data for publication. Prior to public dissemination, the information compiled will be submitted to the Sponsor for review and comment. To ensure that the Sponsor is able to comment and make relevant recommendations, materials for public dissemination should be submitted to the Sponsor for review at least 60 days prior to submission for publication, public communication, or review by the public release committee.

Investigators must agree that all the reasonable comments and requests for edits from the Sponsor will be incorporated into the article by investigators.

When reviewing the article for publication, the Sponsor has the right to postpone publication in order for the Sponsor to protect their patent information. Any materials related to the study cannot be published without the Sponsor's written approval. Unless mandated by law, neither the Sponsor nor the PI can disclose the study results to any third party before reaching a bilateral agreement on data analysis and interpretation.

16. Clinical Study Milestones

Estimated enrollment date of the first subject: April 2020.

Estimated enrollment date of the last subject: December 2022.

Estimated end date of the study: October 2024.

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Annex 1 Performance Status Score Criterion (ECOG)

Eastern Cooperative Oncology Group (ECOG) Performance Status Score

Score	Description
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all self-care but unable to carry out any work activities; up and about more than 50% of waking hours
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair
5	Dead

References:

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Annex 2 Response Evaluation Criteria in Solid Tumors (RECIST 1.1)

Response Evaluation Criteria for Solid Tumors Version 1.1 (Excerpts)

1 Background

Omit

2 Objective

Omit

3 Measurability of tumor at baseline

3.1 Definition

At baseline, tumor lesions/lymph nodes will be categorized as measurable or non-measurable using the following criteria:

3.1.1 Measurable lesions

Measurable Tumor lesions: Must be accurately measured in at least one dimension (the longest diameter in the plane of measurement is to be recorded) with a minimum size of:

- 10 mm when assessed with a CT scan (slice thickness no greater than 5 mm).
- 10 mm when assessed through caliper measurements during a clinical examination (lesions that cannot be accurately measured with calipers should be recorded as non-measurable).
- 20 mm when assessed with a chest X-ray.
- Malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be 15 mm in short axis when assessed with a CT scan (the recommended slice thickness is no greater than 5 mm). At baseline and in follow-up sessions, only the short axis will be measured and monitored.

3.1.2 Non-measurable

All other lesions, including small lesions (longest diameter <10 mm or pathological lymph nodes with 10 to <15 mm short axis) and non-measurable lesions. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of the skin or lung, abdominal masses/abdominal organomegaly identified during a physical examination that

is not measurable by reproducible imaging techniques.

3.1.3 Special considerations regarding lesion measurability

Bone lesions, cystic lesions, and lesions previously treated with local therapy require particular comment:

Bone lesions:

- Bone scans, PET scans, or plain films are not considered adequate imaging techniques to measure bone lesions. However, these techniques can be used to confirm the presence or disappearance of bone lesions.
- Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, that can be evaluated with cross sectional imaging techniques such as CT or MRI can be considered as measurable lesions if the soft tissue component meets the definition of measurability described above.
- Blastic bone lesions are non-measurable.

Cystic lesions:

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts.
- 'Cystic lesions' thought to represent cystic metastases can be considered as measurable lesions if they meet the definition of measurability described above. However, if non-cystic lesions are present in the same patient, these are preferred for selection as target lesions.

Lesions with prior local treatment:

- Tumor lesions situated in a previously irradiated area or an area subjected to other loco-regional therapy, are usually not considered measurable unless there has been demonstrated progression in the lesion. Study protocols should detail the conditions under which such lesions would be considered measurable.

3.2 Specifications by methods of measurements

3.2.1 Measurement of lesions

All measurements should be recorded in metric notation, using calipers if clinically assessed. All baseline evaluations should be performed as close as possible to the treatment initiation and never more than 28 days (4 weeks) before.

3.2.2 Method of assessment

The same method of assessment and technique should be used to characterize each identified and reported lesion at baseline and during follow-up sessions. Imaging-based evaluation should always be prioritized over clinical examination unless the monitored lesion(s) cannot be imaged but are assessable by clinical examination.

Clinical lesions: Clinical lesions will only be considered measurable when they are superficial and 10 mm in diameter when measured using calipers (e.g., skin nodules). In the case of skin lesions, the recommended method is documentation by color photography, with a ruler to estimate the size of the lesion. As noted above, when lesions can be evaluated through both clinical examination and imaging, imaging evaluation should be chosen since it is more objective and can be reviewed at the end of the study.

Chest X-ray: Chest CT is preferred over chest X-ray, particularly when progression is an important endpoint, since it is more sensitive in identifying new lesions. However, lesions on chest X-rays may be considered measurable if they are clearly defined and surrounded by aerated lung tissue.

CT, MRI: CT is currently the best available and reproducible method to measure lesions selected for response assessment. This guideline has defined the measurability of lesions on CT scans based on the assumption that CT slice thickness is 5 mm or less. When the slice thickness is greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g., for body scans).

Ultrasound: Ultrasound is not useful in assessing lesion size and should not be used as a method of measurement. Ultrasound examinations cannot be reproduced in their entirety for independent review at a later date and, because they are operator-dependent, there is no guarantee that the same technique and measurements will be employed in successive assessments. If new lesions are identified through ultrasound during the study, confirmation

with CT or MRI is advised. If there is concern about CT radiation exposure, MRI may be used instead of CT in select instances.

Endoscopy, laparoscopy: These techniques are not recommended for objective tumor evaluation. However, they can be useful for confirming complete pathological response when biopsies are obtained or determining recurrence in trials where recurrence following complete response or surgical resection is an endpoint.

Tumor markers: Tumor markers alone cannot be used to assess objective tumor response. If markers are initially above the upper normal limit, however, they must normalize for a patient to be considered in complete response. Because tumor markers are disease-specific, instructions for their measurement should be incorporated into protocols on a disease-specific basis. Specific guidelines for both CA-125 response (in recurrent ovarian cancer) and PSA response (in recurrent prostate cancer) have been published. In addition, the Gynecologic Cancer Intergroup has developed CA-125 progression criteria, which are to be integrated with objective tumor assessments for use in first-line ovarian cancer trials.

Cytology, histology: These techniques can be used to differentiate between PR and CR in rare cases if required by protocol (for example, residual lesions in tumor types such as germ cell tumors, where known residual benign tumors can remain). When effusions are known to be a potential adverse effect of the treatment (e.g., with certain taxane compounds or angiogenesis inhibitors), cytological confirmation of the neoplastic origin of any effusion that appears or worsens during treatment can be considered if the measurable tumor has met the criteria for response or stable disease. This helps differentiate between response (or stable disease) and progressive disease.

4 Tumor response evaluation

4.1 Assessment of overall tumor burden and measurable disease

To assess objective response or future progression, it is necessary to estimate the overall tumor burden at baseline and use this as a comparator for subsequent measurements. Only patients with measurable disease at baseline should be included in protocols where objective tumor response is the primary endpoint. Measurable disease is defined as the presence of at

least one measurable lesion. In studies where the primary endpoint is tumor progression (either time to progression or proportion with progression at a fixed date), the protocol must specify if entry is restricted to those with measurable disease or whether patients with only non-measurable disease are also eligible.

4.2 Baseline documentation of 'target' and 'non-target' lesions

When more than one measurable lesion is present at baseline, all lesions up to a maximum of five lesions in total (and a maximum of two lesions per organ) representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline (this means in instances where patients have only one or two organ sites involved, a maximum of two and four lesions will be recorded, respectively).

Target lesions should be selected based on their size (lesions with the longest diameter) and be representative of all organs involved. Additionally, target lesions should lend themselves to reproducible and repeatable measurements. On occasion, the largest lesion may not lend itself to reproducible measurement, in which circumstance the next largest lesion that can be measured reproducibly should be selected.

Lymph nodes merit special mention since they are normal anatomical structures that may be visible by imaging even if there is no tumor involved. Pathological nodes, defined as measurable and may be identified as target lesions, must meet the criterion of a 15 mm short axis in CT scans. Only the short axis of these nodes will contribute to the baseline sum. The short axis of the node is the diameter commonly used by radiologists to judge if a node has any involvement with a solid tumor. Nodal size is usually reported as two dimensions in the plane in which the image is obtained (for CT scans, this is almost always the axial plane; for MRIs, the plane of acquisition may be axial, sagittal, or coronal). The smaller of these measures is the short axis. For example, an abdominal node reported as being 20 mm 30 mm has a short axis of 20 mm and qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement. All other pathological nodes (those with a short axis of 10 mm but <15 mm) should be considered non-target lesions. Nodes that have a short axis <10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for non-nodal lesions, the short axis for nodal lesions) for all target lesions will be calculated and reported as the baseline sum diameters. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added to the sum. The baseline sum diameters will be used as a reference to further characterize any objective tumor regression in the measurable dimension of the disease.

All other lesions (or sites of disease), including pathological lymph nodes, should be identified as non-target lesions and also recorded at baseline. Measurements are not required, and these lesions should be monitored as 'present,' 'absent,' or in rare cases, 'unequivocal progression' (more details to follow). In addition, multiple non-target lesions involving the same organ can be recorded as a single item on the case record form (e.g., 'multiple enlarged pelvic lymph nodes' or 'multiple liver metastases').

4.3 Response criteria

4.3.1 Evaluation of target lesions

Complete Response (CR): The disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target) must have a reduction in the short axis to <10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm (Note: the appearance of one or more new lesions is also considered progression).

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum of diameters while on study.

4.3.2 Special notes on the assessment of target lesions

Lymph nodes: Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that if lymph

nodes are included as target lesions, they cannot be considered as completely disappeared even if complete response criteria are met since a normal lymph node is defined as having a short axis of < 10 mm. Target nodal lesions will be specifically recorded in the CRF or other modalities at specific locations: for CR, all nodes must have a short axis < 10 mm; for PR, SD, and PD, the actual short axis measurement of the target node will be included in the sum of target lesion diameters.

Target lesions that become 'too small to measure': While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (e.g., 2 mm). However, sometimes lesions or lymph nodes recorded as target lesions at baseline become so faint on CT scans that the radiologist may not feel comfortable assigning an exact measure and may report them as being 'too small to measure.' When this occurs, a value must be recorded on the case report form. If the radiologist determines that the lesion has likely disappeared, it should also be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned (note: it is unlikely that a lymph node will be present, since it usually has a definable size when normal and is frequently surrounded by adipose tissue, such as in the retroperitoneum; however, if a lymph node is present, a default value of 5 mm should be assigned). This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). The measurement of these lesions is potentially non-reproducible, therefore providing this default value will prevent false responses or progressions based on measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce during treatment: When non-nodal lesions 'fragment,' the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the 'coalesced lesion.'

4.3.3 Evaluation of non-target lesions

This section provides the criteria used to determine tumor response for non-target lesions. While some non-target lesions may actually be measurable, they do not need to be measured and should instead be assessed only qualitatively at the time points specified in the protocol.

Complete Response (CR): The disappearance of all non-target lesions and normalization of the tumor marker level. All lymph nodes must be non-pathological in size (<10 mm short axis).

Non-CR/Non-PD: The persistence of one or more non-target lesion(s) and/or maintenance of tumor marker levels above the normal limits.

Progressive Disease (PD): The unequivocal progression (see comments below) of existing non-target lesions. Note: the appearance of one or more new lesions is also considered progression.

4.3.4 Special notes on assessment of progression of non-target disease

The progression of non-target disease requires further clarification:

When the patient also has measurable disease: In this setting, achieving 'unequivocal progression' on the basis of the non-target disease requires a substantial overall exacerbation in the non-target disease such that, even if the target disease displays SD or PR, the overall tumor burden has increased sufficiently to merit discontinuation of therapy. A modest 'increase' in the size of one or more non-target lesions is usually not sufficient to qualify for unequivocal progression status. Therefore, designating overall progression solely based on changes in non-target disease while target disease shows SD or PR will be extremely rare.

When the patient has only non-measurable disease: This circumstance arises in some phase III trials when it is not a criterion of study entry to have a measurable disease. The same general concepts apply here as noted above, although in this instance, there is no measurable disease assessment to factor into the interpretation of an increase in non-measurable disease burden. Since the exacerbation of a non-target disease cannot be easily assessed (by definition: if all non-target lesions are truly unmeasurable), an effective method of measurement is needed to assess when the change in non-target disease results in an

increase in overall disease burden to the extent that it corresponds to disease progression in the target disease. If described as an increase in tumor burden, it corresponds to an additional 73% increase in volume (which corresponds to a 20% increase in the diameter of a measurable lesion). Examples include peritoneal exudate ranging from "trace" to "large"; lymphangitic disease ranging from "localized" to "widespread dissemination"; or described as "sufficient to change therapy" in protocols. Examples include pleural effusions ranging from trace to large, lymphatic involvement spreading from the primary site to distant sites, or may be described in protocols as "necessary to change in treatment". If ‘unequivocal progression’ is observed, the patient should be considered to have experienced overall PD at that point. While it would be ideal to have objective criteria to apply to non-measurable disease, the very nature of that disease makes it impossible to do so; therefore, the increase must be substantial.

4.3.5 New lesions

The appearance of new malignant lesions denotes disease progression; therefore, comments on the detection of new lesions are necessary. There are no specific criteria for radiographic lesions; however, the finding of a new lesion should be unequivocal. For example, progression cannot be attributed to differences in imaging techniques, changes in imaging modality, or lesions other than tumors (e.g., some so-called new bone lesions are actually attributed to the healing or recurrence of the original lesion). This is particularly important when the patient’s baseline lesions show partial or complete response. For example, necrosis of a liver lesion may be reported on a CT scan report as a ‘new’ cystic lesion, which it is not.

A lesion identified during a follow-up assessment in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is a patient with visceral disease at baseline who, while on study, undergoes a brain CT or MRI that reveals metastases. In this situation, the patient’s brain metastases are considered to be evidence of PD even if he/she did not undergo brain imaging at baseline.

If a new lesion is equivocal (e.g., due to its small size), continued therapy and follow-up evaluations will clarify if it truly represents a new disease. If repeat scans definitively

confirm that there is a new lesion, then progression should be declared using the date of the initial scan.

While FDG-PET response assessments require additional study, it is sometimes reasonable to incorporate the use of FDG-PET scanning to complement CT scanning in the assessment of progression (particularly possible 'new' disease). New lesions on the basis of FDG-PET imaging can be identified according to the following:

Negative FDG-PET at baseline with a positive FDG-PET at follow-up is a sign of PD based on a new lesion.

No FDG-PET at baseline and a positive FDG-PET at follow-up:

If the positive FDG-PET at follow-up corresponds to a new site of disease confirmed by CT, this is PD.

If the positive FDG-PET at follow-up is not confirmed as a new site of disease on CT, additional follow-up CT scans are needed to determine if there is truly progression occurring at that site (if so, the date of PD will be the date of the initial abnormal FDG-PET scan).

If the positive FDG-PET at follow-up corresponds to a pre-existing site of disease on CT that is not progressing based on the anatomic images, this is not PD.

4.4 Evaluation of best overall response

The best overall response is the best response recorded from the start of the study treatment until the EOT, taking into account any requirement for confirmation. On occasion, a response may not be documented until after the end of therapy, so protocols must clarify whether post-treatment assessments should be considered when determining the best overall response. Protocols must specify how any new therapy introduced before progression will affect the best response designation. The patient's best overall response assignment will depend on the findings of both target and non-target diseases and will also take into consideration the appearance of new lesions. Furthermore, depending on the nature of the study and the protocol requirements, it may also require confirmatory measurement. Specifically, in non-randomized trials where response is the primary endpoint, confirmation of PR or CR is needed to deem either one the 'best overall response'.

4.4.1 Time point response

It is assumed that at each protocol-specified time point, a response assessment occurs. Table 1 on the next page provides a summary of the overall response status calculation at each time point for patients who have measurable disease at baseline.

Table 1 Time Point Response: Subjects with Target Lesions (Including or Excluding Non-Target Lesions)

Target lesions	Non-target lesions	New lesions	Overall response
CR	CR	No	CR
CR	Non-CR/non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response, PR = partial response, SD = stable disease, PD = progressive disease, and NE = non-evaluable.

When patients only have non-measurable (therefore non-target) disease, refer to Table 2.

Table 2 Time Point Response - Subjects with Non-Target Lesions Only

Non-target lesions	New lesions	Overall response
CR	No	CR
Non-CR/non-PD	No	Non-CR or non-PD[a]
Not all evaluated	No	Not evaluated
Unequivocal PD	Yes or No	PD
Any	Yes	PD

a: For non-target lesions, 'non-CR/non-PD' refers to efficacy superior to SD. Since SD is increasingly used as an endpoint to assess efficacy, a response of non-CR/non-PD is established to target when no lesions are measurable.

4.4.2 Missing assessments and non-evaluable designation

When no imaging/measurement is performed at a particular time point, the patient is not evaluable (NE) at that time. If only a subset of lesion measurements is taken at an assessment, usually the case is also considered NE at that time point, unless a convincing

argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. This would most likely happen in the case of PD. For example, if a patient has a baseline sum of 50 mm with three measured lesions and at follow-up, only two lesions are assessed, but those give a sum of 80 mm, the patient will have achieved PD status, regardless of the contribution of the missing lesion.

4.4.3 Best overall response: all time points

The best overall response is determined once all the data for the patient is known.

Best response determination in trials where confirmation of complete or partial response IS NOT required: The best response in these trials is defined as the best response across all time points (for example, a patient who has SD at the first assessment, PR at the second assessment, and PD on the last assessment has a best overall response of PR). When SD is believed to be the best response, it must also meet the protocol-specified minimum time from baseline. If the minimum time is not met when SD is otherwise the best time point response, the patient's best response depends on the subsequent assessments. For example, a patient who has SD at the first assessment, PD at the second, and does not meet the minimum duration for SD, will have a best response of PD. If the same patient is lost to follow-up after the first SD assessment, they would be considered non-evaluable.

Best response determination in trials where confirmation of complete or partial response IS required: Complete or partial responses may be claimed only if the criteria for each are met at a subsequent time point as specified in the protocol (generally 4 weeks later). In this circumstance, the best overall response can be interpreted as outlined in Table 3.

Table **3** Best Overall Response Requiring Confirmation for **CR** and **PR**

Overall response (First time point)	Overall response (Subsequent time point)	Best overall response
CR	CR	CR
CR	PR	SD, PD, or PR[a]
CR	SD	SD provided minimum criteria for SD duration met. Otherwise, PD
CR	PD	SD provided minimum criteria for SD duration met. Otherwise, PD
CR	NE	SD provided minimum criteria for SD duration met. Otherwise, NE

PR	CR	PR
PR	PR	PR
PR	SD	SD
PR	PD	SD provided minimum criteria for SD duration met. Otherwise, PD
PR	NE	SD provided minimum criteria for SD duration met. Otherwise, NE
NE	NE	NE

CR = complete response, PR = partial response, SD = stable disease, PD = progressive disease, and NE = non-evaluable.

If a CR is truly reached at the first time point, then any disease seen at a subsequent time point, even a disease meeting PR criterion relative to baseline, is considered PD at that point (since the disease must have reappeared after CR). The best response would depend on whether the minimum duration for SD has been reached. However, sometimes 'CR' may be claimed when subsequent scans suggest small lesions were likely still present and, in fact, the subject had PR rather than CR at the first time point. Under these circumstances, the original CR should be changed to PR, and the best response is PR.

4.4.4 Special notes on response assessment

When nodal disease is included in the sum of target lesions and the nodes decrease to a 'normal' size (<10 mm), they may still have a measurement reported on scans. This measurement should be recorded even if the nodes appear normal in order to not overstate progression should it be based on an increase in node size. As noted earlier, this means that subjects with CR may not have a total sum of 'zero' on the case report form (CRF).

In trials where a confirmation of response is required, repeated 'NE' time point assessments may complicate the best response determination. The trial's analysis plan must outline how missing data/assessments will be addressed when determining response and progression. For example, in most trials, it is reasonable to consider a subject with time point responses of PR-NE-PR as a confirmed response.

Subjects with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as 'symptomatic deterioration.' Every effort should be made to document objective progression even after the discontinuation of treatment. Symptomatic deterioration is not a descriptor of an objective response—it is a reason for stopping study therapy. The objective response status of such subjects should be determined by evaluating target and non-target

disease, as shown in Table s 1–3.

Conditions that define ‘early progression, early death, and non-evaluable are study specific and should be clearly described in each protocol (depending on treatment duration and periodicity).

In some circumstances, it may be difficult to distinguish residual disease from normal tissue. When the evaluation of complete response depends upon this determination, it is recommended that the residual lesion be investigated (fine-needle aspirate/biopsy) before assigning a status of complete response. FDG-PET may be used to upgrade a response to a CR, similar to a biopsy, when a residual radiographic abnormality is thought to represent fibrosis or scarring. The use of FDG-PET in this circumstance should be prospectively described in the protocol and supported by disease-specific medical literature for the indication. However, it must be acknowledged that both approaches may lead to false positive CR due to limitations of FDG-PET and biopsy resolution/sensitivity.

For equivocal findings of progression (e.g., very small and uncertain new lesions; cystic changes or necrosis in existing lesions), treatment may continue until the next scheduled assessment. If progression is confirmed at the next scheduled assessment, the date of progression should be the earlier date when progression was suspected.

4.5 Frequency of tumor re-evaluation

Frequency of tumor re-evaluation while on treatment should be protocol specific and adapted to the type and schedule of treatment. However, in the context of phase II studies where the therapy’s beneficial effects are unknown, follow-up assessments scheduled every 6–8 weeks (timed to coincide with the end of a cycle) is reasonable. Smaller or greater time intervals than these could be justified in specific regimens or circumstances. The protocol should specify which organ sites are to be evaluated at baseline (usually those most likely to be involved with metastatic disease for the tumor type under study) and how often evaluations are repeated. Normally, all target and non-target sites are evaluated at each assessment. In select circumstances, certain non-target organs may be evaluated less frequently. For example, bone scans may need to be repeated only when complete response is identified in the target disease or when progression in bone is suspected.

After the end of the treatment, the need for repetitive tumor evaluations depends on whether the trial's goal is the response rate or the time to an event (progression/death). If 'time to an event' (e.g., TTP/DFS[1]/PFS) is the main endpoint of the study, then a routine re-evaluation of protocol-specified disease sites is warranted. In randomized comparative trials, in particular, the scheduled assessments should be performed as identified on a calendar schedule (for example: every 6–8 weeks during treatment or every 3–4 months after treatment) and should not be affected by delays in therapy, holidays, or any other events that might lead to an imbalance in the timing of disease assessment between treatment arms.

4.6 Confirmatory measurement/duration of response

4.6.1 Confirmation

In non-randomized trials where response is the primary endpoint, a confirmation of PR and CR is required to ensure that identified responses are not the result of measurement error. This will also allow for the appropriate interpretation of results in the context of historical data, where response has traditionally required confirmation in such trials. However, in all other circumstances, i.e., in randomized trials (phase II or III) or studies where stable disease or progression are the primary endpoints, a confirmation of response is not required since it will not add value to the interpretation of trial results. However, eliminating the requirement for response confirmation may increase the importance of central review to protect against bias, particularly in studies that are not blinded.

In the case of SD, measurements must have met the SD criteria at least once after study entry at a minimum interval (in general, no less than 6–8 weeks) that is defined in the study protocol.

4.6.2 Duration of overall response

The duration of overall response is measured from the moment when the measurement criteria for CR/PR (whichever is first recorded) are first met until the first date that recurrent or progressive disease is objectively documented (taking as reference for progressive disease the smallest measurements recorded on study).

[1]

4.6.3 Duration of stable disease

Stable disease is measured from the start of the treatment (in randomized trials, from the randomization date) until the criteria for progression are met, taking as reference the smallest sum on study (if the baseline sum is the smallest, this is the reference for calculating PD). The clinical relevance of the duration of stable disease varies in different studies and diseases. If the proportion of subjects achieving stable disease for a minimum period of time is an endpoint of importance in a particular trial, the protocol should specify the minimal time interval required between two measurements to determine stable disease.

Note: The duration of response and stable disease, as well as the progression-free survival, are influenced by the frequency of follow-up sessions after baseline evaluation. It is not in the scope of this guideline to define a standard follow-up frequency. The frequency should take into account many parameters, including disease types and stages, treatment periodicity, and standard practice. However, these limitations in the precision of the measured endpoint should be taken into account when making comparisons between trials.

4.7 Progression-free survival/proportion progression-free

4.7.1 Phase II trials

This guideline is focused primarily on the use of objective response endpoints for phase II trials. In some circumstances, ‘response rate’ may not be the optimal method to assess the potential anticancer activity of new agents/regimens. In such cases, ‘progression-free survival’ (PFS) or the ‘proportion progression-free’ at landmark time points might be considered as appropriate alternatives to initially signal the biologic effects of new agents. It is clear, however, that in an uncontrolled trial, these measures are subject to criticism since a seemingly promising observation may be related to biological factors such as subject selection and not the impact of the intervention. Thus, phase II screening trials utilizing these endpoints are best designed with a randomized control. Exceptions may exist where the behavior patterns of certain cancers are so consistent (and usually consistently poor) that a non-randomized trial is justifiable. in these cases, However, due to the lack of an active control, the evidence of efficacy requires careful documentation when assessing the

expected PFS or PPF2.